

Topological Energy Condensate Theory (TECT): 17

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I. ACTUAL MICROSCOPIC CARRIER AUDIT AS A FINITE BASIS-OVERLAP PROBLEM

A. Problem statement

The remaining microscopic problem is not the reduced Dirac theorem itself. That part is already closed once the shell witnesses

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

are nonzero in the required pattern. The actual open question is whether the *full microscopic TECT linearization* around the relevant defect/BCC background contains a nontrivial shell-relevant transport sector carrying both:

1. one spatial derivative index,
2. one reduced internal shell-tangent index.

Equivalently, one must determine whether the true microscopic transport space contains a nonzero projection onto the minimal shell seed sector generating

$$(\lambda_{\parallel}, \alpha, \beta).$$

B. Ambient finite-dimensional formulation

Let

$$W = \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}$$

be the ambient shell-enriched tensor space, and let

$$V_{\text{TECT}} \subset W$$

denote the actual symmetry-allowed microscopic transport space extracted from the full TECT linearization around the chosen defect/BCC background.

Fix shell data

$$G_* \neq 0, \quad \hat{n} := \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\} \text{ orthonormal.}$$

The minimal shell-adapted tensor family is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (1)$$

This is exactly the smallest seed family capable of generating one longitudinal channel and the transverse E/O doublet.

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C. Shell target covectors and witness functionals

Define the three shell target covectors

$$T_{ija}^{(\parallel)}, \quad T_{ija}^{(I)}, \quad T_{ija}^{(J)},$$

adapted to the three required reduced channels. The corresponding linear witness functionals are

$$L_\alpha(N) := \langle T^{(\alpha)}, N \rangle, \quad \alpha \in \{\parallel, I, J\}. \quad (2)$$

Their exact logical meaning is:

$$L_\alpha|_{V_{\text{TECT}}} \equiv 0 \iff P_V T^{(\alpha)} = 0,$$

where $P_V : W \rightarrow V_{\text{TECT}}$ is the orthogonal projector.

Thus:

$$P_V T^{(\alpha)} \neq 0$$

means that the actual microscopic TECT linearization contains a shell-relevant nontrivial carrier direction in channel α .

D. Basis-level audit

Choose an arbitrary basis

$$\{E_A\}_{A=1}^d$$

of the actual microscopic transport space V_{TECT} . Define the overlap coefficients

$$\ell_{\parallel A} := \langle T^{(\parallel)}, E_A \rangle, \quad \ell_{IA} := \langle T^{(I)}, E_A \rangle, \quad \ell_{JA} := \langle T^{(J)}, E_A \rangle. \quad (3)$$

Then for a general element

$$N = \sum_{A=1}^d x_A E_A \in V_{\text{TECT}},$$

the witness functionals become

$$L_{\parallel}(N) = \sum_A x_A \ell_{\parallel A}, \quad L_I(N) = \sum_A x_A \ell_{IA}, \quad L_J(N) = \sum_A x_A \ell_{JA}. \quad (4)$$

Therefore:

$$L_\alpha|_{V_{\text{TECT}}} \equiv 0 \iff \ell_{\alpha A} = 0 \quad \forall A.$$

Hence the microscopic carrier audit succeeds if and only if both of the following hold:

$$\exists A : \ell_{\parallel A} \neq 0, \quad (5)$$

and

$$\exists B : (\ell_{IB}, \ell_{JB}) \neq (0, 0). \quad (6)$$

This is the exact basis-level completion criterion.

E. Theorem: basis-level carrier-audit theorem

[Basis-level carrier-audit theorem] Let $V_{\text{TTECT}} \subset W$ be the actual microscopic transport space and $\{E_A\}_{A=1}^d$ any basis of V_{TTECT} . Then the microscopic carrier audit succeeds if and only if:

1. there exists at least one basis element E_A with

$$\ell_{\parallel A} \neq 0;$$

2. there exists at least one basis element E_B with

$$(\ell_{IB}, \ell_{JB}) \neq (0, 0).$$

Equivalently,

$$P_V T^{(\parallel)} \neq 0 \quad \text{and} \quad (P_V T^{(I)}, P_V T^{(J)}) \neq (0, 0).$$

By definition,

$$L_\alpha(N) = \sum_A x_A \ell_{\alpha A}$$

for every $N = \sum_A x_A E_A \in V_{\text{TTECT}}$. Hence $L_\alpha|_{V_{\text{TTECT}}} \equiv 0$ if and only if every coefficient $\ell_{\alpha A}$ vanishes.

Therefore the longitudinal channel survives if and only if there exists at least one basis element with $\ell_{\parallel A} \neq 0$. Likewise, the transverse sector survives if and only if at least one basis element has nonzero overlap with one of the two transverse targets. Since the reduced Weyl/Dirac witness requires one longitudinal channel and at least one transverse channel, the two stated conditions are necessary and sufficient.

F. Shell-adapted decomposition

Let

$$V_{\min} = \text{span}\{B_{\parallel}, B_I, B_J\}$$

be the minimal shell seed space. Decompose each basis element as

$$E_A = a_A B_{\parallel} + b_A B_I + c_A B_J + R_A, \quad R_A \perp V_{\min}. \quad (7)$$

Because the witness functionals vanish on R_A , one gets

$$\ell_{\parallel A} = \mu_{\parallel} a_A, \quad \ell_{IA} = \mu_I b_A, \quad \ell_{JA} = \mu_J c_A,$$

with nonzero shell-normalization constants

$$\mu_{\parallel} \neq 0, \quad \mu_I \neq 0, \quad \mu_J \neq 0.$$

Hence the carrier audit is equivalently:

$$\boxed{\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0).} \quad (8)$$

So the audit has an immediate physical meaning:

1. at least one actual microscopic transport basis vector must contain a nonzero longitudinal seed component $B_{\parallel}$;
2. at least one actual microscopic transport basis vector must contain a nonzero transverse seed component in the B_I/B_J plane.

G. Immediate reduced Dirac consequence

Once the basis-level carrier audit succeeds, the reduced first-order symbol takes the form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

with

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

Equivalently, in the shell-adapted notation,

$$\det \Lambda = \Lambda_{\parallel} (\Lambda_E^2 + \Lambda_O^2) \neq 0.$$

Hence the local shell block is a genuine anisotropic Weyl symbol, and opposite-valley doubling yields the corresponding Dirac block.

H. What is now closed and what remains open

What is now closed is the exact reduction of the microscopic carrier problem to a finite, explicit basis-overlap audit:

$$\boxed{\exists A : \ell_{\parallel A} \neq 0, \quad \exists B : (\ell_{IB}, \ell_{JB}) \neq (0, 0).}$$

What remains open is no longer a conceptual PDE issue. It is the execution of this finite audit for the final complete microscopic TECT linearization.

In particular, the transverse half is already structurally realized inside the explicit seed construction, whereas the honest remaining frontier is the actual embedding/survival of the primitive longitudinal seed inside the true full microscopic transport space.

II. FINAL SUMMARY

The actual microscopic carrier problem has now been reduced to a finite-dimensional basis-overlap audit on the full microscopic transport space

$$V_{\text{TECT}} \subset \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}.$$

For any basis $\{E_A\}_{A=1}^d$ of V_{TECT} , define

$$\ell_{\parallel A} = \langle T^{(\parallel)}, E_A \rangle, \quad \ell_{IA} = \langle T^{(I)}, E_A \rangle, \quad \ell_{JA} = \langle T^{(J)}, E_A \rangle.$$

Then the exact completion criterion is

$$\boxed{\exists A : \ell_{\parallel A} \neq 0, \quad \exists B : (\ell_{IB}, \ell_{JB}) \neq (0, 0).}$$

Equivalently, the full microscopic TECT transport space must contain at least:

1. one admissible longitudinal seed direction,
2. one admissible transverse seed direction.

Once this holds, the reduced first-order symbol has the anisotropic Weyl form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

with

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),$$

and opposite-valley doubling yields the local Dirac block.

Therefore the actual remaining frontier is no longer conceptual. It is the explicit execution of the finite basis audit for the final full microscopic TECT linearization.

III. ACTUAL MICROSCOPIC CARRIER AUDIT

IV. PROBLEM STATEMENT

The remaining microscopic question is not whether the shell-level Weyl/Dirac witness logic is correct. That part is already closed inside the explicit seed/Class II realization. The actual remaining question is:

$$\boxed{\text{Does the true microscopic TECT linearization contain a nontrivial shell-relevant } (i, a)\text{-carrier sector?}} \quad (9)$$

Equivalently, after linearization around the relevant defect/BCC background and shell/defect projection, does the microscopic transport space contain at least one longitudinal seed direction and at least one transverse seed direction?

V. WHAT THE WITNESS ACTUALLY NEEDS

The shell-level local Weyl/Dirac witness is controlled by an enriched tensor

$$N_{ij}^a,$$

through the shell-adapted coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta.$$

The minimal shell-adapted tensor family is

$$\boxed{N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}.} \quad (10)$$

Hence the microscopic carrier must simultaneously supply:

1. one spatial derivative index i or j ,
2. one reduced internal shell-tangent index a .

This is the structural meaning of an (i, a) -carrier sector.

VI. EXPLICIT SEED SPACE

Define the explicit seed space

$$\boxed{V_{\text{seed}} = \text{span} \left\{ N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)} \right\}.} \quad (11)$$

Inside this seed space, for

$$N = c_{\parallel} N_{\text{seed}}^{(\parallel)} + c_E N_{\text{seed}}^{(\perp, E)} + c_O N_{\text{seed}}^{(\perp, O)},$$

the reduced witness coefficients are exactly

$$\boxed{\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}.} \quad (12)$$

Since

$$I_{\theta} \neq 0, \quad I_{\theta'^3} \neq 0,$$

the reduced symbol is nondegenerate if and only if

$$\boxed{c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).} \quad (13)$$

Therefore the witness problem is already exactly closed inside the explicit seed ansatz.

VII. FINITE-DIMENSIONAL CARRIER AUDIT

Let

$$W = \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}$$

be the ambient enriched tensor space, and let

$$V_{\text{TECT}} \subset W$$

be the actual microscopic transport space extracted from the full microscopic TECT linearization around the relevant defect/BCC background.

For each channel $\alpha \in \{\parallel, I, J\}$, define the witness functional

$$L_\alpha(N) = \langle T^{(\alpha)}, N \rangle. \quad (14)$$

Then the exact projection criterion is

$$P_V T^{(\alpha)} = 0 \iff L_\alpha|_{V_{\text{TECT}}} \equiv 0, \quad (15)$$

and

$$P_V T^{(\alpha)} \neq 0 \iff L_\alpha \text{ is generically nonzero on } V_{\text{TECT}}. \quad (16)$$

Thus the carrier problem is not an infinite-dimensional PDE mystery. It is an explicit finite projection / seed-embedding audit.

VIII. BASIS-LEVEL EXECUTABLE CRITERION

Choose any basis

$$\{E_A\}_{A=1}^d$$

of the actual microscopic transport space V_{TECT} .

Define the overlap coefficients

$$\ell_{\parallel A} := \langle T^{(\parallel)}, E_A \rangle, \quad \ell_{IA} := \langle T^{(I)}, E_A \rangle, \quad \ell_{JA} := \langle T^{(J)}, E_A \rangle. \quad (17)$$

Then the carrier audit succeeds if and only if

$$\boxed{\exists A : \ell_{\parallel A} \neq 0, \quad \exists B : (\ell_{IB}, \ell_{JB}) \neq (0, 0).} \quad (18)$$

Equivalently, if each basis element has the invariant shell-reduced decomposition

$$E_A^{\text{inv}} = a_A B_{\parallel} + b_A B_I + c_A B_J, \quad (19)$$

then

$$a_A \neq 0 \iff \ell_{\parallel A} \neq 0, \quad (b_A, c_A) \neq (0, 0) \iff (\ell_{IA}, \ell_{JA}) \neq (0, 0),$$

and the exact executable criterion becomes

$$\boxed{\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0).} \quad (20)$$

IX. THE TRANSVERSE HALF IS ALREADY CLOSED

The explicit transverse seed

$$V_{(\perp, E)}^\mu(\xi) = \theta'(\xi) E^\mu{}_a \sigma_a \quad (21)$$

gives

$$\lambda_{\parallel} = 0, \quad \alpha = I_\theta \neq 0, \quad \beta = 0. \quad (22)$$

Optionally,

$$V_{(\perp, O)}^\mu(\xi) = \theta'(\xi) J^\mu{}_a \sigma_a \quad (23)$$

gives

$$\lambda_{\parallel} = 0, \quad \alpha = 0, \quad \beta = I_\theta \neq 0. \quad (24)$$

Hence the transverse nondegeneracy condition

$$\boxed{(\alpha, \beta) \neq (0, 0)} \quad (25)$$

is already explicitly realizable. So the transverse half of the carrier audit is no longer open.

X. THE ONLY REMAINING MICROSCOPIC CARRIER QUESTION

The explicit longitudinal seed is

$$\boxed{V_{(\parallel)}^\mu(\xi) = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3,} \quad (26)$$

which gives

$$\boxed{\lambda_{\parallel} = \frac{1}{2} I_{\theta^3} \neq 0, \quad \alpha = \beta = 0.} \quad (27)$$

Therefore the unresolved microscopic question is not whether a longitudinal witness can exist in principle. It can. The remaining question is only:

$$\boxed{\text{does the full microscopic TECT linearization actually realize the primitive longitudinal channel } \theta'(\xi)^3 \sigma_3 ?} \quad (28)$$

XI. NON-FORBIDDANCE STATEMENT

At the shell-reduced symmetry level, the longitudinal channel is not symmetry-forbidden. Equivalently, the residual shell symmetry does not force

$$P_V T^{(\parallel)} = 0.$$

Hence, if the longitudinal channel vanishes in the final complete microscopic realization, the reason must be stronger than the minimal residual shell symmetry. It would require one of:

1. an additional microscopic symmetry,
2. a dynamical cancellation among actual basis elements,
3. or the absence of any genuine (i, a) -carrier sector.

So the longitudinal failure is not automatic. It would itself require a concrete microscopic explanation.

XII. CLASS-BY-CLASS AUDIT MEANING

At lowest derivative order, the admissible minimal microscopic carrier classes are:

$$\boxed{\text{Class I,} \quad \text{Class II,} \quad \text{Class III.}}$$

Among them, Class II is the strongest current explicit class because it realizes the three shell coordinates simultaneously. Its shell reduction gives

$$\delta A \propto c_{\parallel}, \quad \delta B \propto c_I, \quad \delta C \propto c_J,$$

so inside the explicit Class II truncation the carrier answer is already yes.

Therefore the strongest honest conclusion is:

$$\boxed{\text{inside the explicit Class II truncation, the carrier problem is solved;}} \quad (29)$$

what remains open is only whether the final complete microscopic TECT realization contains the same nontrivial carrier sector in the physically relevant defect background.

XIII. OPERATIONAL AUDIT ALGORITHM

The final executable microscopic audit is:

a. Step 1. Linearize the complete refined locked TECT equations around

$$\Psi_0 = \phi_0 z_0, \quad P_0 = z_0 z_0^\dagger.$$

b. Step 2. Extract the shell-relevant first-order transport operator

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}.$$

c. Step 3. Choose the low-mode shell basis at the opposite shell pair $\pm G_*$, and compute

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.$$

d. Step 4. Decompose the induced shell tensor in the basis

$$(N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)}),$$

equivalently extract

$$(A, B, C) \quad \text{or} \quad (\lambda_{\parallel}, \alpha, \beta).$$

e. Step 5. Audit the coefficients:

$$A \neq 0, \quad (B, C) \neq (0, 0).$$

Equivalently, at the basis level,

$$\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0).$$

XIV. STRONGEST HONEST THEOREM-LEVEL STATUS

We can now state the sharpest honest result.

[Actual microscopic carrier audit: sharp honest form] Assume:

1. the full refined locked TECT linearization admits a finite shell-relevant transport space V_{TECT} ;
2. the transverse odd-gradient extraction survives, so that $I_\theta \neq 0$;

- the longitudinal primitive channel survives if and only if the final basis audit finds at least one nonzero longitudinal coefficient.

Then:

- the transverse carrier is explicitly realized;
- the transverse nondegeneracy condition

$$(\alpha, \beta) \neq (0, 0)$$

is closed;

- the only remaining microscopic carrier question is the longitudinal primitive odd singlet channel,

$$\theta'(\xi)^3 \sigma_3;$$

- the full local Weyl/Dirac witness audit succeeds if and only if

$$\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0).$$

The transverse statement follows from the explicit transverse seed extraction, which already produces

$$\alpha = I_\theta \neq 0 \quad \text{or} \quad \beta = I_\theta \neq 0.$$

So the transverse half is explicitly realized.

The longitudinal statement follows because the longitudinal witness is linear on the finite-dimensional transport space. Thus it survives if and only if it is nonzero on at least one basis element, equivalently if and only if at least one invariant basis coefficient a_A is nonzero.

Combining the two gives the stated necessary and sufficient audit criterion.

XV. FINAL SUMMARY

The actual microscopic carrier audit is no longer an infinite-dimensional PDE problem. It has been reduced to a finite projection / basis-overlap audit on the shell-relevant microscopic transport space

$$V_{\text{T ECT}} \subset \mathbb{R}_{\text{out}}^3 \otimes \mathbb{R}_{\text{grad}}^3 \otimes U_{\text{int}}.$$

The shell witness requires an enriched tensor

$$N_{ij}^a,$$

whose minimal shell-adapted form is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}.$$

Inside the explicit seed space

$$V_{\text{seed}} = \text{span} \left\{ N_{\text{seed}}^{(\parallel)}, N_{\text{seed}}^{(\perp, E)}, N_{\text{seed}}^{(\perp, O)} \right\},$$

the reduced witness coefficients are exactly

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_\theta, \quad \beta = c_O I_\theta.$$

Since

$$I_\theta \neq 0, \quad I_{\theta'^3} \neq 0,$$

the reduced symbol is nondegenerate if and only if

$$c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).$$

For an actual microscopic basis $\{E_A\}$, with invariant shell-reduced decomposition

$$E_A^{\text{inv}} = a_A B_{\parallel} + b_A B_I + c_A B_J,$$

the exact executable audit criterion is

$$\boxed{\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0).}$$

The transverse half is already explicitly solved. Hence the only remaining microscopic carrier question is whether the full complete TECT linearization actually realizes the primitive longitudinal odd singlet channel

$$\theta'(\xi)^3 \sigma_3.$$

Therefore the strongest honest current status is:

$$\boxed{\text{transverse solved, longitudinal remaining.}}$$

The final frontier is not conceptual. It is the explicit computation of the invariant coefficients

$$(a_A, b_A, c_A)$$

for the basis elements of the full microscopic transport space.

XVI. ACTUAL MICROSCOPIC CARRIER AUDIT IN THE ONE-MEDIATOR BRANCH-B TRUNCATION

XVII. LOGICAL STARTING POINT

The minimal locked isotropic complex-triplet Brazovskii equation does not by itself generate the shell-enriched Pauli-sector witness N_{ij}^a . In particular, the linear quartic term gives no direct enriched first-order source, the isotropic kinetic first-order shell transport cancels exactly at the Brazovskii shell, and the locked zeroth-order cross-shell quartic projection vanishes. Therefore the microscopic carrier problem cannot be closed inside the minimal locked isotropic PDE alone.

Hence the correct microscopic route is the enriched explicit UV/Class-II channel.

XVIII. ONE-MEDIATOR TRUNCATION

Consider the refined one-mediator truncation

$$\mathcal{L}_{\text{trunc}} = \mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_E X_i^a \mathcal{E}_i^a[P, \psi] + \gamma_O X_i^a \mathcal{O}_i^a[P, \psi]. \quad (30)$$

Here:

1. $J_i^a[\Psi]$ is the microscopic longitudinal carrier current;
2. $K_i^a[P, \psi]$ is the defect-tangent odd primitive channel;
3. $\mathcal{E}_i^a[P, \psi]$ and $\mathcal{O}_i^a[P, \psi]$ are the two transverse parity-distinguished shell channels.

The actual detailed tensor realization of these operators is model-dependent, but at the shell-reduced level only their projections onto the three seed channels matter.

XIX. INTEGRATING OUT THE MEDIATOR

Since X_i^a appears quadratically and algebraically, its equation of motion is

$$M_X^2 X_i^a + \alpha_X J_i^a + \beta_X K_i^a + \gamma_E \mathcal{E}_i^a + \gamma_O \mathcal{O}_i^a = 0, \quad (31)$$

hence

$$X_i^a = -\frac{1}{M_X^2} \left(\alpha_X J_i^a + \beta_X K_i^a + \gamma_E \mathcal{E}_i^a + \gamma_O \mathcal{O}_i^a \right). \quad (32)$$

Substituting back gives the effective interaction

$$\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \left(\alpha_X J_i^a + \beta_X K_i^a + \gamma_E \mathcal{E}_i^a + \gamma_O \mathcal{O}_i^a \right)^2. \quad (33)$$

Expanding the square and keeping only the terms relevant to the three first-order shell witnesses yields

$$\mathcal{L}_{\text{eff}}^{(1)} = -\frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a - \frac{\beta_X \gamma_E}{M_X^2} K_i^a \mathcal{E}_i^a - \frac{\beta_X \gamma_O}{M_X^2} K_i^a \mathcal{O}_i^a + \dots. \quad (34)$$

Thus the carrier problem reduces to the shell projection of these three mixed bilinears.

XX. LOCAL DEFECT FRAME AND TANGENT PAULI BASIS

Let the local defect projector background be

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \quad z_B^\dagger z_B = 1, \quad (35)$$

with tangent bundle

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp. \quad (36)$$

Choose the explicit local real frame

$$z_B(\xi) = \begin{pmatrix} \cos \theta(\xi) \\ 0 \\ \sin \theta(\xi) \end{pmatrix}, \quad e_1(\xi) = \begin{pmatrix} -\sin \theta(\xi) \\ 0 \\ \cos \theta(\xi) \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \quad (37)$$

Then the tangent Pauli operators $\Sigma_1, \Sigma_2, \Sigma_3$ are the Pauli matrices acting on the local tangent doublet in the basis (e_1, e_2) .

XXI. PRIMITIVE SEED DECOMPOSITION

At the shell-reduced level, only three seed channels matter:

$$B_{\parallel} \sim \theta'(\xi)^3 \hat{n}_i \hat{n}_j \Sigma_3, \quad (38)$$

$$B_E \sim \theta'(\xi) \hat{n}_i e_{1j} \Sigma_1 + \theta'(\xi) \hat{n}_i e_{2j} \Sigma_2, \quad (39)$$

$$B_O \sim -\theta'(\xi) \hat{n}_i e_{2j} \Sigma_1 + \theta'(\xi) \hat{n}_i e_{1j} \Sigma_2. \quad (40)$$

Equivalently, the minimal shell-adapted tensor family is

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (41)$$

The reduced Weyl/Dirac witnesses are exactly the coefficients of these three channels.

XXII. SHELL PAIR AND LOW-MODE BASIS

Fix the selected shell pair $\pm G_*$, with

$$\hat{n} := \frac{G_*}{|G_*|}, \quad p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2. \quad (42)$$

Let $\{u_a^+\}$ and $\{u_b^-\}$ be the two low-mode tangent doublet bases at the two opposite valleys. The induced first-order shell matrix element of a microscopic carrier V^μ is

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle. \quad (43)$$

The projected first-order symbol is then

$$A_1(p) = p_\mu M^\mu. \quad (44)$$

Decomposing M^μ in the tangent Pauli basis yields

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (45)$$

XXIII. EVALUATION OF THE THREE COEFFICIENTS

The shell projection of the three mixed terms in (34) produces three overlap constants:

$$C_{\parallel}, \quad C_E, \quad C_O, \quad (46)$$

defined as the purely local tangent-matrix normalizations of the three seed projections.

The defect-profile dependence is carried by the two overlap integrals

$$I_3 := \int d\xi w(\xi) \theta'(\xi)^3, \quad J_1 := \int d\xi w(\xi) \theta'(\xi) \cos \theta(\xi), \quad (47)$$

with $w(\xi) = f_+(\xi) f_-(\xi) \geq 0$ the product of the two shell-localized envelope profiles.

Then the three shell witnesses are

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3, \quad (48)$$

$$\tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1, \quad (49)$$

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1. \quad (50)$$

These are the actual shell-projected carrier coefficients in the one-mediator Branch-B truncation.

XXIV. CANONICAL NORMALIZATION

If one chooses the shell tangent basis and operator normalization so that

$$C_{\parallel} = C_E = C_O = 1, \quad (51)$$

then

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} I_3, \quad \tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} J_1, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} J_1. \quad (52)$$

Thus the reduced first-order symbol becomes

$$A_1(p) = -\frac{\alpha_X \beta_X}{2M_X^2} I_3 p_{\parallel} \sigma_3 - \frac{\beta_X \gamma_E}{M_X^2} J_1 (p_1 \sigma_1 + p_2 \sigma_2) - \frac{\beta_X \gamma_O}{M_X^2} J_1 (-p_2 \sigma_1 + p_1 \sigma_2). \quad (53)$$

XXV. EXACT NONDEGENERACY CRITERION

The reduced anisotropic Weyl symbol is nondegenerate if and only if

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0).} \quad (54)$$

Using (48)–(50), this is equivalent to

$$\boxed{\alpha_X \beta_X C_{\parallel} I_3 \neq 0, \quad \beta_X J_1 (\gamma_E C_E, \gamma_O C_O) \neq (0, 0).} \quad (55)$$

Hence, assuming

$$I_3 \neq 0, \quad J_1 \neq 0, \quad \beta_X \neq 0,$$

the exact carrier audit closes if and only if

$$\boxed{\alpha_X C_{\parallel} \neq 0, \quad (\gamma_E C_E, \gamma_O C_O) \neq (0, 0).} \quad (56)$$

XXVI. BRANCH-B INTERPRETATION

Inside the local isotropic TECT algebra, the linear longitudinal selector is absent and the first surviving longitudinal primitive is cubic in θ' . Therefore the surviving longitudinal coefficient is precisely the Branch-B coefficient

$$\lambda_{\parallel} \propto I_3,$$

not a Branch-A linear coefficient.

Thus the one-mediator truncation realizes the exact Branch-B structure:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (57)$$

XXVII. INFRARED VELOCITIES

After the standard identification of the reduced PDE coefficients,

$$v_{\parallel} = \lambda_{\parallel}, \quad v_E = \tilde{\Lambda}_E, \quad v_O = \tilde{\Lambda}_O, \quad (58)$$

one obtains

$$v_{\perp} = \sqrt{v_E^2 + v_O^2} = \frac{|\beta_X| |J_1|}{M_X^2} \sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}. \quad (59)$$

Hence the anisotropy ratio is

$$\rho = \frac{v_{\parallel}}{v_{\perp}} = -\frac{\alpha_X C_{\parallel}}{2\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} \frac{I_3}{J_1}. \quad (60)$$

In the small-defect regime,

$$I_3 \sim \frac{3}{16} (\theta_0 \kappa)^2 J_1,$$

so

$$\boxed{\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X C_{\parallel}}{\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} (\theta_0 \kappa)^2.} \quad (61)$$

This is the leading Branch-B anisotropy formula.

XXVIII. WHAT IS ACTUALLY CLOSED HERE

The calculation above closes the carrier audit at the truncation level:

1. the mediator integration generates exactly the three shell-relevant mixed operators;
2. their shell projections produce the three witness coefficients $\lambda_{\parallel}, \tilde{\Lambda}_E, \tilde{\Lambda}_O$;
3. the reduced symbol is the exact Branch-B anisotropic Weyl symbol;
4. the full low-energy anisotropy ratio ρ follows explicitly.

XXIX. WHAT IS STILL NOT UNCONDITIONAL

What remains not yet unconditional is not the algebraic reduction itself. The unresolved issue is the final microscopic value of the three local normalization overlaps

$$C_{\parallel}, \quad C_E, \quad C_O,$$

for the actual complete refined locked microscopic TECT realization.

Thus the remaining frontier is a calibration problem: compute the exact projected matrix elements (43) in the final full microscopic Bloch linearization and verify

$$C_{\parallel} \neq 0, \quad (C_E, C_O) \neq (0, 0).$$

XXX. FINAL SUMMARY

The actual microscopic carrier audit in the refined one-mediator Branch-B truncation is now reduced to an explicit shell-projection calculation.

Starting from

$$\mathcal{L}_{\text{trunc}} = \mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a + \beta_X X_i^a K_i^a + \gamma_E X_i^a \mathcal{E}_i^a + \gamma_O X_i^a \mathcal{O}_i^a,$$

integrating out X_i^a generates the three relevant mixed effective operators

$$-\frac{\alpha_X \beta_X}{M_X^2} J \cdot K, \quad -\frac{\beta_X \gamma_E}{M_X^2} K \cdot \mathcal{E}, \quad -\frac{\beta_X \gamma_O}{M_X^2} K \cdot \mathcal{O}.$$

After shell/defect projection, these produce the exact Branch-B witness coefficients

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3, \quad \tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.$$

Hence the reduced first-order symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

The exact nondegeneracy criterion is

$$\lambda_{\parallel} \neq 0, \quad (\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0).$$

Therefore the remaining microscopic frontier is not conceptual: it is the explicit determination of the three shell normalization overlaps

$$C_{\parallel}, \quad C_E, \quad C_O$$

from the final complete projected Bloch linearization.

In canonical normalization,

$$C_{\parallel} = C_E = C_O = 1,$$

the anisotropy ratio becomes

$$\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}} (\theta_0 \kappa)^2.$$

Thus the actual carrier audit is reduced completely to a finite calibration problem.

XXXI. MATRIX-ELEMENT AUDIT FOR THE ACTUAL MICROSCOPIC CARRIER COEFFICIENTS

XXXII. GOAL

The remaining microscopic calibration problem is to extract the three shell normalization overlaps

$$C_{\parallel}, \quad C_E, \quad C_O$$

from the projected 2×2 shell matrix elements.

The reduced first-order symbol is generated from

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle, \quad a, b = 1, 2, \quad (62)$$

where $\{u_a^+\}$ and $\{u_b^-\}$ are the low-mode tangent-doublet bases at the opposite shell pair $\pm G_*$.

The effective first-order shell operator is

$$A_1(p) = p_\mu M^\mu. \quad (63)$$

The task is therefore to decompose M^μ into the unique shell-adapted Pauli basis and read off the coefficients.

XXXIII. SHELL-ADAPTED PAULI DECOMPOSITION

Fix the shell direction

$$\hat{n} := \frac{G_*}{|G_*|},$$

and choose an orthonormal frame

$$\{e_1, e_2, \hat{n}\}.$$

Decompose momentum as

$$p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2. \quad (64)$$

Since each M^μ is a 2×2 Hermitian matrix acting on the shell tangent doublet, it admits the unique decomposition

$$M^\mu = m_0^\mu \mathbf{1}_2 + m_1^\mu \sigma_1 + m_2^\mu \sigma_2 + m_3^\mu \sigma_3, \quad (65)$$

with real coefficients

$$m_A^\mu = \frac{1}{2} \text{Tr}(\sigma_A M^\mu), \quad A = 0, 1, 2, 3, \quad (66)$$

where $\sigma_0 := \mathbf{1}_2$.

Therefore

$$A_1(p) = p_\mu m_0^\mu \mathbf{1}_2 + p_\mu m_1^\mu \sigma_1 + p_\mu m_2^\mu \sigma_2 + p_\mu m_3^\mu \sigma_3. \quad (67)$$

XXXIV. BRANCH-B SHELL-ADAPTED NORMAL FORM

The exact Branch-B reduced symbol has the shell-adapted form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \tilde{\Lambda}_E(p_1 \sigma_1 + p_2 \sigma_2) + \tilde{\Lambda}_O(-p_2 \sigma_1 + p_1 \sigma_2), \quad (68)$$

up to the removable scalar drift term $p_\mu m_0^\mu \mathbf{1}_2$.

Comparing (67) with (68), one obtains the shell-adapted coefficient identities

$$m_3^{\parallel} = \lambda_{\parallel}, \quad m_3^1 = m_3^2 = 0, \quad (69)$$

and

$$\begin{pmatrix} m_1^1 & m_1^2 \\ m_2^1 & m_2^2 \end{pmatrix} = \begin{pmatrix} \tilde{\Lambda}_E & -\tilde{\Lambda}_O \\ \tilde{\Lambda}_O & \tilde{\Lambda}_E \end{pmatrix}. \quad (70)$$

Thus the transverse part is exactly a 2×2 matrix of the form

$$\tilde{\Lambda}_E \mathbf{1}_2 + \tilde{\Lambda}_O J, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

which is the standard E/O decomposition.

XXXV. DIRECT EXTRACTION FORMULAS

The three reduced carrier coefficients are therefore read directly from M^μ as

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}), \quad (71)$$

$$\tilde{\Lambda}_E = \frac{1}{2} \text{Tr}(\sigma_1 M^1) = \frac{1}{2} \text{Tr}(\sigma_2 M^2), \quad (72)$$

$$\tilde{\Lambda}_O = \frac{1}{2} \text{Tr}(\sigma_2 M^1) = -\frac{1}{2} \text{Tr}(\sigma_1 M^2). \quad (73)$$

Consistency of the Branch-B form further requires

$$\text{Tr}(\sigma_1 M^1) = \text{Tr}(\sigma_2 M^2), \quad \text{Tr}(\sigma_2 M^1) = -\text{Tr}(\sigma_1 M^2), \quad (74)$$

and the absence of forbidden extra components means

$$\text{Tr}(\sigma_3 M^1) = \text{Tr}(\sigma_3 M^2) = 0. \quad (75)$$

XXXVI. DEFINITION OF THE NORMALIZATION OVERLAPS

Now let the microscopic carrier split into the three relevant pieces

$$V^\mu = V_{(\parallel)}^\mu + V_{(E)}^\mu + V_{(O)}^\mu + \cdots, \quad (76)$$

where the ellipsis denotes shell-irrelevant or higher-order terms.

Define the three projected matrix elements

$$M_{(\parallel)}^\mu := \langle u^+, V_{(\parallel)}^\mu u^- \rangle, \quad M_{(E)}^\mu := \langle u^+, V_{(E)}^\mu u^- \rangle, \quad M_{(O)}^\mu := \langle u^+, V_{(O)}^\mu u^- \rangle. \quad (77)$$

The shell normalization overlaps are then defined by the Pauli traces

$$C_{\parallel} := \frac{2}{I_3} \left(-\frac{M_X^2}{\alpha_X \beta_X} \right) \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}), \quad (78)$$

$$C_E := \frac{1}{J_1} \left(-\frac{M_X^2}{\beta_X \gamma_E} \right) \frac{1}{2} \text{Tr}(\sigma_1 M^1), \quad (79)$$

$$C_O := \frac{1}{J_1} \left(-\frac{M_X^2}{\beta_X \gamma_O} \right) \frac{1}{2} \text{Tr}(\sigma_2 M^1). \quad (80)$$

Equivalently, using the transverse consistency relations,

$$C_E = \frac{1}{J_1} \left(-\frac{M_X^2}{\beta_X \gamma_E} \right) \frac{1}{2} \text{Tr}(\sigma_2 M^2), \quad (81)$$

$$C_O = -\frac{1}{J_1} \left(-\frac{M_X^2}{\beta_X \gamma_O} \right) \frac{1}{2} \text{Tr}(\sigma_1 M^2). \quad (82)$$

These are the exact matrix-element definitions of the three microscopic carrier overlaps.

XXXVII. EQUIVALENCE WITH THE ONE-MEDIATOR COEFFICIENTS

In the one-mediator Branch-B truncation, the shell-projected coefficients were

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3, \quad \tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1. \quad (83)$$

Substituting (71)–(73) into (83) gives the exact identification

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}) = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3,} \quad (84)$$

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_1 M^1) = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1,} \quad (85)$$

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_2 M^1) = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.} \quad (86)$$

Thus the entire carrier audit is reduced to three trace evaluations of projected shell matrices.

XXXVIII. CANONICAL NORMALIZATION AND ORTHONORMAL SHELL BASIS

If the shell bases $\{u_a^{\pm}\}$ are chosen so that the three seed operators have unit Pauli projection, then one may impose canonical normalization

$$C_{\parallel} = C_E = C_O = 1. \quad (87)$$

In this convention, the matrix-element audit simplifies to

$$\frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}) = -\frac{\alpha_X \beta_X}{2M_X^2} I_3, \quad (88)$$

$$\frac{1}{2} \text{Tr}(\sigma_1 M^1) = -\frac{\beta_X \gamma_E}{M_X^2} J_1, \quad (89)$$

$$\frac{1}{2} \text{Tr}(\sigma_2 M^1) = -\frac{\beta_X \gamma_O}{M_X^2} J_1. \quad (90)$$

Hence the reduced symbol is completely calibrated once the shell basis is orthonormalized.

XXXIX. EXECUTABLE AUDIT TABLE

The actual microscopic computation can therefore be organized as the following table:

channel	projected matrix	Pauli trace	extracted coefficient
longitudinal	$M^\parallel$	$\frac{1}{2} \text{Tr}(\sigma_3 M^\parallel)$	$\lambda_\parallel$
transverse E	M^1, M^2	$\frac{1}{2} \text{Tr}(\sigma_1 M^1) = \frac{1}{2} \text{Tr}(\sigma_2 M^2)$	$\tilde{\Lambda}_E$
transverse O	M^1, M^2	$\frac{1}{2} \text{Tr}(\sigma_2 M^1) = -\frac{1}{2} \text{Tr}(\sigma_1 M^2)$	$\tilde{\Lambda}_O$

Equivalently, one reads

$$\lambda_\parallel \neq 0 \iff \text{Tr}(\sigma_3 M^\parallel) \neq 0, \quad (91)$$

and

$$(\tilde{\Lambda}_E, \tilde{\Lambda}_O) \neq (0, 0) \iff (\text{Tr}(\sigma_1 M^1), \text{Tr}(\sigma_2 M^1)) \neq (0, 0). \quad (92)$$

XL. EXACT NONDEGENERACY IN MATRIX FORM

The exact reduced Weyl nondegeneracy criterion becomes

$$\boxed{\text{Tr}(\sigma_3 M^\parallel) \neq 0, \quad (\text{Tr}(\sigma_1 M^1), \text{Tr}(\sigma_2 M^1)) \neq (0, 0).} \quad (93)$$

So the final full microscopic carrier problem is not conceptual: it is the explicit evaluation of these traces in the projected Bloch linearization.

XLI. WHAT IS CLOSED AND WHAT REMAINS OPEN

What is now closed is the exact matrix-element dictionary: the three shell normalization overlaps $C_\parallel, C_E, C_O$ are completely equivalent to three Pauli traces of the projected 2×2 shell matrices.

What remains open is only the actual microscopic numerical/symbolic evaluation of

$$M^\parallel, \quad M^1, \quad M^2$$

for the final complete refined locked TECT Bloch operator.

XLII. FINAL SUMMARY

The microscopic carrier calibration problem has now been reduced to an explicit 2×2 shell matrix audit. For the projected shell matrices

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle,$$

the first-order symbol is

$$A_1(p) = p_\mu M^\mu.$$

Each M^μ admits the unique Pauli decomposition

$$M^\mu = m_0^\mu \mathbf{1}_2 + m_1^\mu \sigma_1 + m_2^\mu \sigma_2 + m_3^\mu \sigma_3.$$

Comparing with the exact Branch-B normal form

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \tilde{\Lambda}_E (p_1 \sigma_1 + p_2 \sigma_2) + \tilde{\Lambda}_O (-p_2 \sigma_1 + p_1 \sigma_2),$$

gives the direct extraction formulas

$$\lambda_\parallel = \frac{1}{2} \text{Tr}(\sigma_3 M^\parallel), \quad \tilde{\Lambda}_E = \frac{1}{2} \text{Tr}(\sigma_1 M^1), \quad \tilde{\Lambda}_O = \frac{1}{2} \text{Tr}(\sigma_2 M^1).$$

Equivalently, the shell normalization overlaps are exactly the corresponding rescaled Pauli traces:

$$C_{\parallel} \leftrightarrow \text{Tr}(\sigma_3 M^{\parallel}), \quad C_E \leftrightarrow \text{Tr}(\sigma_1 M^1), \quad C_O \leftrightarrow \text{Tr}(\sigma_2 M^1).$$

Therefore the exact Weyl nondegeneracy criterion is

$$\text{Tr}(\sigma_3 M^{\parallel}) \neq 0, \quad (\text{Tr}(\sigma_1 M^1), \text{Tr}(\sigma_2 M^1)) \neq (0, 0).$$

Thus the final microscopic carrier problem is reduced completely to three explicit trace evaluations of the projected Bloch shell matrices.

XLIII. OPERATOR-LEVEL AUDIT OF THE ACTUAL MICROSCOPIC CARRIER

XLIV. GOAL

The remaining microscopic task is to compute the projected shell matrices

$$(M^{\mu})_{ab} = \langle \Psi_a^+, V^{\mu} \Psi_b^- \rangle, \quad a, b = 1, 2,$$

for the three relevant first-order carrier operators

$$V_{(\parallel)}^{\mu}, \quad V_{(E)}^{\mu}, \quad V_{(O)}^{\mu}.$$

The purpose of this section is to show that, once the shell-projected operator ansatz is fixed, the reduced coefficients

$$\lambda_{\parallel}, \quad \tilde{\Lambda}_E, \quad \tilde{\Lambda}_O$$

follow by direct trace evaluation.

XLV. LOW-MODE SHELL BASIS

Fix the selected opposite shell pair $\pm G_*$, with

$$\hat{n} := \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\} \text{ orthonormal.}$$

Let the shell-localized low modes be

$$\Psi_a^{\pm}(\xi, y) = f_{\pm}(\xi) e^{\pm i G_* \cdot y} u_a^{\pm}(\xi), \quad a = 1, 2, \quad (94)$$

where:

1. ξ is the defect-normal coordinate,
2. y denotes the slow coordinates tangent to the shell frame,
3. $f_{\pm}(\xi)$ are the two valley envelope profiles,
4. $u_a^{\pm}(\xi)$ are orthonormal tangent-doublet basis vectors.

We assume the opposite-valley basis has already been matched so that the tangent Pauli basis is the same on both sides.

Define the overlap weight

$$w(\xi) := f_+(\xi) f_-(\xi). \quad (95)$$

The two basic overlap integrals are

$$I_3 := \int d\xi w(\xi) \theta'(\xi)^3, \quad J_1 := \int d\xi w(\xi) \theta'(\xi) \cos \theta(\xi). \quad (96)$$

XLVI. PROJECTED OPERATOR ANSATZ

After integrating out the mediator X_i^a , the relevant effective first-order operator is

$$V_{\text{eff}}^\mu = -\frac{\alpha_X \beta_X}{M_X^2} V_{JK}^\mu - \frac{\beta_X \gamma_E}{M_X^2} V_{KE}^\mu - \frac{\beta_X \gamma_O}{M_X^2} V_{KO}^\mu. \quad (97)$$

At the shell-reduced level, we only need the projections of these three pieces onto the tangent-doublet sector. The exact Branch-B ansatz is

$$\Pi_D V_{JK}^\mu \Pi_D = \frac{C_\parallel}{2} \theta'(\xi)^3 \hat{n}^\mu \sigma_3, \quad (98)$$

$$\Pi_D V_{KE}^\mu \Pi_D = C_E \theta'(\xi) \cos \theta(\xi) (e_1^\mu \sigma_1 + e_2^\mu \sigma_2), \quad (99)$$

$$\Pi_D V_{KO}^\mu \Pi_D = C_O \theta'(\xi) \cos \theta(\xi) (-e_2^\mu \sigma_1 + e_1^\mu \sigma_2). \quad (100)$$

Here $C_\parallel, C_E, C_O$ are the purely local shell/tangent normalization overlaps. They encode the actual microscopic Bloch-frame calibration.

XLVII. DIRECT COMPUTATION OF THE PROJECTED SHELL MATRICES

Insert (98)–(100) into (97), and evaluate on the low-mode basis (94).

Because the oscillatory shell phases are already matched between opposite valleys, the projected matrix element reduces to the ξ -integral:

$$M^\mu = \int d\xi w(\xi) \left[-\frac{\alpha_X \beta_X}{M_X^2} \Pi_D V_{JK}^\mu \Pi_D - \frac{\beta_X \gamma_E}{M_X^2} \Pi_D V_{KE}^\mu \Pi_D - \frac{\beta_X \gamma_O}{M_X^2} \Pi_D V_{KO}^\mu \Pi_D \right]. \quad (101)$$

Using (96), one obtains

$$M_{(\parallel)}^\mu = -\frac{\alpha_X \beta_X}{2M_X^2} C_\parallel I_3 \hat{n}^\mu \sigma_3, \quad (102)$$

$$M_{(E)}^\mu = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1 (e_1^\mu \sigma_1 + e_2^\mu \sigma_2), \quad (103)$$

$$M_{(O)}^\mu = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 (-e_2^\mu \sigma_1 + e_1^\mu \sigma_2). \quad (104)$$

Therefore the total shell matrix is

$$M^\mu = -\frac{\alpha_X \beta_X}{2M_X^2} C_\parallel I_3 \hat{n}^\mu \sigma_3 - \frac{\beta_X \gamma_E}{M_X^2} C_E J_1 (e_1^\mu \sigma_1 + e_2^\mu \sigma_2) - \frac{\beta_X \gamma_O}{M_X^2} C_O J_1 (-e_2^\mu \sigma_1 + e_1^\mu \sigma_2). \quad (105)$$

XLVIII. COMPONENT FORM

Project M^μ onto the shell frame directions $\hat{n}, e_1, e_2$:

$$M^\parallel := \hat{n}_\mu M^\mu, \quad M^1 := e_{1\mu} M^\mu, \quad M^2 := e_{2\mu} M^\mu. \quad (106)$$

From (105) we get

$$M^\parallel = -\frac{\alpha_X \beta_X}{2M_X^2} C_\parallel I_3 \sigma_3, \quad (107)$$

$$M^1 = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1 \sigma_1 - \frac{\beta_X \gamma_O}{M_X^2} C_O J_1 \sigma_2, \quad (108)$$

$$M^2 = +\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 \sigma_1 - \frac{\beta_X \gamma_E}{M_X^2} C_E J_1 \sigma_2. \quad (109)$$

This is the exact 2×2 shell block required for the Branch-B carrier audit.

XLIX. READING OFF THE REDUCED SYMBOL

The first-order shell symbol is

$$A_1(p) = p_\mu M^\mu = p_\parallel M^\parallel + p_1 M^1 + p_2 M^2. \quad (110)$$

Substituting (107)–(109),

$$A_1(p) = -\frac{\alpha_X \beta_X}{2M_X^2} C_\parallel I_3 p_\parallel \sigma_3 \quad (111)$$

$$- \frac{\beta_X \gamma_E}{M_X^2} C_E J_1 (p_1 \sigma_1 + p_2 \sigma_2) \quad (112)$$

$$- \frac{\beta_X \gamma_O}{M_X^2} C_O J_1 (-p_2 \sigma_1 + p_1 \sigma_2). \quad (113)$$

Hence the exact reduced coefficients are

$$\lambda_\parallel = -\frac{\alpha_X \beta_X}{2M_X^2} C_\parallel I_3, \quad (114)$$

$$\tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1, \quad (115)$$

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1. \quad (116)$$

So the operator-level audit reproduces exactly the expected Branch-B shell symbol.

L. TRACE TEST

Now compute the Pauli traces directly.
From (107)–(109),

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_3 M^\parallel) = -\frac{\alpha_X \beta_X}{2M_X^2} C_\parallel I_3 = \lambda_\parallel,} \quad (117)$$

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_1 M^1) = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1 = \tilde{\Lambda}_E,} \quad (118)$$

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_2 M^1) = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 = \tilde{\Lambda}_O.} \quad (119)$$

Consistency also gives

$$\frac{1}{2} \text{Tr}(\sigma_2 M^2) = \tilde{\Lambda}_E, \quad -\frac{1}{2} \text{Tr}(\sigma_1 M^2) = \tilde{\Lambda}_O. \quad (120)$$

Thus the full carrier audit reduces to three explicit trace evaluations.

LI. EXACT NONDEGENERACY CRITERION IN OPERATOR FORM

The reduced Weyl symbol is nondegenerate if and only if

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_3 M^\parallel) \neq 0, \quad \left(\frac{1}{2} \text{Tr}(\sigma_1 M^1), \frac{1}{2} \text{Tr}(\sigma_2 M^1) \right) \neq (0, 0).} \quad (121)$$

Equivalently,

$$\boxed{\alpha_X \beta_X C_\parallel I_3 \neq 0, \quad \beta_X J_1 (\gamma_E C_E, \gamma_O C_O) \neq (0, 0).} \quad (122)$$

Assuming

$$I_3 \neq 0, \quad J_1 \neq 0, \quad \beta_X \neq 0,$$

this becomes simply

$$\boxed{\alpha_X C_\parallel \neq 0, \quad (\gamma_E C_E, \gamma_O C_O) \neq (0, 0).} \quad (123)$$

LII. CANONICAL NORMALIZATION

If the shell tangent basis is orthonormalized so that

$$C_\parallel = C_E = C_O = 1, \quad (124)$$

then

$$M^\parallel = -\frac{\alpha_X \beta_X}{2M_X^2} I_3 \sigma_3, \quad M^1 = -\frac{\beta_X \gamma_E}{M_X^2} J_1 \sigma_1 - \frac{\beta_X \gamma_O}{M_X^2} J_1 \sigma_2, \quad (125)$$

$$M^2 = +\frac{\beta_X \gamma_O}{M_X^2} J_1 \sigma_1 - \frac{\beta_X \gamma_E}{M_X^2} J_1 \sigma_2. \quad (126)$$

Therefore

$$A_1(p) = -\frac{\alpha_X \beta_X}{2M_X^2} I_3 p_\parallel \sigma_3 - \frac{\beta_X \gamma_E}{M_X^2} J_1 (p_1 \sigma_1 + p_2 \sigma_2) - \frac{\beta_X \gamma_O}{M_X^2} J_1 (-p_2 \sigma_1 + p_1 \sigma_2). \quad (127)$$

This is the explicit canonical Branch-B shell block.

LIII. ANISOTROPY RATIO

Identifying

$$v_{\parallel} = \lambda_{\parallel}, \quad v_E = \tilde{\Lambda}_E, \quad v_O = \tilde{\Lambda}_O,$$

gives

$$v_{\perp} = \sqrt{v_E^2 + v_O^2} = \frac{|\beta_X| |J_1|}{M_X^2} \sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}. \quad (128)$$

Hence

$$\rho = \frac{v_{\parallel}}{v_{\perp}} = -\frac{\alpha_X C_{\parallel}}{2\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} \frac{I_3}{J_1}. \quad (129)$$

In the small-defect regime,

$$I_3 \sim \frac{3}{16} (\theta_0 \kappa)^2 J_1,$$

so

$$\boxed{\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X C_{\parallel}}{\sqrt{\gamma_E^2 C_E^2 + \gamma_O^2 C_O^2}} (\theta_0 \kappa)^2.} \quad (130)$$

LIV. WHAT IS NOW FULLY CLOSED

The operator-level carrier audit is now algebraically closed:

1. the projected operator ansatz (98)–(100) fixes the shell block;
2. the low-mode projection gives the explicit 2×2 matrices $M^{\parallel}, M^1, M^2$;
3. the Pauli traces directly reproduce

$$\lambda_{\parallel}, \quad \tilde{\Lambda}_E, \quad \tilde{\Lambda}_O;$$

4. the exact Branch-B nondegeneracy criterion becomes the explicit trace test (121);
5. the anisotropy ratio follows immediately.

LV. WHAT IS STILL NOT UNCONDITIONAL

What remains not yet unconditional is not the operator algebra.

The only remaining issue is the final microscopic calibration: compute the actual projected Bloch-frame overlaps and determine the numerical values of

$$C_{\parallel}, \quad C_E, \quad C_O$$

for the complete refined locked TECT linearization.

Thus the remaining frontier is a final microscopic Bloch calibration problem, not a conceptual obstruction.

LVI. FINAL SUMMARY

The operator-level microscopic carrier audit is now algebraically closed.
For the low-mode shell basis

$$\Psi_a^\pm(\xi, y) = f_\pm(\xi) e^{\pm i G_{*} \cdot y} u_a^\pm(\xi),$$

the projected effective carrier operator produces the explicit shell matrices

$$\begin{aligned} M^\parallel &= -\frac{\alpha_X \beta_X}{2M_X^2} C_\parallel I_3 \sigma_3, \\ M^1 &= -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1 \sigma_1 - \frac{\beta_X \gamma_O}{M_X^2} C_O J_1 \sigma_2, \\ M^2 &= +\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 \sigma_1 - \frac{\beta_X \gamma_E}{M_X^2} C_E J_1 \sigma_2. \end{aligned}$$

Therefore the reduced Branch-B first-order symbol is

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \tilde{\Lambda}_E(p_1 \sigma_1 + p_2 \sigma_2) + \tilde{\Lambda}_O(-p_2 \sigma_1 + p_1 \sigma_2),$$

with

$$\lambda_\parallel = -\frac{\alpha_X \beta_X}{2M_X^2} C_\parallel I_3, \quad \tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.$$

Equivalently, the coefficients are extracted by the direct Pauli traces

$$\lambda_\parallel = \frac{1}{2} \text{Tr}(\sigma_3 M^\parallel), \quad \tilde{\Lambda}_E = \frac{1}{2} \text{Tr}(\sigma_1 M^1), \quad \tilde{\Lambda}_O = \frac{1}{2} \text{Tr}(\sigma_2 M^1).$$

Hence the exact nondegeneracy criterion is

$$\frac{1}{2} \text{Tr}(\sigma_3 M^\parallel) \neq 0, \quad \left(\frac{1}{2} \text{Tr}(\sigma_1 M^1), \frac{1}{2} \text{Tr}(\sigma_2 M^1) \right) \neq (0, 0).$$

Thus the final remaining frontier is not conceptual. It is the explicit Bloch-frame calibration of the three microscopic overlaps

$$C_\parallel, \quad C_E, \quad C_O.$$

LVII. FINAL BLOCH CALIBRATION AUDIT

LVIII. GOAL

The remaining frontier is the explicit microscopic calibration of the three shell overlaps

$$C_\parallel, \quad C_E, \quad C_O$$

for the complete refined locked Bloch linearization.

At this stage, the problem is no longer conceptual. The reduced Branch-B symbol is already known to take the form

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \tilde{\Lambda}_E(p_1 \sigma_1 + p_2 \sigma_2) + \tilde{\Lambda}_O(-p_2 \sigma_1 + p_1 \sigma_2),$$

and the only remaining task is to determine the three calibration overlaps from the actual projected Bloch operator.

LIX. COMPLETE PROJECTED BLOCH OPERATOR

Let the complete refined locked microscopic linearization around the chosen defect/BCC background be

$$\mathcal{L}_{\text{mic}}(p). \quad (131)$$

Fix the selected opposite shell pair $\pm G_*$, and let

$$\Pi_D$$

be the projector onto the shell-relevant doubled local block

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2.$$

Then the first-order shell operator is defined by

$$\mathcal{L}_D^{(1)}(p) := \Pi_D \mathcal{L}_{\text{mic}}(p) \Pi_D \Big|_{\text{linear in } p}. \quad (132)$$

The Bloch calibration problem is to compare $\mathcal{L}_D^{(1)}(p)$ with the canonical Branch-B basis.

LX. NORMALIZED LOW-MODE BASIS

Choose orthonormal low-mode states

$$\Psi_a^\pm(\xi, y) = f_\pm(\xi) e^{\pm i G_* \cdot y} u_a^\pm(\xi), \quad a = 1, 2, \quad (133)$$

with normalization

$$\langle \Psi_a^\pm, \Psi_b^\pm \rangle = \delta_{ab}, \quad \langle \Psi_a^+, \Psi_b^- \rangle = 0. \quad (134)$$

Assume the opposite-valley tangent frames have been matched so that the same Pauli basis

$$\sigma_1, \sigma_2, \sigma_3$$

acts on both local doublets.

Define the defect-profile overlaps

$$I_3 = \int d\xi w(\xi) \theta'(\xi)^3, \quad J_1 = \int d\xi w(\xi) \theta'(\xi) \cos \theta(\xi), \quad w(\xi) := f_+(\xi) f_-(\xi). \quad (135)$$

LXI. CANONICAL SEED OPERATORS

Define the three canonical normalized seed operators by

$$S_\parallel^\mu := \frac{1}{2I_3} \theta'(\xi)^3 \hat{n}^\mu \sigma_3, \quad (136)$$

$$S_E^\mu := \frac{1}{J_1} \theta'(\xi) \cos \theta(\xi) (e_1^\mu \sigma_1 + e_2^\mu \sigma_2), \quad (137)$$

$$S_O^\mu := \frac{1}{J_1} \theta'(\xi) \cos \theta(\xi) (-e_2^\mu \sigma_1 + e_1^\mu \sigma_2). \quad (138)$$

These are normalized so that their reduced shell projections directly reproduce the canonical Pauli structures.

LXII. DEFINITION OF THE CALIBRATION OVERLAPS

Let the actual shell-projected microscopic carrier pieces be

$$\Pi_D V_{JK}^\mu \Pi_D, \quad \Pi_D V_{KE}^\mu \Pi_D, \quad \Pi_D V_{KO}^\mu \Pi_D.$$

The calibration overlaps are defined by their projections onto the normalized seed operators:

$$C_{\parallel} := \frac{\langle \Psi^+, \Pi_D V_{JK}^\mu \Pi_D \Psi^- \rangle}{\langle \Psi^+, S_{\parallel}^\mu \Psi^- \rangle}, \quad (139)$$

$$C_E := \frac{\langle \Psi^+, \Pi_D V_{KE}^\mu \Pi_D \Psi^- \rangle}{\langle \Psi^+, S_E^\mu \Psi^- \rangle}, \quad (140)$$

$$C_O := \frac{\langle \Psi^+, \Pi_D V_{KO}^\mu \Pi_D \Psi^- \rangle}{\langle \Psi^+, S_O^\mu \Psi^- \rangle}. \quad (141)$$

Equivalently, at the 2×2 shell matrix level,

$$C_{\parallel} = \frac{\frac{1}{2} \text{Tr}(\sigma_3 M_{JK}^{\parallel})}{-\frac{\alpha_X \beta_X}{2M_X^2} I_3}, \quad (142)$$

$$C_E = \frac{\frac{1}{2} \text{Tr}(\sigma_1 M_{KE}^1)}{-\frac{\beta_X \gamma_E}{M_X^2} J_1}, \quad (143)$$

$$C_O = \frac{\frac{1}{2} \text{Tr}(\sigma_2 M_{KO}^1)}{-\frac{\beta_X \gamma_O}{M_X^2} J_1}. \quad (144)$$

These formulas are the exact Bloch-frame calibration definitions.

LXIII. PROJECTOR-LEVEL YES/NO TEST

The carrier audit now reduces to the following three projector tests:

a. Longitudinal test.

$$C_{\parallel} \neq 0 \iff \Pi_D V_{JK}^\mu \Pi_D \text{ has nonzero projection onto } S_{\parallel}^\mu. \quad (145)$$

b. Transverse E test.

$$C_E \neq 0 \iff \Pi_D V_{KE}^\mu \Pi_D \text{ has nonzero projection onto } S_E^\mu. \quad (146)$$

c. Transverse O test.

$$C_O \neq 0 \iff \Pi_D V_{KO}^\mu \Pi_D \text{ has nonzero projection onto } S_O^\mu. \quad (147)$$

Thus the final exact nondegeneracy criterion is

$$C_{\parallel} \neq 0, \quad (C_E, C_O) \neq (0, 0). \quad (148)$$

LXIV. EQUIVALENT BASIS-OVERLAP FORMULATION

Let

$$\{E_A\}_{A=1}^d$$

be any basis of the shell-relevant microscopic transport space. Write the invariant shell decomposition

$$E_A^{\text{inv}} = a_A B_{\parallel} + b_A B_E + c_A B_O. \quad (149)$$

Then

$$C_{\parallel} \neq 0 \iff \exists A : a_A \neq 0,$$

and

$$(C_E, C_O) \neq (0, 0) \iff \exists B : (b_B, c_B) \neq (0, 0).$$

Hence the final Bloch calibration audit is exactly equivalent to the earlier basis-audit theorem.

LXV. CANONICAL NORMALIZATION

If the low-mode basis and seed operators are chosen so that the canonical seed projections have unit overlap, then

$$\boxed{C_{\parallel} = C_E = C_O = 1.} \quad (150)$$

In this normalization,

$$M^{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} I_3 \sigma_3, \quad (151)$$

$$M^1 = -\frac{\beta_X \gamma_E}{M_X^2} J_1 \sigma_1 - \frac{\beta_X \gamma_O}{M_X^2} J_1 \sigma_2, \quad (152)$$

$$M^2 = +\frac{\beta_X \gamma_O}{M_X^2} J_1 \sigma_1 - \frac{\beta_X \gamma_E}{M_X^2} J_1 \sigma_2. \quad (153)$$

Thus the reduced coefficients are directly

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} I_3, \quad \tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} J_1, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} J_1. \quad (154)$$

LXVI. FINAL CALIBRATION ALGORITHM

The complete executable Bloch calibration audit is:

1. Construct the complete refined locked Bloch linearization $\mathcal{L}_{\text{mic}}(p)$.
2. Project to the shell-relevant doubled block:

$$\mathcal{L}_D^{(1)}(p) = \Pi_D \mathcal{L}_{\text{mic}}(p) \Pi_D \big|_{\text{linear in } p}.$$

3. Choose the orthonormal low-mode basis $\Psi_a^{\pm}$.
4. Compute the shell matrices

$$M_{ab}^{\mu} = \langle \Psi_a^{+}, \mathcal{L}_D^{(1), \mu} \Psi_b^{-} \rangle.$$

5. Evaluate the three Pauli traces

$$\frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}), \quad \frac{1}{2} \text{Tr}(\sigma_1 M^1), \quad \frac{1}{2} \text{Tr}(\sigma_2 M^1).$$

6. Extract $C_{\parallel}, C_E, C_O$ by the ratio formulas (142)–(144).
7. Conclude Weyl/Dirac nondegeneracy iff

$$C_{\parallel} \neq 0, \quad (C_E, C_O) \neq (0, 0).$$

LXVII. STRONGEST HONEST CONCLUSION

The final microscopic obstruction is no longer conceptual. The one-mediator Branch-B carrier mechanism, the projected shell symbol, the trace test, and the anisotropy formula are all algebraically closed.

The only remaining frontier is the explicit numerical/symbolic evaluation of the three Bloch-frame overlaps

$$C_{\parallel}, \quad C_E, \quad C_O$$

in the complete refined locked microscopic TECT realization.

LXVIII. FINAL SUMMARY

The final microscopic carrier problem has been reduced completely to a normalized projector-overlap audit.

For the complete refined locked Bloch linearization $\mathcal{L}_{\text{mic}}(p)$, projected to the shell-relevant doubled block, the three calibration overlaps are defined as the normalized projections of the actual microscopic carriers onto the canonical Branch-B seed operators:

$$C_{\parallel}, \quad C_E, \quad C_O.$$

Equivalently, they are extracted from the three Pauli traces of the projected 2×2 shell matrices:

$$C_{\parallel} \leftrightarrow \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}), \quad C_E \leftrightarrow \frac{1}{2} \text{Tr}(\sigma_1 M^1), \quad C_O \leftrightarrow \frac{1}{2} \text{Tr}(\sigma_2 M^1).$$

The exact final nondegeneracy criterion is

$$\boxed{C_{\parallel} \neq 0, \quad (C_E, C_O) \neq (0, 0).}$$

This is exactly equivalent to the basis-overlap criterion

$$\exists A : a_A \neq 0, \quad \exists B : (b_B, c_B) \neq (0, 0).$$

Therefore the TECT microscopic Dirac carrier problem is now algebraically closed up to a single final calibration task: evaluate the complete Bloch-frame projector overlaps

$$C_{\parallel}, \quad C_E, \quad C_O$$

for the full refined locked microscopic realization.

LXIX. FINAL AUDIT TABLE FOR THE REFINED LOCKED + CLASS II MICROSCOPIC BRANCH

LXX. GOAL

The remaining microscopic task is to organize the full refined locked + Class II Bloch linearization into a finite audit table that tracks which microscopic operator contributes to which shell-reduced seed direction.

The exact reduced seed space is

$$V_{\text{seed}} = \text{span} \{B_{\parallel}, B_E, B_O\}, \tag{155}$$

where:

$$B_{\parallel} \sim \frac{1}{2} \theta'(\xi)^3 \hat{n}^{\mu} \sigma_3, \tag{156}$$

$$B_E \sim \theta'(\xi) (e_1^{\mu} \sigma_1 + e_2^{\mu} \sigma_2), \tag{157}$$

$$B_O \sim \theta'(\xi) (-e_2^{\mu} \sigma_1 + e_1^{\mu} \sigma_2). \tag{158}$$

The microscopic carrier audit succeeds if and only if the complete projected Bloch operator has nonzero projection onto:

$$B_{\parallel} \quad \text{and} \quad \text{at least one of } B_E, B_O.$$

LXXI. MICROSCOPIC STARTING BRANCH

The strongest explicit microscopic branch currently available is

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (159)$$

At the shell-reduced level, the Class II coordinate theorem identifies:

$$X_i^3 \leftrightarrow B_{\parallel}, \quad X_i^1 \leftrightarrow B_E, \quad X_i^2 \leftrightarrow B_O. \quad (160)$$

Hence Class II is the unique currently explicit class that realizes all three invariant directions at once.

LXXII. FINAL AUDIT TABLE

The operator-level finite audit can be summarized as follows.

Microscopic object	Required structure	Shell seed projection	Reduced witness	Audit
Pure scalar corrections	no (i, a) -carrier	0	$(0, 0, 0)$	excluded
Pure projector corrections	no (i, a) -carrier	0	$(0, 0, 0)$	excluded
Amplitude-only derivative corrections	no internal tangent index	0	$(0, 0, 0)$	excluded
$\theta'(\xi)^2 \sigma_3$	even singlet	0	$a = 0$	excluded
$V_{(\perp, E)}^{\mu}$	(i, a) -carrier, transverse-even	B_E	$\alpha \neq 0, \beta = 0$	explicit
$V_{(\perp, O)}^{\mu}$	(i, a) -carrier, transverse-odd	B_O	$\alpha = 0, \beta \neq 0$	explicit
$V_{(\parallel)}^{\mu}$	(i, a) -carrier, primitive odd singlet	$B_{\parallel}$	$\lambda_{\parallel} \neq 0$	minimal long
Class I	direct (i, a) -tensor insertion	$B_{\parallel}, B_E, B_O$	a, b, c possible	admissible
Class II	X_i^a mediator	$B_{\parallel}, B_E, B_O$	a, b, c explicit	strongest
Class III	A_i^a -like carrier	$B_{\parallel}, B_E, B_O$	a, b, c possible	plausible

LXXIII. REDUCED COEFFICIENT DICTIONARY

Inside the shell seed basis,

$$B_{\parallel} \leftrightarrow a, \quad B_E \leftrightarrow b, \quad B_O \leftrightarrow c.$$

Equivalently,

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}. \quad (161)$$

Thus the exact Weyl/Dirac nondegeneracy condition is

$$\boxed{c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0)}. \quad (162)$$

In shell-tensor notation this is simply

$$\boxed{A \neq 0, \quad (B, C) \neq (0, 0)}. \quad (163)$$

LXXIV. PROJECTED BLOCH OPERATOR FORMULATION

Let

$$\mathcal{L}_{\text{mic}}(p)$$

be the complete refined locked Bloch linearization, and let

$$\Pi_D$$

project onto the shell-relevant doubled block. Then the first-order shell operator is

$$\mathcal{L}_D^{(1)}(p) = \Pi_D \mathcal{L}_{\text{mic}}(p) \Pi_D \big|_{\text{linear in } p}. \quad (164)$$

Choose the orthonormal low-mode basis

$$\Psi_a^\pm(\xi, y) = f_\pm(\xi) e^{\pm i G_* \cdot y} u_a^\pm(\xi), \quad a = 1, 2.$$

Define the projected shell matrices

$$(M^\mu)_{ab} = \langle \Psi_a^+, \mathcal{L}_D^{(1), \mu} \Psi_b^- \rangle. \quad (165)$$

Then the audit is completely reduced to the three Pauli traces

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}), \quad (166)$$

$$\tilde{\Lambda}_E = \frac{1}{2} \text{Tr}(\sigma_1 M^1), \quad (167)$$

$$\tilde{\Lambda}_O = \frac{1}{2} \text{Tr}(\sigma_2 M^1). \quad (168)$$

So the final finite yes/no test is

$$\boxed{\text{Tr}(\sigma_3 M^{\parallel}) \neq 0, \quad (\text{Tr}(\sigma_1 M^1), \text{Tr}(\sigma_2 M^1)) \neq (0, 0).} \quad (169)$$

LXXV. MASTER CONTINUATION RULES

From this point onward, the no-loop rule is:

$$\boxed{\text{Do not re-prove the witness theorem.}} \quad (170)$$

The only legitimate next tasks are:

1. construct the complete periodic Bloch operator;
2. project to the shell-relevant doubled block;
3. compute the three traces

$$\text{Tr}(\sigma_3 M^{\parallel}), \quad \text{Tr}(\sigma_1 M^1), \quad \text{Tr}(\sigma_2 M^1);$$

4. extract

$$(\lambda_{\parallel}, \alpha, \beta);$$

5. then move immediately to

$$V(0) = 0, \quad \Delta > 0, \quad \text{mass protection.}$$

LXXVI. STRONGEST HONEST CONCLUSION

The actual microscopic carrier problem is no longer conceptually open.
It has been reduced to the finite projection question:

$$\boxed{\text{Does the complete microscopic TECT Bloch linearization project nontrivially onto } V_{\text{seed}}?} \quad (171)$$

Inside the explicit Class II truncation, the answer is already yes. What remains open is only whether the final complete microscopic TECT realization contains the same nontrivial carrier sector in the physically relevant defect background.

LXXVII. FINAL SUMMARY

The microscopic carrier problem has now been reduced to a finite operator audit on the projected shell seed space

$$V_{\text{seed}} = \text{span}\{B_{\parallel}, B_E, B_O\}.$$

The final audit table is:

- pure scalar / projector / amplitude-only classes are excluded,
- the naive even singlet is excluded by parity,
- the transverse even/odd seed channels are explicitly realized,
- the primitive odd longitudinal singlet is the minimal surviving longitudinal mechanism,
- Class II is the strongest current explicit microscopic class realizing all three invariant coordinates.

The reduced witness dictionary is

$$\lambda_{\parallel} = \frac{1}{2}c_{\parallel}I_{\theta^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta},$$

so the exact local Weyl condition is

$$c_{\parallel} \neq 0, \quad (c_E, c_O) \neq (0, 0).$$

Equivalently, for the projected Bloch shell matrices

$$(M^{\mu})_{ab} = \langle \Psi_a^+, \mathcal{L}_D^{(1),\mu} \Psi_b^- \rangle,$$

the final finite yes/no test is

$$\text{Tr}(\sigma_3 M^{\parallel}) \neq 0, \quad (\text{Tr}(\sigma_1 M^1), \text{Tr}(\sigma_2 M^1)) \neq (0, 0).$$

Therefore the remaining frontier is no longer conceptual. It is the explicit construction and evaluation of the complete projected Bloch operator for the refined locked microscopic TECT branch.

LXXVIII. FOLDED-BRAGG BLOCH-FIBER CONSTRUCTION IN THE LOCKED 12-MODE BCC BRANCH

LXXIX. WORKING CONVENTION

From this point onward, we use the folded Bragg convention

$$\boxed{k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.} \quad (172)$$

This is the correct finite-dimensional Bloch-fiber convention for the shell-pair calculation.

We work in the strongest currently explicit microscopic branch

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (173)$$

LXXX. LOCKED 12-MODE BCC BACKGROUND

Let the first BCC reciprocal shell be

$$\mathcal{S}_{12} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad |\mathcal{S}_{12}| = 12. \quad (174)$$

The locked equal-amplitude background is

$$\phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x} = 2A \sum_{\alpha=1}^6 \cos(Q_\alpha \cdot x), \quad (175)$$

and

$$\Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^\dagger. \quad (176)$$

Choose an internal basis

$$u_0 = z_0, \quad u_1, u_2 \in Q_{z_0}, \quad u_r^\dagger u_s = \delta_{rs}, \quad (177)$$

so that

$$\mathbb{C}^3 = \mathbb{C}z_0 \oplus Q_{z_0}.$$

LXXXI. BLOCH FIBER AND SHELL-PAIR BASIS

Because the linearized operator around Ψ_0 is periodic, it admits Bloch decomposition. At the folded Bragg point $k_* = G_*/2$, the two shell-harmonic states

$$e^{+iG_* \cdot x/2}, \quad e^{-iG_* \cdot x/2}$$

are degenerate modulo reciprocal-lattice folding.

We therefore define the shell-pair basis

$$\Phi_{+,a}(x) = e^{+iG_* \cdot x/2} u_a, \quad \Phi_{-,a}(x) = e^{-iG_* \cdot x/2} u_a, \quad a = 1, 2, \quad (178)$$

which spans the 4-dimensional shell-relevant doubled block

$$\mathcal{H}_* = \text{span}\{\Phi_{+,1}, \Phi_{+,2}, \Phi_{-,1}, \Phi_{-,2}\} \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2.$$

Let P_* denote the orthogonal projector onto $\mathcal{H}_*$.

LXXXII. BLOCH FIBER MATRIX AND PRIMITIVE-CELL FOURIER COEFFICIENTS

Let $L_{\text{act}}(k)$ be the Bloch fiber of the microscopic linearized operator. Because the coefficients are periodic, the matrix elements between shell-pair basis vectors are governed by primitive-cell Fourier coefficients.

For any periodic scalar coefficient $U(x)$,

$$\langle \Phi_{+,a}, U(x) \Phi_{-,b} \rangle = \widehat{U}(G_*) \delta_{ab}, \quad (179)$$

because

$$e^{-iG_* \cdot x/2} U(x) e^{-iG_* \cdot x/2}$$

selects exactly the $Q = G_*$ Fourier mode of U .

Hence the constant intervalley Bragg block is

$$\boxed{M_0 = \widehat{U}_{\text{eff}}(G_*) \mathbf{1}_2.} \quad (180)$$

Likewise, for a traceless enriched coefficient $\mathcal{T}^\alpha(x) \otimes s^\alpha$,

$$\langle \Phi_{+,a}, \mathcal{T}^\alpha(x) \otimes s^\alpha \Phi_{-,b} \rangle = \widehat{\mathcal{T}}^\alpha(G_*) (s^\alpha)_{ab}. \quad (181)$$

Thus the transverse Dirac block is controlled by

$$\boxed{W^\alpha = \widehat{T}_{\text{eff}}^\alpha(G_*) s^\alpha, \quad \alpha = 1, 2.} \quad (182)$$

Therefore the full shell-pair problem is reduced to Fourier extraction at the selected shell vector G_* .

LXXXIII. FIRST-DERIVATIVE EXTRACTION

The local Weyl data are determined by the projected first derivatives of the Bloch fiber:

$$K_i := \partial_{k_i} L_{\text{act}}(k) \Big|_{k=k_*}. \quad (183)$$

The actual shell-reduced test is therefore

$$\boxed{P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta)}. \quad (184)$$

Choose the adapted shell frame

$$\hat{n} = \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\} \text{ orthonormal},$$

and write

$$p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2.$$

Then define

$$K_{\parallel} := \hat{n}_i K_i, \quad K_1 := e_{1i} K_i, \quad K_2 := e_{2i} K_i. \quad (185)$$

LXXXIV. PROJECTED AXIAL ANSATZ

The minimal shell-enriched axial form is

$$K_i = A \hat{n}_i \tau^3 \otimes \mathbf{1}_2 + B (e_{1i} \tau^1 \otimes s_1 + e_{2i} \tau^1 \otimes s_2) + C (-e_{2i} \tau^1 \otimes s_1 + e_{1i} \tau^1 \otimes s_2). \quad (186)$$

Therefore

$$K_{\parallel} = A \tau^3 \otimes \mathbf{1}_2, \quad (187)$$

$$K_1 = B \tau^1 \otimes s_1 + C \tau^1 \otimes s_2, \quad (188)$$

$$K_2 = -C \tau^1 \otimes s_1 + B \tau^1 \otimes s_2. \quad (189)$$

Projecting to the selected positive-valley two-band block $P_+ \subset P_*$,

$$P_+(\tau^3 \otimes \mathbf{1}_2)P_+ \mapsto \sigma^3, \quad P_+(\tau^1 \otimes s_1)P_+ \mapsto \sigma^1, \quad P_+(\tau^1 \otimes s_2)P_+ \mapsto \sigma^2,$$

we obtain

$$h_+(p) = A p_{\parallel} \sigma^3 + B(p_1 \sigma^1 + p_2 \sigma^2) + C(-p_2 \sigma^1 + p_1 \sigma^2) + O(|p|^2). \quad (190)$$

Hence

$$\boxed{\lambda_{\parallel} = A, \quad \alpha = B, \quad \beta = C.} \quad (191)$$

This is exactly the finite extraction dictionary.

LXXXV. DIRECT PROJECTOR-TRACE FORMULAS

The three coefficients are extracted directly by projector traces:

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma^3 P_* K_{\parallel} P_*), \quad (192)$$

$$\alpha = \frac{1}{4} \sum_{a=1}^2 \text{Tr}_{\mathcal{H}_*}(\sigma^a P_*(e_{ai} K_i) P_*), \quad (193)$$

$$\beta = \frac{1}{4} [\text{Tr}_{\mathcal{H}_*}(\sigma^2 P_*(e_{1i} K_i) P_*) - \text{Tr}_{\mathcal{H}_*}(\sigma^1 P_*(e_{2i} K_i) P_*)]. \quad (194)$$

Equivalently, the Weyl matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad \det M = \lambda_{\parallel}(\alpha^2 + \beta^2). \quad (195)$$

Thus local Weyl nondegeneracy is exactly

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (196)$$

LXXXVI. 12-MODE BCC SHELL-SUM INPUT

Inside the locked equal-amplitude 12-mode BCC branch, the selected-shell combinatorics satisfy

$$N(G_*) = \#\{(Q_1, Q_2) \in S_{\text{BCC}}^2 : Q_1 + Q_2 = G_*\} = 4,$$

so the first-shell convolution is nontrivial. In particular, the 12-mode shell sum already shows that the witness does not disappear upon lifting the single-mode representative to the locked BCC shell sum. :contentReference[oaicite:7]index=7

This means the actual remaining bottleneck is no longer “does a shell witness exist at all?” but rather:

$$\text{compute the actual } P_* K_i P_* \text{ of the final refined locked Bloch operator.} \quad (197)$$

LXXXVII. HONEST STATUS

What is now closed:

1. the correct Bragg convention is fixed:

$$k_* = \frac{G_*}{2};$$

2. the shell-relevant Bloch basis is fixed:

$$\Phi_{\pm,a}(x) = e^{\pm i G_* \cdot x / 2} u_a;$$

3. constant and enriched Bragg blocks are read from the $Q = G_*$ primitive-cell Fourier coefficients;
4. the local Weyl data are reduced to the finite derivative blocks

$$P_* K_i P_*;$$

5. the exact yes/no test is the triple

$$(\lambda_{\parallel}, \alpha, \beta)$$

extracted by (192)–(194).

What is still open:

the explicit symbolic/numerical evaluation of $K_i = \partial_{k_i} L_{\text{act}}(k) _{k=k_*}$ for the final complete refined locked TECT/Class II operator	(198)
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This is now a finite operator-construction problem, not a conceptual one.

LXXXVIII. FINAL SUMMARY

Using the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p,$$

the locked 12-mode BCC microscopic linearization reduces to a finite shell-pair Bloch problem on

$$\mathcal{H}_* = \text{span}\{\Phi_{+,a}, \Phi_{-,a}\}_{a=1,2}, \quad \Phi_{\pm,a}(x) = e^{\pm i G_* \cdot x/2} u_a.$$

The constant intervalley Bragg block and the enriched transverse blocks are controlled by the primitive-cell Fourier coefficients at $Q = G_*$:

$$M_0 = \widehat{U}_{\text{eff}}(G_*) \mathbf{1}_2, \quad W^\alpha = \widehat{T}_{\text{eff}}^\alpha(G_*) s^\alpha.$$

The local Weyl data are extracted from the projected first derivatives

$$K_i = \partial_{k_i} L_{\text{act}}(k)|_{k=k_*}, \quad P_* K_i P_*.$$

In the minimal axial shell-adapted ansatz,

$$K_i = A \hat{n}_i \tau^3 \otimes \mathbf{1}_2 + B (e_{1i} \tau^1 \otimes s_1 + e_{2i} \tau^1 \otimes s_2) + C (-e_{2i} \tau^1 \otimes s_1 + e_{1i} \tau^1 \otimes s_2),$$

so that

$$\lambda_{\parallel} = A, \quad \alpha = B, \quad \beta = C.$$

Equivalently,

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}_{\mathcal{H}_*}(\sigma^3 P_* K_{\parallel} P_*),$$

$$\alpha = \frac{1}{4} \sum_{a=1}^2 \text{Tr}_{\mathcal{H}_*}(\sigma^a P_*(e_{ai} K_i) P_*),$$

$$\beta = \frac{1}{4} [\text{Tr}_{\mathcal{H}_*}(\sigma^2 P_*(e_{1i} K_i) P_*) - \text{Tr}_{\mathcal{H}_*}(\sigma^1 P_*(e_{2i} K_i) P_*)].$$

Thus the final microscopic Weyl test is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

So the remaining frontier is completely localized: compute the actual projected derivative blocks $P_* K_i P_*$ for the final refined locked TECT/Class II Bloch operator.

LXXXIX. EXPLICIT COEFFICIENT BOOKKEEPING FOR THE FOLDED-BRAGG 12-MODE BLOCH AUDIT

XC. IMMEDIATE GOAL

The next concrete task is not to introduce a new mechanism, but to write the actual periodic coefficient extraction formulas for

$$\widehat{U}_{\text{eff}}(G_*), \quad \widehat{T}_{\text{eff}}^1(G_*), \quad \widehat{T}_{\text{eff}}^2(G_*), \quad v_{\parallel}.$$

These are the four microscopic quantities from which the shell data

$$M_0, \quad W^1, \quad W^2, \quad v_{\parallel}$$

are obtained in the folded Bragg convention.

XCI. FOLDED-BRAGG SHELL DICTIONARY

Work at

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.$$

Then the shell-relevant Bloch basis is

$$\Phi_{+,a}(x) = e^{+iG_* \cdot x/2} u_a, \quad \Phi_{-,a}(x) = e^{-iG_* \cdot x/2} u_a, \quad a = 1, 2.$$

The exact shell quantities are:

$$\boxed{M_0 = B^{\text{eff}}(E_*, 0),} \tag{199}$$

$$\boxed{W^i = \partial_{p_i} B^{\text{eff}}(E_*, p)|_{p=0}, \quad i = 1, 2,} \tag{200}$$

$$\boxed{v_{\parallel} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \frac{E_+^{\text{eff}}(E_*, p) - E_-^{\text{eff}}(E_*, p)}{2} \Big|_{p=0}.} \tag{201}$$

Equivalently, in the primitive-cell Fourier language,

$$\boxed{M_0 = \widehat{U}_{\text{eff}}(G_*) \mathbf{1}_2, \quad W^\alpha = \widehat{T}_{\text{eff}}^\alpha(G_*) s^\alpha, \quad \alpha = 1, 2.} \tag{202}$$

XCII. THREE-WAY DECOMPOSITION OF THE EFFECTIVE COEFFICIENTS

The exact bookkeeping decomposition is

$$\boxed{\widehat{U}_{\text{eff}}(G_*) = \widehat{U}_{\text{bare}}(G_*) + \widehat{U}_{(II)}(G_*) + \widehat{U}_{(\text{sp})}(G_*),} \tag{203}$$

$$\boxed{\widehat{T}_{\text{eff}}^\alpha(G_*) = \widehat{T}_{\text{bare}}^\alpha(G_*) + \widehat{T}_{(II)}^\alpha(G_*) + \widehat{T}_{(\text{sp})}^\alpha(G_*), \quad \alpha = 1, 2,} \tag{204}$$

$$\boxed{v_{\parallel} = v_{\parallel}^{\text{bare}} + \delta v_{\parallel}^{(II)} + \delta v_{\parallel}^{(\text{sp})}.} \tag{205}$$

So the actual full-PDE work is now reduced to computing exactly these four quantities. This reduction is already fixed in the project notes.

XCIII. BARE PERIODIC-CORE CONTRIBUTION

Let the $\Psi\Psi$ -block of the Bloch operator be

$$H_{\Psi\Psi}(k) = H_0(k) + U^{(\text{per})} + U^{(\text{lock})}.$$

Its harmonic matrix elements are

$$(H_{\Psi\Psi}(k))_{(G,r),(G',s)} = \delta_{GG'} \delta_{rs} h_r(k+G) + U_{rs}^{(\text{per})}(G-G') + U_{rs}^{(\text{lock})}(G-G').$$

Then the direct periodic-core Bragg coefficients are:

$$\widehat{U}_{\text{bare}}(G_*) = [U^{(\text{per})}(G_*) + U^{(\text{lock})}(G_*)]_{\text{scalar on } Q_{z_0}}, \quad (206)$$

$$\widehat{T}_{\text{bare}}^\alpha(G_*) = [U^{(\text{per})}(G_*) + U^{(\text{lock})}(G_*)]_{\text{traceless}, \alpha}, \quad \alpha = 1, 2, \quad (207)$$

where “scalar” means proportional to $\mathbf{1}_2$ on the tangent doublet, and “traceless, α ” means the coefficient of the Pauli matrix s^α .

For the longitudinal slope, the direct periodic-core contribution is

$$v_{\parallel}^{\text{bare}} = \frac{1}{2} \partial_{p_{\parallel}} \left. \text{tr}_{\text{int}} \frac{E_+^{\text{bare}}(p) - E_-^{\text{bare}}(p)}{2} \right|_{p=0}. \quad (208)$$

As already emphasized in the notes, the constant Bragg channel M_0 and the longitudinal slope $v_{\parallel}$ can have direct periodic-core contributions. :contentReference[oaicite:4]index=4

XCIV. CLASS II SELF-ENERGY CONTRIBUTION

Let

$$\mathcal{C}(k) = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k), \quad H_{XX}(k) = M_X^2 \mathbf{1} + \Delta^{XX}(k),$$

and

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \mathcal{C}(k) H_{XX}(k)^{-1} \mathcal{C}(k)^\dagger.$$

At leading heavy-mass order,

$$H_{\text{eff}}(k) = H_{\Psi\Psi}(k) - \frac{1}{M_X^2} \mathcal{C}(k) \mathcal{C}(k)^\dagger + O(M_X^{-4}).$$

Therefore the Class II contributions to the Bragg Fourier coefficients are

$$\widehat{U}_{(II)}(G_*) = -\frac{1}{M_X^2} \widehat{(\mathcal{C}\mathcal{C}^\dagger)}_{\text{scalar}}(G_*), \quad (209)$$

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{1}{M_X^2} \widehat{(\mathcal{C}\mathcal{C}^\dagger)}_{\text{traceless}, \alpha}(G_*), \quad \alpha = 1, 2. \quad (210)$$

Likewise the longitudinal derivative correction is

$$\delta v_{\parallel}^{(II)} = -\frac{1}{M_X^2} \partial_{p_{\parallel}} \left. \text{tr}_{\text{int}} \frac{(\mathcal{C}\mathcal{C}^\dagger)_+(p) - (\mathcal{C}\mathcal{C}^\dagger)_-(p)}{4} \right|_{p=0}. \quad (211)$$

At the shell-block derivative level this is exactly

$$K_i^{(II)} = -\frac{1}{M_X^2} \Pi_{S_*} \left[(\partial_{k_i} \mathcal{C}) \mathcal{C}^\dagger + \mathcal{C} (\partial_{k_i} \mathcal{C}^\dagger) \right]_{k=k_*} \Pi_{S_*}. \quad (212)$$

This is the correct leading target because the isotropic on-shell first derivative of the bare Brazovskii kernel vanishes at q_* . :contentReference[oaicite:5]index=5

XCV. ONE-STEP SPECTATOR-SHELL CORRECTION

If a single spectator shell is retained and integrated out, the exact renormalized core blocks are

$$E_{\pm}^{\text{eff}}(E, p) = E_{\pm}(p) - C_{\pm}(p)(H_{ss}(p) - E)^{-1}C_{\pm}(p)^{\dagger},$$

$$B^{\text{eff}}(E, p) = B(p) - C_{+}(p)(H_{ss}(p) - E)^{-1}C_{-}(p)^{\dagger}.$$

Hence the spectator contributions are

$$\widehat{U}_{(\text{sp})}(G_*) = -C_{+}(0) \widehat{(H_{ss}(0) - E_*)}^{-1} C_{-}(0)^{\dagger}_{\text{scalar}}(G_*), \quad (213)$$

$$\widehat{T}_{(\text{sp})}^{\alpha}(G_*) = -C_{+}(0) \widehat{(H_{ss}(0) - E_*)}^{-1} C_{-}(0)^{\dagger}_{\text{traceless}, \alpha}(G_*), \quad \alpha = 1, 2, \quad (214)$$

and

$$\delta v_{\parallel}^{(\text{sp})} = \frac{1}{2} \partial_{p_{\parallel}} \text{tr}_{\text{int}} \frac{\delta E_{+}^{(\text{sp})}(p) - \delta E_{-}^{(\text{sp})}(p)}{2} \Big|_{p=0}. \quad (215)$$

Equivalently, for the transverse matrices,

$$W^i = \partial_{p_i} B - (\partial_{p_i} C_{+}) R_0 C_{-}^{\dagger} - C_{+} R_0 (\partial_{p_i} C_{-}^{\dagger}) + C_{+} R_0 (\partial_{p_i} H_{ss}) R_0 C_{-}^{\dagger} \Big|_{p=0}, \quad R_0 = (H_{ss}(0) - E_*)^{-1}. \quad (216)$$

This exact spectator-shell formula is already in the notes. :contentReference[oaicite:6]index=6

XCVI. TERM-BY-TERM HARMONIC EXTRACTION AT $Q = G_*$

Now fix any periodic operator kernel $F(x)$ acting on the tangent doublet. Its first-shell Bragg coefficient is

$$\widehat{F}(G_*) = \frac{1}{|\Omega|} \int_{\Omega} d^3x e^{-iG_* \cdot x} F(x), \quad (217)$$

where Ω is the primitive cell.

Decompose F on the tangent doublet as

$$F(x) = f_0(x) \mathbf{1}_2 + f_1(x) s^1 + f_2(x) s^2 + f_3(x) s^3.$$

Then

$$\widehat{U}(G_*) = \widehat{f}_0(G_*), \quad \widehat{T}^1(G_*) = \widehat{f}_1(G_*), \quad \widehat{T}^2(G_*) = \widehat{f}_2(G_*). \quad (218)$$

So the actual extraction rule is brutally simple:

$$\boxed{\text{compute the } Q = G_* \text{ Fourier coefficient of each periodic kernel, then split it into } \mathbf{1}_2, s^1, s^2 \text{ parts.}} \quad (219)$$

XCVII. IMMEDIATE PRACTICAL AUDIT TABLE

The practical bookkeeping table is therefore:

microscopic source	$Q = G_*$ coefficient to compute	output
$H_{\Psi\Psi}$	$\hat{U}_{\text{bare}}, \hat{T}_{\text{bare}}^1, \hat{T}_{\text{bare}}^2$	$M_0^{\text{bare}}, W_{\text{bare}}^1, W_{\text{bare}}^2$
$-\frac{1}{M_X^2}\mathcal{C}\mathcal{C}^\dagger$	$\hat{U}_{(II)}, \hat{T}_{(II)}^1, \hat{T}_{(II)}^2$	$M_0^{(II)}, W_{(II)}^1, W_{(II)}^2$
$-C_+R_0C_-^\dagger$	$\hat{U}_{(\text{sp})}, \hat{T}_{(\text{sp})}^1, \hat{T}_{(\text{sp})}^2$	$M_0^{(\text{sp})}, W_{(\text{sp})}^1, W_{(\text{sp})}^2$
$E_+ - E_-$	$\partial_{p_\parallel}$ at $p = 0$	$v_\parallel$

Then assemble

$$M_0 = [\hat{U}_{\text{bare}} + \hat{U}_{(II)} + \hat{U}_{(\text{sp})}] \mathbf{1}_2,$$

$$W^\alpha = [\hat{T}_{\text{bare}}^\alpha + \hat{T}_{(II)}^\alpha + \hat{T}_{(\text{sp})}^\alpha] s^\alpha, \quad \alpha = 1, 2,$$

and

$$v_\parallel = v_\parallel^{\text{bare}} + \delta v_\parallel^{(II)} + \delta v_\parallel^{(\text{sp})}.$$

XCVIII. FINAL LOCAL WEYL TEST

Once the four coefficient sets above are computed, the final shell test is:

$$\boxed{M_0 = 0, \quad v_\parallel \neq 0, \quad \widetilde{W}^1, \widetilde{W}^2 \text{ linearly independent traceless Hermitian matrices.}} \quad (220)$$

Equivalently, in the reduced $S2$ parametrization,

$$\boxed{\lambda_\parallel \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (221)$$

XCIX. WHAT IS HONESTLY CLOSED NOW

At this point, the following is closed:

$$\boxed{\text{the full periodic coefficient-extraction problem has been reduced to a finite Fourier bookkeeping problem at } Q = G_*}.$$

What is not yet closed is numerical or symbolic execution for the final chosen $L_\Psi^{\text{ref}}[\Psi, P]$. That is a real calculation still to be done, but it is no longer conceptually open.

C. FINAL SUMMARY

The full periodic folded-Bragg Bloch audit has now been reduced to the explicit extraction of four microscopic quantities:

$$\hat{U}_{\text{eff}}(G_*), \quad \hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*), \quad v_\parallel.$$

Each of them splits as

$$\text{effective} = \text{bare} + \text{Class II} + \text{spectator}.$$

The bare contribution comes from the periodic $\Psi\Psi$ -block, the Class II contribution from the self-energy

$$-\frac{1}{M_X^2}\mathcal{C}\mathcal{C}^\dagger,$$

and the spectator contribution from

$$-C_+R_0C_-^\dagger.$$

Therefore the actual remaining microscopic work is:

1. compute the $Q = G_*$ Fourier coefficient of each periodic kernel;
2. split it into $\mathbf{1}_2, s^1, s^2$ parts on the tangent doublet;
3. assemble $M_0, W^1, W^2, v_{\parallel}$;
4. check

$$M_0 = 0, \quad v_{\parallel} \neq 0, \quad \widetilde{W}^1, \widetilde{W}^2 \text{ independent.}$$

Equivalently, the final local Weyl test is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

So the frontier is no longer conceptual. It is the explicit symbolic or numerical execution of the Fourier bookkeeping for the final chosen refined locked microscopic operator.

CI. HARMONIC SELECTION RULE FOR THE CLASS II BLOCH COUPLING MATRIX

CII. GOAL

The next explicit calculation is to derive the harmonic selection rule for the Class II coupling matrix

$$\mathcal{C}_{(G,r),(G',i,a)}(k),$$

defined by the linearized shell-mediator mixing between the Bloch harmonics of $\delta\Psi$ and δX_i^a .

The purpose of this section is not yet to fix a unique microscopic closed form of

$$J_i^a[\Psi], \quad K_i^a[P, \psi],$$

but to derive the exact Bloch-harmonic bookkeeping rule that any such local periodic coupling must satisfy.

CIII. PERIODIC BACKGROUND AND BLOCH EXPANSIONS

Work in the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.$$

The locked 12-mode BCC background is

$$\Psi_0(x) = z_0 \phi_0(x), \quad \phi_0(x) = \sum_{Q \in \mathcal{S}_{12}} A_Q e^{iQ \cdot x}, \quad P_0 = z_0 z_0^\dagger, \quad (222)$$

with $\mathcal{S}_{12}$ the first BCC reciprocal shell.

Choose the internal basis

$$u_0 = z_0, \quad u_1, u_2 \in Q_{z_0}, \quad u_r^\dagger u_s = \delta_{rs}.$$

Expand the microscopic fluctuation in Bloch harmonics:

$$\delta\Psi(x) = \sum_{G \in \Lambda^*} \sum_{r=0}^2 \psi_{G,r}(k) e^{i(k+G) \cdot x} u_r, \quad (223)$$

and the Class II mediator fluctuation as

$$\delta X_i^a(x) = \sum_{G' \in \Lambda^*} X_{G',i}^a(k) e^{i(k+G') \cdot x}. \quad (224)$$

Here Λ^* is the reciprocal lattice generated by the periodic BCC condensate.

CIV. LINEARIZED SHELL-MEDIATOR BLOCK

The linearized Class II mixing comes from

$$\alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi].$$

Around the background (Ψ_0, P_0) , the linearized off-diagonal block is schematically

$$L_{\Psi X}^{\text{lin}} = \alpha_X \delta J_i^a[\Psi_0; \delta\Psi] + \beta_X \delta K_i^a[P_0; \delta\Psi]. \quad (225)$$

Because the background is periodic, the coefficients of this linearized operator are periodic. Therefore the corresponding Bloch matrix elements are translation-covariant and admit a harmonic expansion.

CV. GENERIC LOCAL HARMONIC FORM

Any local first-order shell-mediator coupling generated by the periodic background can be written in the form

$$\delta J_i^a[\Psi_0; \delta\Psi](x) = \sum_{Q \in \Lambda^*} e^{iQ \cdot x} J_{i,r}^a(Q; k + G) \psi_{G,r}(k) e^{i(k+G) \cdot x}, \quad (226)$$

and similarly

$$\delta K_i^a[P_0; \delta\Psi](x) = \sum_{Q \in \Lambda^*} e^{iQ \cdot x} K_{i,r}^a(Q; k + G) \psi_{G,r}(k) e^{i(k+G) \cdot x}. \quad (227)$$

Thus the total Class II coupling takes the harmonic form

$$\delta J_i^a + \delta K_i^a = \sum_{G, Q, r} C_{i,r}^a(Q; k + G) \psi_{G,r}(k) e^{i(k+G+Q) \cdot x}, \quad (228)$$

with

$$C_{i,r}^a(Q; k + G) = \alpha_X J_{i,r}^a(Q; k + G) + \beta_X K_{i,r}^a(Q; k + G). \quad (229)$$

CVI. EXACT BLOCH SELECTION RULE

Projecting onto the mediator harmonic $e^{i(k+G') \cdot x}$ gives the exact selection rule

$$\boxed{\mathcal{C}_{(G,r),(G',i,a)}(k) = \sum_{Q \in \Lambda^*} \delta_{G', G+Q} C_{i,r}^a(Q; k + G).} \quad (230)$$

Equivalently,

$$\boxed{\mathcal{C}_{(G,r),(G',i,a)}(k) = C_{i,r}^a(G' - G; k + G).} \quad (231)$$

This is the fundamental harmonic bookkeeping formula: the Class II coupling connects harmonic G to harmonic G' only through the Fourier mode

$$Q = G' - G$$

of the periodic background coefficient.

CVII. SPECIALIZATION TO THE LOCKED 12-MODE BRANCH

In the locked equal-amplitude 12-mode BCC background, the only nonzero fundamental background harmonics at leading shell order are

$$Q \in \mathcal{S}_{12}.$$

Therefore the leading shell-mixing block satisfies

$$\boxed{\mathcal{C}_{(G,r),(G',i,a)}(k) = \sum_{Q \in \mathcal{S}_{12}} \delta_{G', G+Q} \mathcal{C}_{i,r}^a(Q; k+G),} \quad (232)$$

up to higher-harmonic corrections generated by nonlinear products of the background.

Hence, at leading locked-shell order, only differences $G' - G \in \mathcal{S}_{12}$ can contribute.

CVIII. FOLDED SHELL-PAIR INTERVALLEY CHANNEL

For the selected folded shell pair, the relevant two Bloch harmonics are

$$G_+ = +\frac{G_*}{2}, \quad G_- = -\frac{G_*}{2},$$

or equivalently the basis states

$$\Phi_{+,a}(x) = e^{+iG_* \cdot x/2} u_a, \quad \Phi_{-,a}(x) = e^{-iG_* \cdot x/2} u_a.$$

The intervalley block connecting $-$ to $+$ therefore has

$$G_+ - G_- = G_*.$$

So the selection rule (231) reduces immediately to

$$\boxed{\mathcal{C}_{(-,r),(+,i,a)}(k_*) = \mathcal{C}_{i,r}^a(G_*; k_* - \frac{G_*}{2}).} \quad (233)$$

Thus the entire shell-pair carrier audit is controlled by the single first-shell Fourier coefficient

$$Q = G_*.$$

This is exactly why the reduced dictionary uses

$$\hat{U}_{\text{eff}}(G_*), \quad \hat{T}_{\text{eff}}^1(G_*), \quad \hat{T}_{\text{eff}}^2(G_*).$$

CIX. DERIVATIVE CONTENT AND PARITY STRUCTURE

At first derivative order, the local shell-mediator tensors $\mathbf{J}, \mathbf{K}$ decompose into the three shell-adapted seed sectors:

$$\mathbf{C}_{i,r}^a(G_*; \cdot) = c_{\parallel} \hat{n}_i (\Sigma_3)_r^a + c_E (e_{1i} \Sigma_1 + e_{2i} \Sigma_2)_r^a + c_O (-e_{2i} \Sigma_1 + e_{1i} \Sigma_2)_r^a + \cdots, \quad (234)$$

where the omitted terms are shell-irrelevant, higher-derivative, or symmetry-forbidden components.

In particular:

1. the longitudinal primitive odd singlet contributes to $B_{\parallel}$,
2. the transverse even doublet contributes to B_E ,
3. the transverse odd doublet contributes to B_O .

Therefore the reduced witness coefficients are nothing but the three seed coefficients extracted from the $Q = G_*$ harmonic of the Class II mixing block.

CX. INDEX-LEVEL BOOKKEEPING FORMULA

Using the shell-adapted basis u_r , the explicit coefficient extraction is:

$$c_{\parallel} \propto \hat{n}^i(\Sigma_3)^r{}_a C_{i,r}^a(G_*; k_* - \frac{G_*}{2}), \quad (235)$$

$$c_E \propto \frac{1}{2} [e_1^i(\Sigma_1)^r{}_a + e_2^i(\Sigma_2)^r{}_a] C_{i,r}^a(G_*; k_* - \frac{G_*}{2}), \quad (236)$$

$$c_O \propto \frac{1}{2} [-e_2^i(\Sigma_1)^r{}_a + e_1^i(\Sigma_2)^r{}_a] C_{i,r}^a(G_*; k_* - \frac{G_*}{2}). \quad (237)$$

These coefficients then feed directly into

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}.$$

CXI. FINAL PRACTICAL FORM FOR THE CLASS II COUPLING MATRIX

Combining everything, the actual working coupling matrix is

$$\mathcal{C}_{(G,r),(G',i,a)}(k) = \delta_{G'-G \in S_{12}} \left[\alpha_X J_{i,r}^a(G' - G; k + G) + \beta_X K_{i,r}^a(G' - G; k + G) \right] \quad (238)$$

at leading locked-shell order.

For the shell-pair intervalley block relevant to the Dirac audit, this collapses to

$$\mathcal{C}_{r,(i,a)}^{\text{valley}} = \alpha_X J_{i,r}^a(G_*; 0) + \beta_X K_{i,r}^a(G_*; 0), \quad (239)$$

where we used $k_* - \frac{G_*}{2} = 0$.

This is the concrete object whose three shell projections must be evaluated.

CXII. WHAT IS CLOSED AND WHAT REMAINS OPEN

What is now closed:

1. the exact Bloch-harmonic selection rule;
2. the fact that the intervalley shell-pair channel is controlled only by the $Q = G_*$ harmonic;
3. the index-level formula reducing the carrier audit to the three shell projections of

$$C_{i,r}^a(G_*; 0).$$

What remains open is the explicit microscopic closed form of the coefficient functions

$$J_{i,r}^a(G_*; 0), \quad K_{i,r}^a(G_*; 0)$$

for the final chosen refined locked microscopic operator. That is now a concrete tensor-contraction computation, not a conceptual problem.

CXIII. FINAL SUMMARY

For the refined locked + Class II microscopic branch, the Bloch-harmonic shell-mediator coupling matrix satisfies the exact selection rule

$$\mathcal{C}_{(G,r),(G',i,a)}(k) = \mathcal{C}_{i,r}^a(G' - G; k + G),$$

or equivalently

$$\mathcal{C}_{(G,r),(G',i,a)}(k) = \sum_Q \delta_{G', G+Q} \mathcal{C}_{i,r}^a(Q; k + G).$$

In the locked 12-mode BCC branch, the leading background harmonics satisfy $Q \in \mathcal{S}_{12}$, so the intervalley shell-pair channel is controlled only by the single coefficient

$$Q = G_*.$$

Thus the actual microscopic carrier audit reduces to the evaluation of

$$\mathcal{C}_{i,r}^a(G_*; 0) = \alpha_X J_{i,r}^a(G_*; 0) + \beta_X K_{i,r}^a(G_*; 0).$$

Its three shell-adapted projections define the seed coefficients

$$c_{\parallel}, \quad c_E, \quad c_O,$$

which feed into the reduced witnesses

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}.$$

Therefore the remaining frontier is no longer conceptual. It is the explicit tensor-contraction computation of the $Q = G_*$ harmonic coefficients of the linearized Class II currents J_i^a and K_i^a .

CXIV. LOWEST-DERIVATIVE LOCAL ANSATZ FOR $J_i^a[\Psi]$ AND $K_i^a[P, \psi]$

CXV. GOAL

The next concrete task is to write the most general lowest-derivative local ansatz for the Class II currents

$$J_i^a[\Psi], \quad K_i^a[P, \psi],$$

compatible with the refined locked background, and then extract the Bloch-harmonic coefficients

$$J_{i,r}^a(G_*; 0), \quad K_{i,r}^a(G_*; 0)$$

that control the shell-pair intervalley channel.

The purpose is not to fix a unique final complete TECT action, but to reduce the remaining microscopic work to explicit tensor contractions at the single harmonic

$$Q = G_*.$$

CXVI. WORKING MICROSCOPIC BRANCH

We work in the strongest currently explicit branch,

$$\mathcal{L}_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a[\Psi] + \beta X_i^a K_i^a[P, \psi]. \quad (240)$$

The minimal explicit current choices already recorded in the notes are

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T^a \Psi + \Psi^\dagger T^a (\partial_i \Psi), \quad (241)$$

and

$$K_i^a[P, \psi] \sim (T^a \partial_i P) \mathcal{O}[\psi]. \quad (242)$$

A bookkeeping-equivalent lowest-order parametrization is

$$J_i^a[\phi_0, P_0] = m^a(P_0) \partial_i \phi_0, \quad (243)$$

$$K_i^a[\psi] = \mathcal{C}_{ij}^{ab} (-i \partial_j) s_b \psi, \quad (244)$$

with $m^a(P_0)$ a constant internal vector fixed by the locked projector background and $\mathcal{C}_{ij}^{ab}$ a fixed UV tensor kernel. :contentReference[oaicite:4]index=4

CXVII. LOCKED BACKGROUND EXPANSION

Write the refined locked background as

$$\Psi_0(x) = z_0 \phi_0(x), \quad \phi_0(x) = \sum_{Q \in \mathcal{S}_{12}} A_Q e^{iQ \cdot x}, \quad P_0 = z_0 z_0^\dagger. \quad (245)$$

Expand the full field as

$$\Psi = (\phi_0 + \sigma) z_0 + \psi, \quad P = P_0 + \delta P, \quad \delta P \sim z_0 \xi^\dagger + \xi z_0^\dagger. \quad (246)$$

Then

$$\partial_i \Psi = (\partial_i \phi_0) z_0 + \partial_i \psi + (\partial_i \sigma) z_0 + \cdots, \quad (247)$$

while

$$\partial_i P = \partial_i \delta P + \cdots \sim z_0 (\partial_i \xi)^\dagger + (\partial_i \xi) z_0^\dagger + \cdots. \quad (248)$$

Therefore J_i^a is linear in the shell gradient $\partial_i \phi_0$, and K_i^a is linear in the reduced defect/projector derivative sector. This is exactly the required (i, a) -carrier structure. :contentReference[oaicite:6]index=6

CXVIII. MOST GENERAL LOWEST-DERIVATIVE LOCAL ANSATZ

At the lowest local derivative order compatible with the locked background and tangent-doublet structure, the most general linearized current ansatz may be written as follows.

A. Background current

The linearized background current has the form

$$\delta J_i^a[\Psi_0; \delta \Psi] = \mathcal{A}_r^a (\partial_i \phi_0) \psi_r + \mathcal{B}_{jr}^a (\partial_i \phi_0) (-i \partial_j) \psi_r + \mathcal{D}_{ir}^a \psi_r + \cdots, \quad (249)$$

where r labels the shell-reduced internal/tangent basis.

For the first-order shell witness, only the piece with exactly one worldvolume derivative on the test mode is load-bearing. Therefore the relevant part is

$$\boxed{\delta J_i^a[\Psi_0; \delta \Psi]_{\text{lb}} = \mathcal{B}_{jr}^a (\partial_i \phi_0) (-i \partial_j) \psi_r.} \quad (250)$$

B. Projector/test current

Likewise the linearized projector/test current has the general lowest-order form

$$\delta K_i^a[P_0; \delta \Psi] = \mathcal{E}_{ir}^{ab} (\partial_i \xi^a) s_b \psi_r + \mathcal{F}_{ijr}^{ab} (\partial_i \xi^a) (-i \partial_j) s_b \psi_r + \cdots. \quad (251)$$

Again, for the shell witness we retain only the piece with exactly one worldvolume derivative:

$$\boxed{\delta K_i^a[P_0; \delta \Psi]_{\text{lb}} = \mathcal{F}_{ijr}^{ab} (\partial_i \xi^a) (-i \partial_j) s_b \psi_r.} \quad (252)$$

C. Minimal factorized form

A practical minimal factorized specialization consistent with the notes is

$$\boxed{\delta J_i^a[\Psi_0; \delta\Psi]_{\text{lb}} = m^a(P_0) (\partial_i \phi_0) \mathcal{U}_{jr}^b (-i\partial_j) s_b \psi_r,} \quad (253)$$

$$\boxed{\delta K_i^a[P_0; \delta\Psi]_{\text{lb}} = (\partial_i \xi^a) \mathcal{V}_{jr}^b (-i\partial_j) s_b \psi_r.} \quad (254)$$

This is the most useful form for explicit harmonic bookkeeping.

CXIX. BLOCH-HARMONIC EXPANSION

Expand the reduced test mode in Bloch harmonics:

$$\psi_r(x) = \sum_{G \in \Lambda^*} \psi_{G,r}(k) e^{i(k+G) \cdot x}. \quad (255)$$

Expand the locked shell background gradient:

$$\partial_i \phi_0(x) = \sum_{Q \in \mathcal{S}_{12}} iQ_i A_Q e^{iQ \cdot x}. \quad (256)$$

Then (253) gives

$$\delta J_i^a[\Psi_0; \delta\Psi]_{\text{lb}} = m^a(P_0) \sum_{Q \in \mathcal{S}_{12}} iQ_i A_Q e^{iQ \cdot x} \sum_{G,r} \mathcal{U}_{jr}^b(k+G)_j s_b \psi_{G,r}(k) e^{i(k+G) \cdot x} \quad (257)$$

$$= \sum_{G,Q,r} m^a(P_0) \underbrace{iQ_i A_Q \mathcal{U}_{jr}^b(k+G)_j s_b}_{J_{i,r}^a(Q; k+G)} \psi_{G,r}(k) e^{i(k+G+Q) \cdot x}. \quad (258)$$

Thus

$$\boxed{J_{i,r}^a(Q; k+G) = m^a(P_0) iQ_i A_Q \mathcal{U}_{jr}^b(k+G)_j s_b.} \quad (259)$$

Similarly, for (254), expand

$$\partial_i \xi^a(x) = \sum_{Q \in \Lambda^*} \Xi_i^a(Q) e^{iQ \cdot x},$$

to obtain

$$\boxed{K_{i,r}^a(Q; k+G) = \Xi_i^a(Q) \mathcal{V}_{jr}^b(k+G)_j s_b.} \quad (260)$$

CXX. INTERVALLEY SHELL-PAIR REDUCTION

For the folded shell-pair audit, the only relevant harmonic is

$$Q = G_*, \quad k_* - \frac{G_*}{2} = 0.$$

Therefore the intervalley Class II mixing block is controlled by

$$\boxed{J_{i,r}^a(G_*; 0) = m^a(P_0) i(G_*)_i A_{G_*} \mathcal{U}_{jr}^b(0)_j s_b,} \quad (261)$$

$$\boxed{\mathsf{K}_{i,r}^a(G_*; 0) = \Xi_i^a(G_*) \mathcal{V}_{jr}^b(0)_{jsb}.} \quad (262)$$

Hence the actual Class II harmonic block is

$$\boxed{\mathsf{C}_{i,r}^a(G_*; 0) = \alpha_X \mathsf{J}_{i,r}^a(G_*; 0) + \beta_X \mathsf{K}_{i,r}^a(G_*; 0).} \quad (263)$$

This is the concrete object whose shell projections define

$$c_{\parallel}, \quad c_E, \quad c_O.$$

CXXI. SHELL-SEED PROJECTION FORMULAS

Let $(\hat{n}, e_1, e_2)$ be the adapted shell frame and $(\Sigma_1, \Sigma_2, \Sigma_3)$ the tangent-doublet Pauli basis. Then the three seed coefficients are extracted by

$$\boxed{c_{\parallel} = \mathcal{N}_{\parallel} \hat{n}^i(\Sigma_3)^r{}_a \mathsf{C}_{i,r}^a(G_*; 0),} \quad (264)$$

$$\boxed{c_E = \mathcal{N}_E \frac{1}{2} [e_1^i(\Sigma_1)^r{}_a + e_2^i(\Sigma_2)^r{}_a] \mathsf{C}_{i,r}^a(G_*; 0),} \quad (265)$$

$$\boxed{c_O = \mathcal{N}_O \frac{1}{2} [-e_2^i(\Sigma_1)^r{}_a + e_1^i(\Sigma_2)^r{}_a] \mathsf{C}_{i,r}^a(G_*; 0),} \quad (266)$$

with $\mathcal{N}_{\parallel}, \mathcal{N}_E, \mathcal{N}_O$ fixed normalization constants.

Then

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}. \quad (267)$$

CXXII. IMMEDIATE YES/NO CRITERION

Therefore the full remaining microscopic question is reduced to the finite tensor-contraction test:

$$\boxed{\hat{n}^i(\Sigma_3)^r{}_a \mathsf{C}_{i,r}^a(G_*; 0) \neq 0, \quad ([e_1^i \Sigma_1 + e_2^i \Sigma_2]^r{}_a \mathsf{C}_{i,r}^a(G_*; 0), [-e_2^i \Sigma_1 + e_1^i \Sigma_2]^r{}_a \mathsf{C}_{i,r}^a(G_*; 0)) \neq (0, 0).} \quad (268)$$

This is exactly equivalent to

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

CXXIII. WHAT IS CLOSED AND WHAT REMAINS OPEN

What is now closed:

1. the most general lowest-derivative local ansatz for J_i^a and K_i^a ;
2. the explicit Bloch-harmonic formulas for

$$\mathsf{J}_{i,r}^a(Q; k + G), \quad \mathsf{K}_{i,r}^a(Q; k + G);$$

3. the reduction of the intervalley shell-pair channel to the single harmonic

$$Q = G_*;$$

4. the final shell-seed projection formulas producing

$$c_{\parallel}, \quad c_E, \quad c_O.$$

What remains open is only the explicit microscopic evaluation of the background tensors

$$m^a(P_0), \quad \mathcal{U}_{jr}^b(0), \quad \Xi_i^a(G_*), \quad \mathcal{V}_{jr}^b(0)$$

for the final chosen refined locked operator. That is now a direct tensor-contraction computation.

CXXIV. FINAL SUMMARY

The Class II microscopic carrier audit is now reduced to the explicit $Q = G_*$ harmonic coefficients of the lowest-derivative local current ansatz.

The general lowest-order local forms are

$$\delta J_i^a[\Psi_0; \delta\Psi]_{\text{lb}} = m^a(P_0) (\partial_i \phi_0) \mathcal{U}_{jr}^b (-i\partial_j) s_b \psi_r,$$

$$\delta K_i^a[P_0; \delta\Psi]_{\text{lb}} = (\partial_i \xi^a) \mathcal{V}_{jr}^b (-i\partial_j) s_b \psi_r.$$

Their Bloch-harmonic coefficients are

$$\mathcal{J}_{i,r}^a(Q; k+G) = m^a(P_0) iQ_i A_Q \mathcal{U}_{jr}^b(k+G)_j s_b,$$

$$\mathcal{K}_{i,r}^a(Q; k+G) = \Xi_i^a(Q) \mathcal{V}_{jr}^b(k+G)_j s_b.$$

For the folded shell-pair intervalley channel, only

$$Q = G_*, \quad k_* - \frac{G_*}{2} = 0$$

survives, so the relevant coupling is

$$\mathcal{C}_{i,r}^a(G_*; 0) = \alpha_X \mathcal{J}_{i,r}^a(G_*; 0) + \beta_X \mathcal{K}_{i,r}^a(G_*; 0).$$

Its three shell-adapted projections define

$$c_{\parallel}, \quad c_E, \quad c_O,$$

which determine

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}.$$

Therefore the final remaining frontier is the explicit tensor-contraction evaluation of

$$m^a(P_0), \quad \mathcal{U}_{jr}^b(0), \quad \Xi_i^a(G_*), \quad \mathcal{V}_{jr}^b(0)$$

for the final chosen refined locked microscopic operator.

CXXV. CANONICAL NORMALIZATION OF THE CLASS II SHELL COEFFICIENTS

CXXVI. GOAL

The remaining task is to choose a canonical local gauge/basis in the locked background and reduce the tensor contractions

$$c_{\parallel}, \quad c_E, \quad c_O$$

to explicit scalar coefficients multiplying the three shell seeds

$$B_{\parallel}, \quad B_E, \quad B_O.$$

At this stage, the problem is no longer to prove the existence of the carrier mechanism. It is to normalize the internal and shell-tangent bases so that the three reduced coefficients are read directly from the lowest-order UV tensors.

CXXVII. CANONICAL LOCKED BASIS

Take the locked background representative

$$z_0 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad P_0 = z_0 z_0^\dagger = \text{diag}(1, 0, 0). \quad (269)$$

Choose the tangent basis

$$u_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad (270)$$

so that

$$Q_{z_0} = \text{span}_{\mathbb{C}}\{u_1, u_2\}.$$

In this basis, the tangent Pauli matrices are the standard ones:

$$\Sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \Sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \Sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (271)$$

This fixes the internal tangent frame up to a residual $U(2)$ rotation. For the calibration step, we use this basis as the canonical frame.

CXXVIII. CANONICAL SHELL FRAME

Fix the selected shell vector G_* , and define

$$\hat{n} = \frac{G_*}{|G_*|}.$$

Choose a real orthonormal shell frame

$$\{e_1, e_2, \hat{n}\}. \quad (272)$$

Then every spatial tensor X_{ij} relevant to the first-order shell witness decomposes uniquely into the three shell-adapted invariant directions:

$$\hat{n}_i \hat{n}_j, \quad \hat{n}_i e_{1j} + \hat{n}_i e_{2j}, \quad -\hat{n}_i e_{2j} + \hat{n}_i e_{1j}, \quad (273)$$

paired with $\Sigma_3, \Sigma_1, \Sigma_2$, respectively.

CXXIX. CANONICAL CHOICE OF THE BACKGROUND VECTOR $m^a(P_0)$

The background vector $m^a(P_0)$ is an internal object determined only by the locked projector P_0 . In the canonical aligned basis, the unique distinguished tangent-singlet direction is the Σ_3 axis.

Therefore the minimal canonical choice is

$$\boxed{m^a(P_0) = m_{\parallel} \delta^{a3}}, \quad (274)$$

with $m_{\parallel} \in \mathbb{R}$.

This is the precise canonical statement that the longitudinal primitive odd singlet is aligned with the third internal axis.

CXXX. MINIMAL TENSOR DECOMPOSITION OF $\mathcal{U}$

The tensor $\mathcal{U}_{jr}{}^b(0)$ carries one gradient index j , one tangent-doublet index r , and one Pauli label b . At lowest derivative order, in the canonical shell frame, its shell-relevant part decomposes as

$$\boxed{\mathcal{U}_{jr}{}^b(0) = u_{\parallel} \hat{n}_j (\Sigma_3)_r{}^b + u_E e_{1j} (\Sigma_1)_r{}^b + u_E e_{2j} (\Sigma_2)_r{}^b + u_O (-e_{2j}) (\Sigma_1)_r{}^b + u_O e_{1j} (\Sigma_2)_r{}^b + \mathcal{U}_{jr}^{irr b}}, \quad (275)$$

where $\mathcal{U}^{irr}$ is orthogonal to the three shell seeds and therefore does not contribute to the reduced witness.

Equivalently,

$$\mathcal{U}_{jr}{}^b(0) = u_{\parallel} \hat{n}_j (\Sigma_3)_r{}^b + u_E (e_{1j} \Sigma_1 + e_{2j} \Sigma_2)_r{}^b + u_O (-e_{2j} \Sigma_1 + e_{1j} \Sigma_2)_r{}^b + \mathcal{U}_{jr}^{irr b}. \quad (276)$$

CXXXI. MINIMAL TENSOR DECOMPOSITION OF $\mathcal{V}$

Similarly, the projector-sector tensor $\mathcal{V}_{jr}{}^b(0)$ decomposes as

$$\boxed{\mathcal{V}_{jr}{}^b(0) = v_{\parallel} \hat{n}_j (\Sigma_3)_r{}^b + v_E (e_{1j} \Sigma_1 + e_{2j} \Sigma_2)_r{}^b + v_O (-e_{2j} \Sigma_1 + e_{1j} \Sigma_2)_r{}^b + \mathcal{V}_{jr}^{irr b}}, \quad (277)$$

again with $\mathcal{V}^{irr}$ orthogonal to the shell seed sector.

CXXXII. CANONICAL FORM OF THE $Q = G_*$ HARMONIC COEFFICIENTS

From the general formulas already derived,

$$J_{i,r}^a(G_*; 0) = m^a(P_0) i(G_*)_i A_{G_*} \mathcal{U}_{jr}{}^b(0) (0)_j s_b, \quad (278)$$

and

$$K_{i,r}^a(G_*; 0) = \Xi_i^a(G_*) \mathcal{V}_{jr}{}^b(0) (0)_j s_b. \quad (279)$$

For the shell witness one keeps the first nontrivial linear term in the slow momentum p_j . Thus the effective load-bearing coefficient is the linear coefficient multiplying p_j , which we denote by replacing $(0)_j$ with the tangent derivative insertion basis.

Accordingly, the shell-relevant current block becomes

$$\boxed{J_{i,r}^a(G_*; 0) = m_{\parallel} \delta^{a3} i(G_*)_i A_{G_*} \left[u_{\parallel} \hat{n}_j \Sigma_3 + u_E (e_{1j} \Sigma_1 + e_{2j} \Sigma_2) + u_O (-e_{2j} \Sigma_1 + e_{1j} \Sigma_2) \right]_r{}^b}, \quad (280)$$

and similarly

$$\boxed{K_{i,r}^a(G_*; 0) = \Xi_i^a(G_*) \left[v_{\parallel} \hat{n}_j \Sigma_3 + v_E (e_{1j} \Sigma_1 + e_{2j} \Sigma_2) + v_O (-e_{2j} \Sigma_1 + e_{1j} \Sigma_2) \right]_r{}^b}. \quad (281)$$

CXXXIII. CANONICAL DECOMPOSITION OF $\Xi_i^a(G_*)$

The projector-harmonic coefficient $\Xi_i^a(G_*)$ has one spatial index and one internal index. At the shell-relevant level, its most general canonical decomposition is

$$\boxed{\Xi_i^a(G_*) = \xi_{\parallel} \hat{n}_i \delta^{a3} + \xi_E (e_{1i} \delta^{a1} + e_{2i} \delta^{a2}) + \xi_O (-e_{2i} \delta^{a1} + e_{1i} \delta^{a2}) + \Xi_i^{a,irr}}. \quad (282)$$

Again Ξ^{irr} is orthogonal to the shell seed sector.

CXXXIV. EVALUATION OF THE SEED COEFFICIENTS

Now substitute (274), (276), (277), and (282) into

$$C_{i,r}^a(G_*; 0) = \alpha_X J_{i,r}^a(G_*; 0) + \beta_X K_{i,r}^a(G_*; 0).$$

Because the shell projections are orthogonal by construction, only aligned seed pieces survive. Therefore the three reduced coefficients take the canonical form

$$c_{\parallel} = \alpha_X \mu_J m_{\parallel} u_{\parallel} + \beta_X \mu_K \xi_{\parallel} v_{\parallel}, \quad (283)$$

$$c_E = \alpha_X \nu_J m_{\parallel} u_E + \beta_X \nu_K \xi_E v_E, \quad (284)$$

$$c_O = \alpha_X \omega_J m_{\parallel} u_O + \beta_X \omega_K \xi_O v_O. \quad (285)$$

Here $\mu_J, \mu_K, \nu_J, \nu_K, \omega_J, \omega_K$ are fixed numerical normalization factors determined by the chosen orthonormal shell/tangent basis.

In the fully normalized canonical convention, these factors may be set to unity:

$$\mu_J = \mu_K = \nu_J = \nu_K = \omega_J = \omega_K = 1. \quad (286)$$

Then

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel} + \beta_X \xi_{\parallel} v_{\parallel}, \quad (287)$$

$$c_E = \alpha_X m_{\parallel} u_E + \beta_X \xi_E v_E, \quad (288)$$

$$c_O = \alpha_X m_{\parallel} u_O + \beta_X \xi_O v_O. \quad (289)$$

CXXXV. MINIMAL EXPLICIT SPECIALIZATION

The sharpest minimal canonical specialization is obtained by taking the background current to contribute only to the longitudinal singlet and the projector current to contribute dominantly to the transverse doublet:

$$u_E = u_O = 0, \quad v_{\parallel} = 0. \quad (290)$$

Then

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel}, \quad c_E = \beta_X \xi_E v_E, \quad c_O = \beta_X \xi_O v_O. \quad (291)$$

This is the cleanest canonical realization of the Branch-B picture:

1. the longitudinal coefficient comes from the odd singlet background-current channel;
2. the transverse E/O coefficients come from the projector-mediated tangent-doublet channel.

CXXXVI. REDUCED WITNESS COEFFICIENTS

Substituting into the shell witness dictionary gives

$$\lambda_{\parallel} = \frac{1}{2} (\alpha_X m_{\parallel} u_{\parallel} + \beta_X \xi_{\parallel} v_{\parallel}) I_{\theta^3}, \quad (292)$$

$$\alpha = (\alpha_X m_{\parallel} u_E + \beta_X \xi_E v_E) I_{\theta}, \quad (293)$$

$$\beta = (\alpha_X m_{\parallel} u_O + \beta_X \xi_O v_O) I_{\theta}. \quad (294)$$

Under the minimal explicit specialization (290),

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta^3}, \quad \alpha = \beta_X \xi_E v_E I_{\theta}, \quad \beta = \beta_X \xi_O v_O I_{\theta}. \quad (295)$$

CXXXVII. FINAL YES/NO TEST IN CANONICAL NORMALIZATION

Thus the exact local Weyl/Dirac nondegeneracy criterion becomes

$$\alpha_X m_{\parallel} u_{\parallel} + \beta_X \xi_{\parallel} v_{\parallel} \neq 0, \quad (296)$$

and

$$\left(\alpha_X m_{\parallel} u_E + \beta_X \xi_E v_E, \alpha_X m_{\parallel} u_O + \beta_X \xi_O v_O \right) \neq (0, 0). \quad (297)$$

In the minimal explicit canonical split, this simplifies to

$$m_{\parallel} u_{\parallel} \neq 0, \quad (\xi_E v_E, \xi_O v_O) \neq (0, 0). \quad (298)$$

CXXXVIII. WHAT IS CLOSED AND WHAT REMAINS OPEN

What is now closed:

1. the canonical gauge/basis choice for the locked background;
2. the minimal shell-seed decomposition of the UV tensors $\mathcal{U}, \mathcal{V}, \Xi$;
3. the explicit canonical formulas for

$$c_{\parallel}, \quad c_E, \quad c_O;$$

4. the reduced witness coefficients $\lambda_{\parallel}, \alpha, \beta$ in canonical normalization.

What remains open is only the explicit microscopic evaluation of the scalar coefficients

$$m_{\parallel}, \quad u_{\parallel}, u_E, u_O, \quad \xi_{\parallel}, \xi_E, \xi_O, \quad v_{\parallel}, v_E, v_O$$

for the final chosen refined locked operator. That is now a finite coefficient-matching problem.

CXXXIX. FINAL SUMMARY

With the canonical locked basis

$$z_0 = (1, 0, 0)^T, \quad u_1 = (0, 1, 0)^T, \quad u_2 = (0, 0, 1)^T,$$

the tangent Pauli basis is fixed canonically.

The background vector aligns with the singlet axis as

$$m^a(P_0) = m_{\parallel} \delta^{a3}.$$

The shell-relevant UV tensors decompose into the three invariant seed directions:

$$\mathcal{U} = u_{\parallel} B_{\parallel} + u_E B_E + u_O B_O + \cdots,$$

$$\mathcal{V} = v_{\parallel} B_{\parallel} + v_E B_E + v_O B_O + \cdots,$$

$$\Xi = \xi_{\parallel} B_{\parallel} + \xi_E B_E + \xi_O B_O + \cdots.$$

Therefore the reduced shell coefficients become

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel} + \beta_X \xi_{\parallel} v_{\parallel},$$

$$c_E = \alpha_X m_{\parallel} u_E + \beta_X \xi_E v_E,$$

$$c_O = \alpha_X m_{\parallel} u_O + \beta_X \xi_O v_O.$$

Hence

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}.$$

Under the minimal explicit canonical split

$$u_E = u_O = 0, \quad v_{\parallel} = 0,$$

this simplifies to

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel}, \quad c_E = \beta_X \xi_E v_E, \quad c_O = \beta_X \xi_O v_O.$$

Thus the remaining frontier is reduced to the finite coefficient-matching problem for

$$m_{\parallel}, \quad u_{\parallel}, u_E, u_O, \quad \xi_{\parallel}, \xi_E, \xi_O, \quad v_{\parallel}, v_E, v_O.$$

CXL. SUBMISSION-READY MASTER FORMULA FOR THE MINIMAL EXPLICIT CANONICAL SPLIT

CXLI. CANONICAL SPLIT

We adopt the minimal explicit canonical split:

$$\boxed{u_E = u_O = 0, \quad v_{\parallel} = 0.} \tag{299}$$

This means:

1. the longitudinal primitive odd singlet is carried entirely by the background-current J -sector;
2. the transverse even/odd doublets are carried entirely by the projector/test-current K -sector.

Equivalently, the shell-relevant coefficients reduce to

$$\boxed{c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel}, \quad c_E = \beta_X \xi_E v_E, \quad c_O = \beta_X \xi_O v_O.} \tag{300}$$

CXLII. REDUCED WITNESS COEFFICIENTS

Using the exact shell witness dictionary,

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}, \quad (301)$$

we obtain

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}, \quad (302)$$

$$\alpha = \beta_X \xi_E v_E I_{\theta}, \quad (303)$$

$$\beta = \beta_X \xi_O v_O I_{\theta}. \quad (304)$$

CXLIII. REDUCED FIRST-ORDER SYMBOL

Therefore the exact reduced Branch-B first-order symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2). \quad (305)$$

Substituting (302)–(304),

$$A_1(p) = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3} p_{\parallel} \sigma_3 + \beta_X \xi_E v_E I_{\theta} (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X \xi_O v_O I_{\theta} (-p_2 \sigma_1 + p_1 \sigma_2). \quad (306)$$

CXLIV. EXACT LOCAL WEYL/DIRAC CRITERION

The exact nondegeneracy condition becomes

$$m_{\parallel} u_{\parallel} \neq 0, \quad (\xi_E v_E, \xi_O v_O) \neq (0, 0), \quad (307)$$

assuming

$$\alpha_X \neq 0, \quad \beta_X \neq 0, \quad I_{\theta'^3} \neq 0, \quad I_{\theta} \neq 0.$$

Equivalently,

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (308)$$

This is the complete submission-ready local Weyl condition for the minimal explicit canonical split.

CXLV. INFRARED VELOCITIES

Identifying the reduced coefficients with the infrared velocities,

$$v_{\parallel}^{\text{IR}} = \lambda_{\parallel}, \quad v_E^{\text{IR}} = \alpha, \quad v_O^{\text{IR}} = \beta, \quad (309)$$

we get

$$v_{\parallel}^{\text{IR}} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}, \quad (310)$$

$$v_{\perp}^{\text{IR}} = \sqrt{\alpha^2 + \beta^2} = |\beta_X I_{\theta}| \sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}. \quad (311)$$

CXLVI. ANISOTROPY RATIO

The anisotropy ratio is therefore

$$\rho = \frac{v_{\parallel}^{\text{IR}}}{v_{\perp}^{\text{IR}}} = \frac{\frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}}{|\beta_X I_{\theta}| \sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}}. \quad (312)$$

Equivalently,

$$\rho = \frac{\alpha_X}{\beta_X} \frac{m_{\parallel} u_{\parallel}}{2 \sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}} \frac{I_{\theta'^3}}{I_{\theta}}, \quad (313)$$

up to the overall sign convention of I_{θ} and β_X .

CXLVII. SMALL-DEFECT REGIME

In the natural small-defect regime,

$$I_{\theta'^3} \sim \frac{3}{16} (\theta_0 \kappa)^2 I_{\theta},$$

so the leading-order anisotropy becomes

$$\rho_{\text{LO}} = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{m_{\parallel} u_{\parallel}}{\sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}} (\theta_0 \kappa)^2. \quad (314)$$

Thus the Branch-B cone is naturally strongly transverse-dominated unless the longitudinal coefficient is parametrically enhanced.

CXLVIII. ASPECT RATIO AND INVERSE ESTIMATE

Define the cone aspect ratio

$$A := \frac{1}{|\rho|} = \frac{v_{\perp}^{\text{IR}}}{|v_{\parallel}^{\text{IR}}|}. \quad (315)$$

Then

$$A_{\text{LO}} = \frac{32}{3} \frac{\beta_X}{\alpha_X} \frac{\sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}}{|m_{\parallel} u_{\parallel}|} (\theta_0 \kappa)^{-2}. \quad (316)$$

Hence the microscopic defect-strength combination is inferred as

$$\theta_0 \kappa = \sqrt{\frac{32}{3A} \frac{\beta_X}{\alpha_X} \frac{\sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}}{|m_{\parallel} u_{\parallel}|}}. \quad (317)$$

CXLIX. TRACE FORMULATION

In the projected 2×2 shell block language, the same formulas are read from

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M^{\parallel}), \quad \alpha = \frac{1}{2} \text{Tr}(\sigma_1 M^1), \quad \beta = \frac{1}{2} \text{Tr}(\sigma_2 M^1). \quad (318)$$

Under the minimal explicit canonical split, these become

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_3 M^\parallel) = \frac{1}{2} \alpha_X m_\parallel u_\parallel I_{\theta'^3}}, \quad (319)$$

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_1 M^1) = \beta_X \xi_E v_E I_\theta}, \quad (320)$$

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_2 M^1) = \beta_X \xi_O v_O I_\theta}. \quad (321)$$

CL. SUBMISSION-READY THEOREM

[Minimal explicit canonical Branch-B theorem] Assume:

1. the refined locked microscopic branch admits the Class II carrier sector;
2. the canonical split (299) holds;
3. the overlap integrals satisfy

$$I_{\theta'^3} \neq 0, \quad I_\theta \neq 0;$$

4. the UV couplings satisfy

$$\alpha_X \neq 0, \quad \beta_X \neq 0.$$

Then the reduced first-order shell symbol is

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

with

$$\lambda_\parallel = \frac{1}{2} \alpha_X m_\parallel u_\parallel I_{\theta'^3}, \quad \alpha = \beta_X \xi_E v_E I_\theta, \quad \beta = \beta_X \xi_O v_O I_\theta.$$

Hence the local Weyl/Dirac phase exists if and only if

$$m_\parallel u_\parallel \neq 0, \quad (\xi_E v_E, \xi_O v_O) \neq (0, 0).$$

Moreover, the anisotropy ratio is

$$\rho = \frac{\frac{1}{2} \alpha_X m_\parallel u_\parallel I_{\theta'^3}}{|\beta_X I_\theta| \sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}},$$

and in the small-defect regime satisfies

$$\rho_{\text{LO}} = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{m_\parallel u_\parallel}{\sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}} (\theta_0 \kappa)^2.$$

CLI. FINAL SUMMARY

Under the minimal explicit canonical split,

$$u_E = u_O = 0, \quad v_\parallel = 0,$$

the shell coefficients are

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel}, \quad c_E = \beta_X \xi_E v_E, \quad c_O = \beta_X \xi_O v_O.$$

Therefore the reduced Branch-B coefficients are

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}, \quad \alpha = \beta_X \xi_E v_E I_{\theta}, \quad \beta = \beta_X \xi_O v_O I_{\theta}.$$

The exact local Weyl/Dirac condition is

$$m_{\parallel} u_{\parallel} \neq 0, \quad (\xi_E v_E, \xi_O v_O) \neq (0, 0).$$

The anisotropy ratio is

$$\rho = \frac{\frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}}{|\beta_X I_{\theta}| \sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}},$$

and in the small-defect regime becomes

$$\rho_{\text{LO}} = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{m_{\parallel} u_{\parallel}}{\sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}} (\theta_0 \kappa)^2.$$

Thus the entire remaining microscopic frontier is reduced to a finite coefficient-matching problem for the scalar quantities

$$m_{\parallel}, \quad u_{\parallel}, \quad \xi_E v_E, \quad \xi_O v_O.$$

CLII. FINITE COEFFICIENT-MATCHING PROBLEM

CLIII. STATEMENT OF THE PROBLEM

After the full reduction to the refined locked + Class II branch, the remaining microscopic problem is no longer an infinite-dimensional PDE problem. It is a finite coefficient-matching problem for the shell-seed sector.

Under the minimal explicit canonical split

$$u_E = u_O = 0, \quad v_{\parallel} = 0, \tag{322}$$

the shell coefficients reduce to

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel}, \quad c_E = \beta_X \xi_E v_E, \quad c_O = \beta_X \xi_O v_O. \tag{323}$$

Therefore the reduced witness coefficients are

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}, \tag{324}$$

$$\alpha = \beta_X \xi_E v_E I_{\theta}, \tag{325}$$

$$\beta = \beta_X \xi_O v_O I_{\theta}. \tag{326}$$

Hence the exact local Weyl condition becomes

$$\boxed{m_{\parallel} u_{\parallel} \neq 0, \quad (\xi_E v_E, \xi_O v_O) \neq (0, 0),} \tag{327}$$

assuming

$$\alpha_X \neq 0, \quad \beta_X \neq 0, \quad I_{\theta'^3} \neq 0, \quad I_{\theta} \neq 0.$$

CLIV. NATURAL REPACKAGING INTO TWO FINITE INVARIANTS

Define the longitudinal scalar

$$L_* := m_{\parallel} u_{\parallel}, \quad (328)$$

and the transverse two-vector

$$T_* := (T_E, T_O) := (\xi_E v_E, \xi_O v_O). \quad (329)$$

Then the shell coefficients become

$$c_{\parallel} = \alpha_X L_*, \quad (c_E, c_O) = \beta_X T_*, \quad (330)$$

and the reduced witness coefficients are

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta^3}, \quad (\alpha, \beta) = \beta_X I_{\theta} T_*. \quad (331)$$

Thus the entire local carrier problem is reduced to:

$$\boxed{L_* \neq 0, \quad T_* \neq 0.} \quad (332)$$

CLV. POLAR FORM OF THE TRANSVERSE SECTOR

It is useful to write the transverse vector in polar form:

$$R_{\perp} := \sqrt{T_E^2 + T_O^2}, \quad T_E = R_{\perp} \cos \varphi, \quad T_O = R_{\perp} \sin \varphi. \quad (333)$$

Then

$$\alpha = \beta_X I_{\theta} R_{\perp} \cos \varphi, \quad \beta = \beta_X I_{\theta} R_{\perp} \sin \varphi, \quad (334)$$

and

$$v_{\perp}^{\text{IR}} = \sqrt{\alpha^2 + \beta^2} = |\beta_X I_{\theta}| R_{\perp}. \quad (335)$$

So the transverse sector is fully controlled by two finite numbers: its magnitude $R_{\perp}$ and the orientation angle φ .

CLVI. MINIMAL MASTER FORMULA

With

$$L_* = m_{\parallel} u_{\parallel}, \quad R_{\perp} = \sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2},$$

the reduced first-order symbol is

$$\boxed{A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta^3} p_{\parallel} \sigma_3 + \beta_X I_{\theta} R_{\perp} \left[\cos \varphi (p_1 \sigma_1 + p_2 \sigma_2) + \sin \varphi (-p_2 \sigma_1 + p_1 \sigma_2) \right]}. \quad (336)$$

This is the submission-ready master formula for the finite coefficient-matching problem.

CLVII. EXACT WEYL/DIRAC CRITERION

The exact nondegeneracy condition is now simply

$$\boxed{L_* \neq 0, \quad R_{\perp} > 0.} \quad (337)$$

Equivalently,

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (338)$$

So the local Weyl/Dirac phase depends only on two finite matching conditions: one longitudinal scalar and one transverse norm.

CLVIII. ANISOTROPY RATIO

The infrared longitudinal velocity is

$$v_{\parallel}^{\text{IR}} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}, \quad (339)$$

and the transverse velocity is

$$v_{\perp}^{\text{IR}} = |\beta_X I_{\theta}| R_{\perp}. \quad (340)$$

Hence

$$\rho = \frac{v_{\parallel}^{\text{IR}}}{v_{\perp}^{\text{IR}}} = \frac{\alpha_X}{\beta_X} \frac{L_*}{2R_{\perp}} \frac{I_{\theta'^3}}{I_{\theta}}, \quad (341)$$

up to the overall sign convention.

In the small-defect regime,

$$I_{\theta'^3} \sim \frac{3}{16} (\theta_0 \kappa)^2 I_{\theta},$$

so

$$\rho_{\text{LO}} = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{L_*}{R_{\perp}} (\theta_0 \kappa)^2. \quad (342)$$

Therefore the entire anisotropy prediction is reduced to the finite ratio

$$\frac{L_*}{R_{\perp}}.$$

CLIX. WHAT IS PROVED, WHAT IS ASSUMED

At this point the honest status is:

a. Proved algebraically. If the microscopic Class II branch reduces to the minimal explicit canonical split, then the entire shell witness problem is equivalent to the finite matching condition

$$L_* \neq 0, \quad R_{\perp} > 0.$$

b. Not yet computed microscopically. The actual values of

$$L_* = m_{\parallel} u_{\parallel}, \quad T_E = \xi_E v_E, \quad T_O = \xi_O v_O$$

have not yet been obtained from the final complete refined locked Bloch operator.

c. Therefore. The remaining frontier is not conceptual. It is the explicit coefficient-matching computation of the three scalar combinations above.

CLX. FINAL FINITE-MATCHING THEOREM

[Finite coefficient-matching theorem] Assume:

1. the refined locked microscopic branch admits the Class II carrier sector;
2. the canonical split

$$u_E = u_O = 0, \quad v_{\parallel} = 0$$

holds;

3. the overlap integrals satisfy

$$I_{\theta'^3} \neq 0, \quad I_\theta \neq 0;$$

4. the UV couplings satisfy

$$\alpha_X \neq 0, \quad \beta_X \neq 0.$$

Define

$$L_* = m_\parallel u_\parallel, \quad T_* = (\xi_E v_E, \xi_O v_O), \quad R_\perp = \|T_*\|.$$

Then the reduced first-order shell symbol is

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'^3} p_\parallel \sigma_3 + \beta_X I_\theta T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_\theta T_O (-p_2 \sigma_1 + p_1 \sigma_2),$$

and the local Weyl/Dirac phase exists if and only if

$$L_* \neq 0, \quad R_\perp > 0.$$

Moreover,

$$\rho = \frac{\alpha_X}{\beta_X} \frac{L_*}{2R_\perp} \frac{I_{\theta'^3}}{I_\theta},$$

and in the small-defect regime,

$$\rho_{\text{LO}} = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{L_*}{R_\perp} (\theta_0 \kappa)^2.$$

CLXI. FINAL SUMMARY

Under the minimal explicit canonical split, the microscopic carrier problem reduces from four scalar coefficients to two finite invariants:

$$L_* := m_\parallel u_\parallel, \quad R_\perp := \sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}.$$

The reduced Branch-B shell symbol is

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'^3} p_\parallel \sigma_3 + \beta_X I_\theta T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_\theta T_O (-p_2 \sigma_1 + p_1 \sigma_2),$$

where

$$T_E = \xi_E v_E, \quad T_O = \xi_O v_O.$$

The exact local Weyl/Dirac condition is

$$L_* \neq 0, \quad R_\perp > 0.$$

The anisotropy ratio is

$$\rho = \frac{\alpha_X}{\beta_X} \frac{L_*}{2R_\perp} \frac{I_{\theta'^3}}{I_\theta},$$

and in the small-defect regime becomes

$$\rho_{\text{LO}} = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{L_*}{R_\perp} (\theta_0 \kappa)^2.$$

Thus the entire remaining microscopic frontier is reduced to a finite coefficient-extraction problem for

$$L_*, \quad T_E, \quad T_O.$$

CLXII. FINITE COEFFICIENT-MATCHING PROBLEM: FINAL CLOSED FORM

CLXIII. SHORT KOREAN EXPLANATION

At the present stage, the remaining microscopic TECT frontier is no longer an infinite-dimensional PDE problem. It is a finite coefficient-matching problem for the shell-reduced carrier sector.

Under the minimal explicit canonical split,

$$u_E = u_O = 0, \quad v_{\parallel} = 0,$$

the shell coefficients reduce to three scalar combinations:

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel}, \quad c_E = \beta_X \xi_E v_E, \quad c_O = \beta_X \xi_O v_O.$$

Therefore the whole local Weyl/Dirac question is controlled only by one longitudinal scalar and one transverse norm.

CLXIV. FULL DETAILED LATEX DERIVATION

A. Minimal explicit canonical split

Adopt the minimal explicit canonical split

$$u_E = u_O = 0, \quad v_{\parallel} = 0. \quad (343)$$

Then the shell-seed coefficients become

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel}, \quad c_E = \beta_X \xi_E v_E, \quad c_O = \beta_X \xi_O v_O. \quad (344)$$

Using the exact seed dictionary,

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \quad \alpha = c_E I_{\theta}, \quad \beta = c_O I_{\theta}, \quad (345)$$

we obtain

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3},} \quad (346)$$

$$\boxed{\alpha = \beta_X \xi_E v_E I_{\theta},} \quad (347)$$

$$\boxed{\beta = \beta_X \xi_O v_O I_{\theta}.} \quad (348)$$

B. Two invariant combinations

Define

$$L_* := m_{\parallel} u_{\parallel}, \quad (349)$$

and

$$T_* := (T_E, T_O) := (\xi_E v_E, \xi_O v_O). \quad (350)$$

Let

$$R_{\perp} := \sqrt{T_E^2 + T_O^2}. \quad (351)$$

Then the shell coefficients are simply

$$c_{\parallel} = \alpha_X L_*, \quad (c_E, c_O) = \beta_X T_*, \quad (352)$$

and the reduced witness coefficients become

$$\boxed{\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}, \quad (\alpha, \beta) = \beta_X I_{\theta} T_*.} \quad (353)$$

C. Submission-ready master symbol

Hence the reduced first-order shell symbol is

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'^3} p_{\parallel} \sigma_3 + \beta_X I_{\theta} T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_{\theta} T_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (354)$$

Equivalently, if we write

$$T_E = R_{\perp} \cos \varphi, \quad T_O = R_{\perp} \sin \varphi,$$

then

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'^3} p_{\parallel} \sigma_3 + \beta_X I_{\theta} R_{\perp} \left[\cos \varphi (p_1 \sigma_1 + p_2 \sigma_2) + \sin \varphi (-p_2 \sigma_1 + p_1 \sigma_2) \right]. \quad (355)$$

D. Exact Weyl/Dirac condition

The exact local Weyl/Dirac nondegeneracy condition is therefore

$$L_* \neq 0, \quad R_{\perp} > 0, \quad (356)$$

assuming

$$\alpha_X \neq 0, \quad \beta_X \neq 0, \quad I_{\theta'^3} \neq 0, \quad I_{\theta} \neq 0.$$

This is equivalent to

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (357)$$

E. Infrared velocities and anisotropy

Identify

$$v_{\parallel}^{\text{IR}} = \lambda_{\parallel}, \quad v_{\perp}^{\text{IR}} = \sqrt{\alpha^2 + \beta^2}. \quad (358)$$

Then

$$v_{\parallel}^{\text{IR}} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}, \quad v_{\perp}^{\text{IR}} = |\beta_X I_{\theta}| R_{\perp}. \quad (359)$$

Hence the anisotropy ratio is

$$\rho = \frac{v_{\parallel}^{\text{IR}}}{v_{\perp}^{\text{IR}}} = \frac{\alpha_X}{\beta_X} \frac{L_*}{2R_{\perp}} \frac{I_{\theta'^3}}{I_{\theta}}, \quad (360)$$

up to the overall sign convention.

In the natural small-defect regime,

$$I_{\theta'^3} \sim \frac{3}{16} (\theta_0 \kappa)^2 I_{\theta},$$

so

$$\rho_{\text{LO}} = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{L_*}{R_{\perp}} (\theta_0 \kappa)^2. \quad (361)$$

F. What is proved and what is not

What is now proved algebraically is:

If the refined locked Class II branch reduces to the minimal canonical split, then the entire shell witness problem is equivalent to

(362)

What is not yet computed from the final full microscopic Bloch operator is:

$$\boxed{L_* = m_{\parallel} u_{\parallel}, \quad T_E = \xi_E v_E, \quad T_O = \xi_O v_O.} \quad (363)$$

Therefore the remaining frontier is a finite coefficient-extraction problem, not a conceptual one.

CLXV. FINAL SUMMARY (LATEX)

The finite coefficient-matching problem is now reduced to two invariant quantities:

$$L_* := m_{\parallel} u_{\parallel}, \quad R_{\perp} := \sqrt{(\xi_E v_E)^2 + (\xi_O v_O)^2}.$$

Under the minimal explicit canonical split,

$$u_E = u_O = 0, \quad v_{\parallel} = 0,$$

the reduced Branch-B shell symbol is

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'^3} p_{\parallel} \sigma_3 + \beta_X I_{\theta} T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_{\theta} T_O (-p_2 \sigma_1 + p_1 \sigma_2),$$

where

$$T_E = \xi_E v_E, \quad T_O = \xi_O v_O.$$

The exact local Weyl/Dirac condition is

$$L_* \neq 0, \quad R_{\perp} > 0.$$

The anisotropy ratio is

$$\rho = \frac{\alpha_X}{\beta_X} \frac{L_*}{2R_{\perp}} \frac{I_{\theta'^3}}{I_{\theta}},$$

and in the small-defect regime becomes

$$\rho_{\text{LO}} = \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{L_*}{R_{\perp}} (\theta_0 \kappa)^2.$$

Thus the entire remaining microscopic frontier is reduced to the explicit extraction of the finite scalar combinations

$$L_*, \quad T_E, \quad T_O$$

from the final refined locked Bloch operator.

CLXVI. TECT MASTER CONTEXT (FINITE COEFFICIENT-MATCHING STAGE)

A. Project goal

The goal of the TECT program is to derive spacetime fermionic carriers from a microscopic topological energy condensate background.

In particular, the program aims to show that a Dirac/Weyl sector emerges as a shell-reduced Bloch mode near a BCC condensate background.

B. Locked condensate background

The energy condensate background is assumed to form a locked BCC structure.

$$\Psi_0(x) = z_0 \phi_0(x)$$

$$\phi_0(x) = \sum_{Q \in \mathcal{S}_{12}} A_Q e^{iQ \cdot x}$$

with the locked internal projector

$$P_0 = z_0 z_0^\dagger.$$

C. Defect sector

Topological defects correspond to CP^2 excitations

$$P(x) = z(x)z(x)^\dagger.$$

The tangent sector

$$Q_{z_0}$$

produces the two-component fermionic mode.

D. Bloch shell reduction

Near the resonant shell

$$k_* = \frac{G_*}{2}$$

the Bloch expansion is

$$\Psi(x) = \sum_G \psi_G e^{i(k+G) \cdot x}.$$

The Dirac valley basis is

$$\Phi_\pm = e^{\pm i G_* x / 2}.$$

Define the Dirac projector

$$\Pi_D.$$

E. Microscopic operator

The microscopic linearized operator is

$$\mathcal{L}_{\text{mic}}(k).$$

The projected Dirac block is

$$M(k) = \Pi_D \mathcal{L}_{\text{mic}}(k) \Pi_D.$$

F. Momentum expansion

Near the valley point

$$k = k_* + p$$

we expand

$$M(p) = M_0 + M_i p_i + O(p^2).$$

G. Reduced first-order symbol

The shell-reduced Dirac symbol takes the universal form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

H. Seed dictionary

The reduced coefficients are

$$\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta^3}$$

$$\alpha = c_E I_{\theta}$$

$$\beta = c_O I_{\theta}.$$

Under the minimal canonical split

$$u_E = u_O = 0, \quad v_{\parallel} = 0$$

the seed coefficients reduce to

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel}$$

$$c_E = \beta_X \xi_E v_E$$

$$c_O = \beta_X \xi_O v_O.$$

I. Finite invariants

Define

$$L_* = m_{\parallel} u_{\parallel}$$

$$T_E = \xi_E v_E$$

$$T_O = \xi_O v_O.$$

Let

$$R_{\perp} = \sqrt{T_E^2 + T_O^2}.$$

J. Master symbol

The reduced Dirac symbol becomes

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'^3} p_{\parallel} \sigma_3 + \beta_X I_{\theta} T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_{\theta} T_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

K. Weyl/Dirac condition

The exact local Weyl condition is

$$L_* \neq 0$$

$$R_{\perp} > 0.$$

L. Infrared velocities

$$v_{\parallel}^{IR} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}$$

$$v_{\perp}^{IR} = |\beta_X I_{\theta}| R_{\perp}.$$

M. Anisotropy ratio

$$\rho = \frac{v_{\parallel}^{IR}}{v_{\perp}^{IR}} = \frac{\alpha_X}{\beta_X} \frac{L_*}{2R_{\perp}} \frac{I_{\theta'^3}}{I_{\theta}}.$$

N. Remaining microscopic frontier

The entire remaining microscopic problem reduces to the extraction of three scalar coefficients from the Bloch operator:

$$L_*, \quad T_E, \quad T_O.$$

These are obtained from

$$M_i = \partial_{p_i} (\Pi_D \mathcal{L}_{\text{mic}}(p) \Pi_D)_{p=0}.$$

Explicitly

$$\lambda_{\parallel} = \frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel})$$

$$\alpha = \frac{1}{2} \text{Tr}(\sigma_1 M_1)$$

$$\beta = \frac{1}{2} \text{Tr}(\sigma_2 M_1).$$

Thus

$$L_*, T_E, T_O$$

are directly read from the projected microscopic Bloch operator.

CLXVII. FINAL SUMMARY

The TECT microscopic carrier problem has been reduced to a finite coefficient-matching problem. All microscopic structure enters only through three scalar quantities

$$L_*, \quad T_E, \quad T_O.$$

The reduced Dirac symbol is

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'3} p_{\parallel} \sigma_3 + \beta_X I_{\theta} T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_{\theta} T_O (-p_2 \sigma_1 + p_1 \sigma_2).$$

The Weyl/Dirac condition is

$$L_* \neq 0, \quad R_{\perp} > 0.$$

Thus the remaining microscopic frontier is the explicit extraction of

$$L_*, \quad T_E, \quad T_O$$

from the Bloch operator

$$\Pi_D \mathcal{L}_{\text{mic}} \Pi_D.$$

CLXVIII. ACTUAL MICROSCOPIC COEFFICIENT EXTRACTION IN THE FINITE COEFFICIENT-MATCHING STAGE

A. Goal

The remaining microscopic task is to extract the three shell coefficients

$$\lambda_{\parallel}, \quad \alpha, \quad \beta$$

from the projected first derivative of the actual Bloch-linearized microscopic operator.

Equivalently, the extraction chain is

$$L_{\text{mic}} \longrightarrow L_{\text{Bloch}}(k) \longrightarrow K_i := \partial_{k_i} L_{\text{Bloch}}(k) \Big|_{k=k_*} \longrightarrow P_* K_i P_* \longrightarrow (\lambda_{\parallel}, \alpha, \beta).$$

This is precisely the finite operator audit form of the remaining TECT microscopic frontier.

B. Folded Bragg convention

We work in the folded Bloch convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.$$

Let the shell-adapted frame be

$$\hat{n} = \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\} \text{ orthonormal,}$$

and decompose

$$p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2.$$

Let

$$P_*$$

denote the rank-two projector onto the shell-relevant Bloch valley subspace. Then the projected first-order symbol is

$$A_1(p) = P_* (L_{\text{Bloch}}(k_* + p) - L_{\text{Bloch}}(k_*)) P_* + O(|p|^2).$$

C. Projected derivative matrices

Define

$$K_i := \partial_{k_i} L_{\text{Bloch}}(k) \Big|_{k=k_*}, \quad M_i := P_* K_i P_*.$$

Then

$$A_1(p) = p_i M_i + O(|p|^2).$$

In the shell-adapted basis, write

$$M_{\parallel} := \hat{n}_i M_i, \quad M_1 := e_{1i} M_i, \quad M_2 := e_{2i} M_i.$$

The reduced S2 branch has the universal form

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

Hence the coefficients are extracted by Pauli traces:

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}), \tag{364}$$

$$\alpha = \frac{1}{2}(\sigma_1 M_1) = \frac{1}{2}(\sigma_2 M_2), \tag{365}$$

$$\beta = \frac{1}{2}(\sigma_2 M_1) = -\frac{1}{2}(\sigma_1 M_2). \tag{366}$$

Therefore the entire nondegeneracy problem is reduced to finite projector-trace contractions.

D. Concrete Class II microscopic branch

Take as working microscopic branch

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi]. \quad (367)$$

Integrating out the heavy mediator X_i^a at quadratic order gives the effective enriched transport contribution

$$\delta L_{\text{eff}}^{(II)} \sim -\frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a. \quad (368)$$

At shell-reduced level, this generates the enriched microscopic tensor

$$\delta N_{ij}^a.$$

Inside the shell-adapted seed family one has

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (369)$$

For the explicit Class II reduction,

$$\delta A = -\Gamma c_{\parallel}, \quad \delta B = -\Gamma c_I, \quad \delta C = -\Gamma c_J, \quad \Gamma = \frac{\lambda g_X m}{M_X^2}, \quad (370)$$

so that

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

Therefore the shell coefficients become

$$\lambda_{\parallel} = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3, \quad (371)$$

$$\alpha = \frac{\kappa_* |G_*|}{2} \left(e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2 \right), \quad (372)$$

$$\beta = \frac{\kappa_* |G_*|}{2} \left(e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2 \right). \quad (373)$$

Substituting (369) gives

$$\lambda_{\parallel} = \kappa_* |G_*| A, \quad \alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C. \quad (374)$$

Hence, after the Class II correction,

$$\lambda_{\parallel}^{(II)} \propto c_{\parallel}, \quad \alpha^{(II)} \propto c_I, \quad \beta^{(II)} \propto c_J. \quad (375)$$

E. Identification with the finite invariants

Now compare the general finite-coefficient dictionary

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}, \quad \alpha = \beta_X I_{\theta} T_E, \quad \beta = \beta_X I_{\theta} T_O,$$

with the concrete Class II witness formulas

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta}.$$

This gives the exact identification

$$g_c = \alpha_X L_*, \quad g_E = \beta_X T_E, \quad g_O = \beta_X T_O. \quad (376)$$

Equivalently,

$$L_* = \frac{g_c}{\alpha_X}, \quad T_E = \frac{g_E}{\beta_X}, \quad T_O = \frac{g_O}{\beta_X}. \quad (377)$$

Inside the minimal canonical split,

$$L_* = m_{\parallel} u_{\parallel}, \quad T_E = \xi_E v_E, \quad T_O = \xi_O v_O.$$

So the actual microscopic coefficient audit reduces to computing

$$g_c, \quad g_E, \quad g_O,$$

or equivalently the projected traces (364)–(366).

F. Explicit concrete normalization

In the exact concrete Class II transport normalization one has

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad (378)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa. \quad (379)$$

Therefore

$$g_c = -\frac{512}{3\sqrt{3}\pi \theta_0^3 \kappa^3} \lambda_{\parallel}, \quad (380)$$

$$g_E = -\frac{16}{\sqrt{3}\pi \theta_0 \kappa} \alpha, \quad g_O = -\frac{16}{\sqrt{3}\pi \theta_0 \kappa} \beta. \quad (381)$$

Combining with (377),

$$L_* = -\frac{512}{3\sqrt{3}\pi \alpha_X \theta_0^3 \kappa^3} \lambda_{\parallel}, \quad (382)$$

$$T_E = -\frac{16}{\sqrt{3}\pi \beta_X \theta_0 \kappa} \alpha, \quad T_O = -\frac{16}{\sqrt{3}\pi \beta_X \theta_0 \kappa} \beta. \quad (383)$$

G. Weyl condition in the actual extraction form

The local Weyl condition becomes

$$L_* \neq 0, \quad R_{\perp} := \sqrt{T_E^2 + T_O^2} > 0. \quad (384)$$

Equivalently,

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0), \quad (385)$$

or, in seed variables,

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0). \quad (386)$$

Thus the actual microscopic finite coefficient-matching program is closed at the protocol level:

$$\boxed{L_{\text{Bloch}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta) \rightarrow (g_c, g_E, g_O) \rightarrow (L_*, T_E, T_O).}$$

H. What is closed and what remains open

What is closed rigorously is the extraction protocol and the exact dictionary:

$$(\lambda_{\parallel}, \alpha, \beta) \Longleftrightarrow (g_c, g_E, g_O) \Longleftrightarrow (L_*, T_E, T_O).$$

What remains open is only the final explicit computation of

$$K_i = \partial_{k_i} L_{\text{Bloch}}(k)|_{k=k_*}$$

from the final complete microscopic TECT operator, and the direct evaluation of the projected traces

$$\frac{1}{2}(\sigma_3 M_{\parallel}), \quad \frac{1}{2}(\sigma_1 M_1), \quad \frac{1}{2}(\sigma_2 M_1).$$

CLXIX. FINAL SUMMARY

The finite coefficient-matching problem is now reduced to the explicit shell-projected derivative audit

$$M_i = P_* K_i P_*, \quad K_i = \partial_{k_i} L_{\text{Bloch}}(k)|_{k=k_*}.$$

The three actual microscopic Dirac coefficients are

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}), \quad \alpha = \frac{1}{2}(\sigma_1 M_1), \quad \beta = \frac{1}{2}(\sigma_2 M_1).$$

Inside the concrete Class II realization,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta},$$

with

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

Hence

$$L_* = \frac{g_c}{\alpha_X}, \quad T_E = \frac{g_E}{\beta_X}, \quad T_O = \frac{g_O}{\beta_X}.$$

Therefore the remaining TECT microscopic frontier is no longer structural. It is the explicit computation of the projected Bloch derivative matrices M_i , after which the Weyl/Dirac condition is the finite test

$$L_* \neq 0, \quad \sqrt{T_E^2 + T_O^2} > 0.$$

CLXX. FINITE COEFFICIENT-MATCHING: EXACT CURRENT STATUS

The microscopic Dirac problem is reduced to the shell-projected derivative data

$$M_i = P_* K_i P_*, \quad K_i = \partial_{k_i} L_{\text{Bloch}}(k)|_{k=k_*}.$$

The extracted coefficients are

$$\lambda_{\parallel} = \frac{1}{2}(\sigma_3 M_{\parallel}), \quad \alpha = \frac{1}{2}(\sigma_1 M_1), \quad \beta = \frac{1}{2}(\sigma_2 M_1).$$

In the concrete Class II branch,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta},$$

and

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

Therefore

$$L_* = \frac{g_c}{\alpha_X}, \quad T_E = \frac{g_E}{\beta_X}, \quad T_O = \frac{g_O}{\beta_X},$$

so the Weyl condition is the finite test

$$L_* \neq 0, \quad \sqrt{T_E^2 + T_O^2} > 0.$$

What remains open is only the direct explicit computation of K_i from the final complete microscopic TECT operator.

CLXXI. SHELL-ADAPTED BLOCK CONSTRUCTION OF THE PROJECTED BLOCH DERIVATIVE

A. Goal

The next legitimate microscopic step is to construct the first Bloch-derivative block

$$K_i := \partial_{k_i} L_{\text{Bloch}}(k) \Big|_{k=k_*}$$

in a shell-adapted basis and then to project it to the selected local valley subspace.

This is exactly the finite operator-construction stage:

$$L_{\text{Bloch}}(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

B. Folded Bragg convention

We use the folded Bragg convention

$$k_* = \frac{G_*}{2}, \quad k = k_* + p, \quad |p| \ll |G_*|.$$

Choose the shell-adapted orthonormal frame

$$\hat{n} = \frac{G_*}{|G_*|}, \quad \{e_1, e_2, \hat{n}\}.$$

Then

$$p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2,$$

with

$$p_{\parallel} = \hat{n}_i p_i, \quad p_a = e_{ai} p_i, \quad a = 1, 2.$$

C. First-order Bloch expansion

Assume the Bloch symbol admits the Taylor expansion

$$L_{\text{Bloch}}(k_* + p) = L_* + p_i K_i + O(|p|^2), \quad K_i = \partial_{k_i} L_{\text{Bloch}}(k) \Big|_{k=k_*}.$$

Then the selected local valley block is

$$h_{\text{lin}}^+(p) = P_*(p_i K_i) P_*.$$

Since the projected block is 2×2 Hermitian, it has the Pauli expansion

$$h_{\text{lin}}^+(p) = M_{\mu i} p_i \sigma_{\mu}, \quad M_{\mu i} = \frac{1}{2 H_*} (\sigma_{\mu} P_* K_i P_*).$$

D. Minimal shell-enriched axial ansatz

The correct shell-adapted first-derivative ansatz is the axial A, B, C decomposition:

$$K_i = A \hat{n}_i \tau^3 \otimes \mathbf{1}_2 + B (e_{1i} \tau^1 \otimes s_1 + e_{2i} \tau^1 \otimes s_2) + C (-e_{2i} \tau^1 \otimes s_1 + e_{1i} \tau^1 \otimes s_2). \quad (387)$$

This is exactly the minimal shell-enriched axial form that separates:

1. the longitudinal singlet coefficient A ,
2. the transverse even coefficient B ,
3. the transverse odd coefficient C .

E. Directional components

Contracting (387) with the shell frame gives

$$K_{\parallel} := \hat{n}_i K_i = A \tau^3 \otimes \mathbf{1}_2, \quad (388)$$

$$K_1 := e_{1i} K_i = B \tau^1 \otimes s_1 + C \tau^1 \otimes s_2, \quad (389)$$

$$K_2 := e_{2i} K_i = -C \tau^1 \otimes s_1 + B \tau^1 \otimes s_2. \quad (390)$$

Therefore the full first-order symbol is

$$\begin{aligned} p_i K_i &= p_{\parallel} K_{\parallel} + p_1 K_1 + p_2 K_2 \\ &= A p_{\parallel} \tau^3 \otimes \mathbf{1}_2 + B (p_1 \tau^1 \otimes s_1 + p_2 \tau^1 \otimes s_2) + C (-p_2 \tau^1 \otimes s_1 + p_1 \tau^1 \otimes s_2). \end{aligned} \quad (391)$$

F. Valley projection

Project to the selected two-band valley block with projector P_* , using the reduced identifications

$$P_*(\tau^3 \otimes \mathbf{1}_2)P_* \mapsto \sigma^3, \quad P_*(\tau^1 \otimes s_1)P_* \mapsto \sigma^1, \quad P_*(\tau^1 \otimes s_2)P_* \mapsto \sigma^2.$$

Then (391) becomes

$$h_+^{\text{lin}}(p) = A p_{\parallel} \sigma^3 + B (p_1 \sigma^1 + p_2 \sigma^2) + C (-p_2 \sigma^1 + p_1 \sigma^2). \quad (392)$$

Comparing with the universal S2 witness form

$$H_{\text{wit}}(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2),$$

we read off immediately

$$\boxed{\lambda_{\parallel} = A, \quad \alpha = B, \quad \beta = C.} \quad (393)$$

G. Equivalent Pauli-trace extraction

The same result follows from the projector-trace formulas. From (388)–(390),

$$P_* K_{\parallel} P_* = A \sigma^3, \quad P_* K_1 P_* = B \sigma^1 + C \sigma^2, \quad P_* K_2 P_* = -C \sigma^1 + B \sigma^2.$$

Hence

$$\lambda_{\parallel} = \frac{1}{2} (\sigma^3 P_* K_{\parallel} P_*) = A, \quad (394)$$

$$\alpha = \frac{1}{2} (\sigma^1 P_* K_1 P_*) = \frac{1}{2} (\sigma^2 P_* K_2 P_*) = B, \quad (395)$$

$$\beta = \frac{1}{2} (\sigma^2 P_* K_1 P_*) = -\frac{1}{2} (\sigma^1 P_* K_2 P_*) = C. \quad (396)$$

H. Weyl nondegeneracy matrix

The coefficient matrix in the shell frame is therefore

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}, \quad (397)$$

so that

$$\det M = A(B^2 + C^2). \quad (398)$$

Therefore the exact local Weyl condition is

$$\boxed{A \neq 0, \quad (B, C) \neq (0, 0)}. \quad (399)$$

Equivalently,

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)}. \quad (400)$$

I. Dictionary to the finite coefficient-matching variables

In the finite coefficient-matching language,

$$\lambda_{\parallel} = \frac{1}{2}\alpha_X L_* I_{\theta'^3}, \quad \alpha = \beta_X I_{\theta} T_E, \quad \beta = \beta_X I_{\theta} T_O.$$

Hence (393) gives

$$A = \frac{1}{2}\alpha_X L_* I_{\theta'^3}, \quad B = \beta_X I_{\theta} T_E, \quad C = \beta_X I_{\theta} T_O. \quad (401)$$

Equivalently,

$$L_* = \frac{2A}{\alpha_X I_{\theta'^3}}, \quad T_E = \frac{B}{\beta_X I_{\theta}}, \quad T_O = \frac{C}{\beta_X I_{\theta}}. \quad (402)$$

Inside the concrete Class II branch, one already has

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \quad \alpha = g_E I_{\theta}, \quad \beta = g_O I_{\theta},$$

so

$$A = \frac{g_c}{2} I_{\theta'^3}, \quad B = g_E I_{\theta}, \quad C = g_O I_{\theta}. \quad (403)$$

Thus

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0) \quad (404)$$

is exactly the same condition as

$$A \neq 0, \quad (B, C) \neq (0, 0).$$

J. What is now closed

The following step is now rigorously closed:

1. the shell-adapted first-derivative block K_i has the exact A, B, C decomposition;
2. the directional blocks $K_{\parallel}, K_1, K_2$ are explicit;

3. after valley projection, the reduced Bloch Hamiltonian is exactly

$$h_+^{\text{lin}}(p) = A p_{\parallel} \sigma^3 + B (p_1 \sigma^1 + p_2 \sigma^2) + C (-p_2 \sigma^1 + p_1 \sigma^2);$$

4. therefore

$$\lambda_{\parallel} = A, \quad \alpha = B, \quad \beta = C;$$

5. the local Weyl test is the finite condition

$$A \neq 0, \quad (B, C) \neq (0, 0).$$

K. What remains open

What remains open is not the shell-reduced logic. It is only the explicit computation of the actual microscopic numbers

$$A, \quad B, \quad C$$

from the final complete periodic refined locked + Class II Bloch operator. Equivalently, one must compute

$$K_i = \partial_{k_i} L_{\text{Bloch}}(k) \Big|_{k=k_*}$$

for the actual microscopic branch and then evaluate the projected traces.

CLXXII. SHELL-ADAPTED BLOCH BLOCK: EXACT RESULT

The first projected Bloch derivative is written in the axial shell form

$$K_i = A \hat{n}_i \tau^3 \otimes \mathbf{1}_2 + B (e_{1i} \tau^1 \otimes s_1 + e_{2i} \tau^1 \otimes s_2) + C (-e_{2i} \tau^1 \otimes s_1 + e_{1i} \tau^1 \otimes s_2).$$

Hence

$$K_{\parallel} = A \tau^3 \otimes \mathbf{1}_2, \quad K_1 = B \tau^1 \otimes s_1 + C \tau^1 \otimes s_2, \quad K_2 = -C \tau^1 \otimes s_1 + B \tau^1 \otimes s_2.$$

After projection to the selected two-band valley block,

$$h_+^{\text{lin}}(p) = A p_{\parallel} \sigma^3 + B (p_1 \sigma^1 + p_2 \sigma^2) + C (-p_2 \sigma^1 + p_1 \sigma^2).$$

Therefore

$$\lambda_{\parallel} = A, \quad \alpha = B, \quad \beta = C,$$

and the exact local Weyl condition is

$$A \neq 0, \quad (B, C) \neq (0, 0).$$

Equivalently,

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

Thus the remaining microscopic problem is only to compute the actual values of

$$A, \quad B, \quad C$$

from the final periodic refined locked + Class II Bloch operator.

CLXXIII. 12-MODE BCC BLOCH MATRIX IN THE SHELL-RESTRICTED MODE BASIS

A. Goal

The correct next step is to write the periodic Bloch operator explicitly in the reciprocal-harmonic basis of the locked 12-mode BCC background, isolate the Bragg-resonant finite block at

$$k_* = \frac{G_*}{2},$$

and identify the reduced 2×2 valley operator.

This is the actual finite-dimensional matrix stage of the microscopic program:

$$\Psi_0 \rightarrow H_{\text{Bloch}}(k) \rightarrow H_{\text{Bragg}}(k_* + p) \rightarrow H_{\text{shell}}(p) \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

B. Locked 12-mode BCC background

Take the first BCC reciprocal shell

$$\mathcal{S}_{12} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}, \quad |\mathcal{S}_{12}| = 12.$$

The equal-amplitude locked background is

$$\phi_0(x) = A \sum_{Q \in \mathcal{S}_{12}} e^{iQ \cdot x} = 2A \sum_{\alpha=1}^6 \cos(Q_{\alpha} \cdot x),$$

and

$$\Psi_0(x) = z_0 \phi_0(x), \quad P_0 = z_0 z_0^{\dagger}.$$

C. Bloch-harmonic expansion

For a Bloch quasimomentum k , expand the fluctuation in reciprocal harmonics:

$$\delta \Psi_k(x) = e^{ik \cdot x} \sum_{G \in \mathcal{G}} \psi_G e^{iG \cdot x},$$

where $\mathcal{G}$ is the reciprocal lattice generated by the BCC background. Then the Bloch fiber acts on the coefficient vector (ψ_G) by

$$(H_{\text{Bloch}}(k))_{G,G'} = h_{\text{diag}}(k+G) \delta_{G,G'} + V_{G-G'}. \quad (405)$$

Here:

1. $h_{\text{diag}}(q)$ is the diagonal bare/refined kinetic symbol;
2. V_K are the Fourier coefficients of the periodic linearized background potential / transport kernel.

For the scalar periodic Hessian this reduces to the standard formula

$$(H_{\text{Bloch}}(k))_{G,G'} = h_0(k+G) \delta_{G,G'} + U_{G-G'},$$

and the same reciprocal-convolution structure persists in the refined locked + Class II branch, only with matrix-valued internal blocks replacing the scalar coefficients.

D. The Bragg-resonant shell pair

Fix one selected shell vector $G_* \in \mathcal{S}_{12}$. The Bragg crossing occurs at

$$k_* = \frac{G_*}{2}.$$

The two directly resonant harmonics are

$$|+\rangle := |G = 0\rangle, \quad |-\rangle := |G = -G_*\rangle,$$

since

$$k_* + 0 = \frac{G_*}{2}, \quad k_* - G_* = -\frac{G_*}{2},$$

and therefore the two bare momenta have equal norm.

In the locked triplet branch there is also a reduced internal doublet

$$u_1, u_2 \in Q_{z_0}, \quad u_r^\dagger u_s = \delta_{rs},$$

so the shell-relevant local carrier space is

$$\mathcal{H}_D \cong \text{span}\{|+\rangle, |-\rangle\} \otimes \text{span}\{u_1, u_2\} \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2.$$

A convenient ordered basis is

$$\mathcal{B}_D = \{|+, u_1\rangle, |+, u_2\rangle, |-, u_1\rangle, |-, u_2\rangle\}.$$

E. The finite 4×4 Bragg block

Restrict the full Bloch fiber to $\mathcal{H}_D$. Then the shell-relevant Bragg block has the form

$$H_D(k) = \begin{pmatrix} E_+(k) \mathbf{1}_2 + A_+(k) & M(k) \\ M(k)^\dagger & E_-(k) \mathbf{1}_2 + A_-(k) \end{pmatrix}, \quad (406)$$

where:

1. $E_\pm(k)$ are the diagonal bare dispersions at k and $k - G_*$;
2. $A_\pm(k)$ are internal 2×2 self-block corrections coming from the refined locked/Class II linearization;
3. $M(k)$ is the 2×2 pair-mixing block induced by the Fourier coefficient at momentum transfer G_* and its enriched transport corrections.

At the exact Bragg point $k_* = G_*/2$,

$$E_+(k_*) = E_-(k_*),$$

so the nontrivial shell question is controlled by the off-diagonal and first-order derivative structure.

F. Decomposition of the pair-mixing block

Write the 2×2 pair-mixing block as

$$M(k) = M_0(k) \mathbf{1}_2 + M_a(k) s_a, \quad a = 1, 2, 3. \quad (407)$$

At $k = k_*$, the constant scalar piece

$$M_0 := M_0(k_*)$$

is the constant Bragg mass obstruction. For a naive quadratic scalar periodic coefficient

$$U_{\text{quad}}(x) = u_2 \phi_0(x)^2,$$

its selected-shell Fourier coefficient is

$$\hat{U}_{\text{quad}}(G_*) = 4u_2 A^2,$$

hence

$$M_0^{\text{quad}} = 4u_2 A^2 \mathbf{1}_2 \neq 0 \quad (408)$$

generically.

Therefore a direct Dirac crossing requires an actual microscopic cancellation, symmetry, or protected elimination of this constant scalar pair-mixing piece.

G. Linear expansion around the Bragg point

Now set

$$k = k_* + p, \quad p = p_{\parallel} \hat{n} + p_1 e_1 + p_2 e_2, \quad \hat{n} = \frac{G_*}{|G_*|}.$$

Expand the shell block to first order:

$$H_D(k_* + p) = H_D(k_*) + p_i K_i + O(|p|^2), \quad (409)$$

with

$$K_i = \partial_{k_i} H_D(k) \Big|_{k=k_*}.$$

Inside the shell-adapted reduced branch, the first-order part has the universal axial form

$$K_i = A \hat{n}_i \tau_3 \otimes \mathbf{1}_2 + B (e_{1i} \tau_1 \otimes s_1 + e_{2i} \tau_1 \otimes s_2) + C (-e_{2i} \tau_1 \otimes s_1 + e_{1i} \tau_1 \otimes s_2). \quad (410)$$

Hence

$$p_i K_i = A p_{\parallel} \tau_3 \otimes \mathbf{1}_2 + B (p_1 \tau_1 \otimes s_1 + p_2 \tau_1 \otimes s_2) + C (-p_2 \tau_1 \otimes s_1 + p_1 \tau_1 \otimes s_2). \quad (411)$$

H. Reduced 2×2 valley block

After projection to the selected two-band valley branch,

$$\tau_3 \otimes \mathbf{1}_2 \mapsto \sigma_3, \quad \tau_1 \otimes s_1 \mapsto \sigma_1, \quad \tau_1 \otimes s_2 \mapsto \sigma_2,$$

and one gets

$$h_{\text{val}}(p) = M_0 \mathbf{1}_2 + \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha (p_1 \sigma_1 + p_2 \sigma_2) + \beta (-p_2 \sigma_1 + p_1 \sigma_2) + O(|p|^2), \quad (412)$$

with

$$\lambda_{\parallel} = A, \quad \alpha = B, \quad \beta = C. \quad (413)$$

Therefore the exact local Weyl condition is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad (414)$$

while the exact local Dirac crossing also requires

$$M_0 = 0. \quad (415)$$

I. What the 12-mode locked branch already gives

In the locked equal-amplitude 12-mode BCC shell-sum calculation, the shell-projected witnesses are

$$W_{\parallel}^{(12)} = 4 \frac{gA^2}{q_*^2}, \quad W_I^{(12)} = \frac{4Z}{A}, \quad W_J^{(12)} = 0. \quad (416)$$

Thus the reduced witness determinant is

$$\det M^{(12)} = 64 \kappa_* g Z^2 q_* > 0. \quad (417)$$

So the witness survives the genuine 12-mode shell sum:

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)$$

in this working branch.

However, this does not by itself prove the full unrestricted microscopic theorem, because:

polarization locking is still assumed, and the full multi-mode projector/BdG linearization is still open.

J. Exact status of the matrix construction

The finite shell-relevant matrix problem is now completely identified:

1. build the Bloch matrix $H_{\text{Bloch}}(k)$ in the reciprocal-harmonic basis;
2. restrict it to the resonant 4×4 shell block $\mathcal{H}_D$;
3. separate the constant scalar Bragg block $M_0 \mathbf{1}_2$;
4. extract the first-order coefficients $(\lambda_{\parallel}, \alpha, \beta)$.

The reduced witness part is already structurally and, in the locked 12-mode branch, explicitly nondegenerate. The true remaining microscopic bottleneck is therefore

$$\boxed{M_0 = 0.}$$

CLXXIV. FINAL SUMMARY

For the locked 12-mode BCC background, the periodic Bloch operator in the reciprocal-harmonic basis has the convolution form

$$(H_{\text{Bloch}}(k))_{G,G'} = h_{\text{diag}}(k+G)\delta_{G,G'} + V_{G-G'}.$$

At the Bragg point

$$k_* = \frac{G_*}{2},$$

the shell-relevant local carrier space is

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

giving a finite 4×4 block

$$H_D(k) = \begin{pmatrix} E_+(k)\mathbf{1}_2 + A_+(k) & M(k) \\ M(k)^\dagger & E_-(k)\mathbf{1}_2 + A_-(k) \end{pmatrix}.$$

Its reduced 2×2 valley block has the form

$$h_{\text{val}}(p) = M_0 \mathbf{1}_2 + \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2) + O(|p|^2).$$

Thus:

$$\text{Weyl witness} \iff \lambda_{\parallel} \neq 0, (\alpha, \beta) \neq (0, 0),$$

while

$$\text{actual ungapped crossing} \iff M_0 = 0$$

in addition.

In the locked equal-amplitude 12-mode BCC branch one already has

$$W_{\parallel}^{(12)} = 4 \frac{gA^2}{q_*^2}, \quad W_I^{(12)} = \frac{4Z}{A}, \quad W_J^{(12)} = 0,$$

hence

$$\det M^{(12)} = 64 \kappa_* g Z^2 q_* > 0.$$

So the shell witness survives the first genuine 12-mode BCC shell sum. The honest remaining microscopic obstruction is the constant scalar Bragg mass M_0 .

CLXXV. 12-MODE BCC BLOCH MATRIX: EXACT CURRENT STAGE

The periodic Bloch operator over the locked 12-mode BCC background has the reciprocal-harmonic matrix form

$$(H_{\text{Bloch}}(k))_{G,G'} = h_{\text{diag}}(k+G)\delta_{G,G'} + V_{G-G'}.$$

At the Bragg point

$$k_* = \frac{G_*}{2},$$

the shell-relevant local space is

$$\mathcal{H}_D \cong \mathbb{C}_{\text{valley}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

so the finite shell block is 4×4 :

$$H_D(k) = \begin{pmatrix} E_+(k)\mathbf{1}_2 + A_+(k) & M(k) \\ M(k)^\dagger & E_-(k)\mathbf{1}_2 + A_-(k) \end{pmatrix}.$$

Its reduced valley operator is

$$h_{\text{val}}(p) = M_0\mathbf{1}_2 + \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2) + O(|p|^2).$$

Hence the Weyl witness requires

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0),$$

while an actual ungapped crossing additionally requires

$$M_0 = 0.$$

In the locked equal-amplitude 12-mode BCC branch, one already finds

$$W_{\parallel}^{(12)} = 4 \frac{gA^2}{q_*^2}, \quad W_I^{(12)} = \frac{4Z}{A}, \quad W_J^{(12)} = 0,$$

and therefore

$$\det M^{(12)} = 64 \kappa_* g Z^2 q_* > 0.$$

So the witness survives the first genuine 12-mode shell sum. The remaining microscopic bottleneck is the constant Bragg mass condition $M_0 = 0$.

CLXXVI. EXACT 12-MODE CONVOLUTION LAW FOR THE CONSTANT BRAGG MASS

A. Goal

The true remaining obstruction in the plain shell-pair Bloch route is the constant scalar identity-channel Bragg block

$$M_0 = B_0^{\text{tot}} \mathbf{1}_2.$$

The task is therefore not to re-discuss the derivative witnesses, but to compute the exact 12-mode BCC convolution formula for B_0^{tot} and isolate the precise cancellation law.

B. Shell-relevant scalar sector

Take the most general scalar periodic contribution built from the natural locked first-shell operators

$$U_{\text{tot}}(x) = c_0 \phi_0(x)^2 + c_1 |\nabla \phi_0(x)|^2 + c_2 \phi_0(x)(-\Delta)\phi_0(x), \quad (418)$$

where

$$\phi_0(x) = A \sum_{Q \in S_{12}} e^{iQ \cdot x}$$

is the equal-amplitude locked 12-mode BCC background.

The constant Bragg block at the selected shell pair $(0, -G_*)$ is controlled by the selected-shell Fourier coefficient

$$\widehat{U}_{\text{tot}}(G_*).$$

C. Exact 12-mode BCC Fourier coefficients

In the locked 12-mode BCC branch, the first-shell convolution factors are

$$\widehat{\phi_0^2}(G_*) = 4A^2, \quad (419)$$

$$\widehat{|\nabla \phi_0|^2}(G_*) = 2q_*^2 A^2, \quad (420)$$

$$\widehat{\phi_0(-\Delta)\phi_0}(G_*) = 4q_*^2 A^2. \quad (421)$$

Therefore

$$\begin{aligned} \widehat{U}_{\text{tot}}(G_*) &= c_0 \widehat{\phi_0^2}(G_*) + c_1 \widehat{|\nabla \phi_0|^2}(G_*) + c_2 \widehat{\phi_0(-\Delta)\phi_0}(G_*) \\ &= 4c_0 A^2 + 2c_1 q_*^2 A^2 + 4c_2 q_*^2 A^2 \\ &= 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2). \end{aligned} \quad (422)$$

Hence the projected scalar Bragg block is exactly

$$\boxed{B_0^{\text{tot}} = 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2.} \quad (423)$$

D. Exact cancellation condition

The ungapped plain shell-pair crossing requires

$$M_0 = 0 \quad \Longleftrightarrow \quad B_0^{\text{tot}} = 0.$$

By (423), this is equivalent to

$$\boxed{2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0.} \quad (424)$$

So the constant Bragg-mass problem is reduced to a single finite linear relation among the identity-channel scalar coefficients.

E. Naive scalar square obstruction

For the simplest scalar square sector

$$U_{\text{quad}}(x) = u_2 \phi_0(x)^2,$$

one gets

$$\hat{U}_{\text{quad}}(G_*) = 4u_2 A^2, \quad M_0^{\text{quad}} = 4u_2 A^2 \mathbf{1}_2 \neq 0 \quad (425)$$

generically.

Thus the cancellation (424) is not automatic even in the full 12-mode shell sum.

F. Why canonical Class II does not solve this

The canonical Class II correction lives in the traceless Pauli channel:

$$\delta B_0^{\text{Class II}} \in \text{span}\{s_1, s_2, s_3\},$$

so in the shell-pair block it contributes as

$$\tau_1 \otimes s_a \quad \text{or} \quad \tau_2 \otimes s_a,$$

not as

$$\tau_1 \otimes \mathbf{1}_2.$$

Since the identity and traceless channels are orthogonal,

$$(\mathbf{1}_2 s_a) = 0,$$

canonical Class II cannot directly cancel the scalar identity-channel block

$$B_0^{\text{tot}} \tau_1 \otimes \mathbf{1}_2.$$

Therefore:

Class II renormalizes the derivative witness sector (A, B, C) , but does not eliminate M_0 .

(426)

G. Only two logically distinct routes

The reduced operator analysis leaves exactly two possibilities.

a. Route 1: direct identity-channel cancellation. Introduce an additional microscopic sector whose projected constant pair block contains a nonzero identity component

$$\delta M_0^{\text{id}} = \delta m_0 \mathbf{1}_2 + \delta \widetilde{M}_0, \quad \delta m_0 = -B_0^{\text{tot}}.$$

Then the total constant scalar block vanishes.

b. Route 2: graded 2+2 bypass. Instead of canceling M_0 inside the plain scalar shell-pair block, pass to a J -graded $2+2$ first-order infrared sector. In that reduced operator,

$$D_1(0) = 0,$$

so the dangerous constant scalar Bragg block is absent from the relevant principal Dirac operator from the outset.

Hence the exact classification is

identity-carrying mediator route or graded $2+2$ bypass route.

(427)

H. Spectator-assisted identity-channel cancellation

A concrete identity-channel route is already visible at the effective level. If a spectator sector produces a Schur-complement identity correction, the total scalar block becomes

$$B_0^{\text{tot}} = \left[4uA^2 - \sum_{\eta \in S_{\text{bi}}} \frac{\widehat{V}(\eta) \overline{\widehat{V}(\eta + G_*)}}{\Delta_\eta} \right] \mathbf{1}_2. \quad (428)$$

In the equal-coupling diagonal approximation this reduces to

$$4uA^2 - \frac{4|v_1|^2}{\Delta_{\text{sp}}} A^2 = 0, \quad (429)$$

equivalently

$$u = \frac{|v_1|^2}{\Delta_{\text{sp}}}. \quad (430)$$

More generally, combining this with (423) gives

$$2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) - \sum_{\eta, \zeta \in S_{\text{bi}}} \widehat{V}(\eta)(R_0)_{\eta\zeta} \overline{\widehat{V}(\zeta + G_*)} = 0. \quad (431)$$

I. Honest logical status

The exact 12-mode convolution law for the scalar Bragg block is now closed. What is also closed is the negative result: within the current safe scalar + canonical Class II library, the identity-channel obstruction is not removed automatically.

Therefore the plain shell-pair route is no longer the unique mainline. The honest decision tree is:

(A) add an identity-carrying sector and satisfy (431),

or

(B) move to the graded 2 + 2 bypass route.

CLXXVII. FINAL SUMMARY

For the locked equal-amplitude 12-mode BCC branch, the scalar identity-channel Bragg block is

$$B_0^{\text{tot}} = 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2.$$

Hence the exact cancellation condition is

$$2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0.$$

A naive scalar square sector gives instead

$$M_0^{\text{quad}} = 4u_2 A^2 \mathbf{1}_2 \neq 0,$$

so cancellation is not automatic. Canonical Class II cannot solve this because it lives in the traceless Pauli channel and therefore renormalizes (A, B, C) , not the identity-channel block M_0 .

Thus only two logically distinct routes remain:

identity-channel cancellation or graded 2 + 2 bypass.

The first route is governed by the generalized exact law

$$2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) - \sum_{\eta, \zeta \in S_{\text{bi}}} \widehat{V}(\eta)(R_0)_{\eta\zeta} \overline{\widehat{V}(\zeta + G_*)} = 0,$$

while the second route avoids the scalar Bragg block at the principal infrared level.

CLXXVIII. EXACT STATUS OF THE CONSTANT BRAGG-MASS PROBLEM

In the locked 12-mode BCC branch, the scalar identity-channel Bragg block is exactly

$$B_0^{\text{tot}} = 2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) \mathbf{1}_2.$$

Therefore the exact scalar cancellation condition is

$$2c_0 + q_*^2 c_1 + 2q_*^2 c_2 = 0.$$

A naive scalar square sector gives

$$M_0^{\text{quad}} = 4u_2 A^2 \mathbf{1}_2 \neq 0,$$

so the cancellation is not automatic.

Canonical Class II corrections lie in the traceless Pauli channel and therefore renormalize the derivative witness sector (A, B, C) , but do not remove the scalar identity-channel obstruction M_0 .

Hence only two logically distinct routes remain:

- (i) identity-channel cancellation, (ii) graded $2 + 2$ bypass.

A general identity-channel cancellation law is

$$2A^2(2c_0 + q_*^2 c_1 + 2q_*^2 c_2) - \sum_{\eta, \zeta \in S_{\text{bi}}} \widehat{V}(\eta)(R_0)_{\eta\zeta} \overline{\widehat{V}(\zeta + G_*)} = 0.$$

Thus the plain shell-pair route is no longer conceptually open: its exact bottleneck and its exact possible exits are now fully identified.

CLXXIX. ACTUAL COEFFICIENT-POINT AUDIT IN THE GRADED $2 + 2$ MAINLINE

A. Goal

The plain scalar Bragg-pair route is no longer the correct mainline. The correct mainline is the graded $2 + 2$ bypass route, in which the relevant principal infrared operator is

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_i M_i,$$

with

$$D_1(0) = 0.$$

Therefore the decisive remaining microscopic problem is not the cancellation of B_0 , but the extraction of the actual coefficient point

$$\boxed{(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O)}.$$

B. Graded infrared sector and projector-defect carrier

Let the actual local low-energy carrier be a graded four-dimensional sector

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2,$$

with grading involution

$$J^\dagger = J, \quad J^2 = \mathbf{1}, \quad J|_{K_{\pm}} = \pm 1.$$

Let

$$\Pi_K$$

be the spectral projector onto this isolated infrared sector. Then

$$D_{\text{eff}}(p) = \Pi_K L(p) \Pi_K = D_1(p) + O(|p|^2),$$

and the graded bypass hypothesis is

$$JD_1(p)J^{-1} = -D_1(p).$$

Hence in the J -graded basis,

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(p) = p_i M_i. \quad (432)$$

C. Pauli decomposition of the mixed symbol

Write the 2×2 mixed symbol in Pauli form:

$$A_1(p) = \ell_0(p) \mathbf{1}_2 + \ell_a(p) \sigma_a, \quad \ell_\alpha(p) = C_\alpha^i p_i, \quad \alpha = 0, 1, 2, 3. \quad (433)$$

Then

$$\det A_1(p) = -p_i G_{ij} p_j, \quad G_{ij} = \sum_{a=1}^3 C_a^i C_a^j - C_0^i C_0^j. \quad (434)$$

So the exact graded nondegeneracy test is

$$\boxed{G_{ij} > 0.} \quad (435)$$

D. Natural local S2 branch

For the actual local projector-defect geometry, the natural reduced branch is the S2 singlet $\oplus$ doublet form:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (436)$$

Equivalently,

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\parallel} \end{pmatrix}, \quad \det \Lambda = \lambda_{\parallel} (\alpha^2 + \beta^2). \quad (437)$$

Thus

$$\boxed{G_{ij} > 0 \iff \lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).} \quad (438)$$

E. Actual transport kernel from the full Hessian

The actual microscopic task is to derive the background-dependent transport kernel

$$V^\mu(\xi)$$

from the full TECT Hessian. Let the defect background be $P_B(\xi)$, with

$$Q_B(\xi) = (\mathbb{C} z_B(\xi))^\perp, \quad \dim_{\mathbb{C}} Q_B = 2.$$

Choose an adapted local frame

$$(e_1(\xi), e_2(\xi))$$

of $Q_B(\xi)$, where e_1 is the defect-selected tangent direction and e_2 is the orthogonal spectator direction.

The full microscopic quadratic response restricted to Q_B is

$$R_\xi^{\text{act}} = \Pi_Q \mathcal{R}_\xi^{\text{act}} \Pi_Q \in \text{End}(Q_B). \quad (439)$$

In the adapted frame,

$$R_\xi^{\text{act}} = \begin{pmatrix} r_{11}(\xi) & r_{12}(\xi) \\ r_{12}(\xi) & r_{22}(\xi) \end{pmatrix}, \quad r_{11}, r_{22} \in \mathbb{R}. \quad (440)$$

Its traceless part is

$$S_\xi^{\text{act}} = R_\xi^{\text{act}} - \frac{1}{2}(R_\xi^{\text{act}})\mathbf{1}_2 = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3, \quad (441)$$

with

$$s_3(\xi) = \frac{1}{2}(r_{11}(\xi) - r_{22}(\xi)), \quad s_1(\xi) = \Re r_{12}(\xi), \quad s_2(\xi) = -\Im r_{12}(\xi). \quad (442)$$

F. Longitudinal coefficient-point split

At the graded reduced level there are two longitudinal possibilities:

a. Branch A. The primitive longitudinal singlet survives directly:

$$\Lambda_{\parallel}^{(0)} \neq 0.$$

b. Branch B. The primitive singlet vanishes,

$$\Lambda_{\parallel}^{(0)} = 0,$$

but the concrete Class II longitudinal channel survives through

$$\lambda_c \neq 0.$$

Therefore the exact longitudinal decision tree is

$$\boxed{(1) \Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0, \quad (2) \text{ if not, } \lambda_c \stackrel{?}{\neq} 0.} \quad (443)$$

G. Practical sufficient criterion for longitudinal survival

The sharpest practical sufficient condition for a nonzero longitudinal singlet component is diagonal anisotropy of the actual Q_B -response:

$$\boxed{s_3(\xi) = \frac{1}{2}(\langle e_1, R_\xi^{\text{act}} e_1 \rangle - \langle e_2, R_\xi^{\text{act}} e_2 \rangle).} \quad (444)$$

Hence a sufficient microscopic test is:

$$\boxed{r_{11}(\xi) - r_{22}(\xi) \neq 0 \implies s_3(\xi) \neq 0 \implies J_{\parallel}^{(3)} \neq 0.} \quad (445)$$

This is the exact replacement of the old vague question

“does the complete microscopic linearization produce a longitudinal singlet?”

H. Finite overlap formulas for the S2 coefficients

Let $f_{\pm}(\xi)$ denote the two graded low-mode profiles. Then the actual local overlap formulas for the S2 coefficients are

$$\lambda_{\parallel} = \int d\xi f_{+}(\xi) f_{-}(\xi) \theta'(\xi) h_{\parallel}(\theta(\xi)), \quad (446)$$

$$\alpha = \frac{1}{2} \int d\xi f_{+} f_{-} \theta'(\xi) (h_{11}(\theta(\xi)) + h_{22}(\theta(\xi))), \quad (447)$$

$$\beta = \frac{1}{2} \int d\xi f_{+} f_{-} \theta'(\xi) (h_{21}(\theta(\xi)) - h_{12}(\theta(\xi))). \quad (448)$$

Thus the problem is reduced to deriving the coefficient functions

$$h_{\parallel}(\theta), \quad h_{11}(\theta), \quad h_{12}(\theta), \quad h_{21}(\theta), \quad h_{22}(\theta)$$

from the complete background-dependent TECT Hessian.

I. Dictionary to the concrete Class II branch

Inside the exact concrete Class II transport ansatz one has

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} g_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} g_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} g_O \theta_0 \kappa. \quad (449)$$

Therefore

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0) \quad (450)$$

is exactly the same finite microscopic test as

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0). \quad (451)$$

Moreover, inside the minimal shell-adapted Class II seed family,

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

So explicit witness seeds already exist at the reduced level; what remains open is whether the full complete TECT action realizes them in the actual defect background.

J. Coefficient-point audit protocol

The full microscopic audit can now be written as the following finite protocol.

a. Step 1. Grading audit. Construct the isolated four-dimensional infrared sector $K = K_{+} \oplus K_{-}$ and verify

$$J D_1 J^{-1} = -D_1.$$

b. Step 2. Hessian-to-response audit. Compute the actual Q_B -response

$$R_{\xi}^{\text{act}} = \Pi_Q \mathcal{R}_{\xi}^{\text{act}} \Pi_Q$$

and extract

$$s_1(\xi), \quad s_2(\xi), \quad s_3(\xi).$$

c. *Step 3. Longitudinal branch audit.* Check:

$$\Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0.$$

If not, compute the concrete Class II longitudinal coefficient

$$\lambda_c.$$

d. *Step 4. Transverse audit.* Compute the transverse coefficients

$$\Lambda_E, \quad \Lambda_O$$

or equivalently

$$g_E, \quad g_O.$$

e. *Step 5. Final finite test.* The graded principal operator is Dirac if either

$$\boxed{\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0),} \quad (452)$$

or

$$\boxed{\text{Branch B: } \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (453)$$

K. What is already closed

The following are already theorem-level closed:

1. the plain scalar Bragg route is insufficient;
2. canonical Class II cannot cancel the scalar identity channel;
3. the graded $2 + 2$ bypass route is rigorous;
4. the actual local CP^2 projector-defect geometry belongs naturally to this graded operator class;
5. the natural local branch is S2;
6. the reduced nondegeneracy problem is exactly the finite coefficient-point audit above.

L. What remains honestly open

What remains genuinely open is only:

1. derive $V^\mu(\xi)$ from the *full* microscopic TECT Hessian;
2. prove directly that the actual leading projected perturbation is grading-odd;
3. compute the actual values of

$$\Lambda_{\parallel}^{(0)}, \quad \lambda_c, \quad \Lambda_E, \quad \Lambda_O;$$

4. verify microscopically that the final finite test (452) or (453) holds.

CLXXX. FINAL SUMMARY

The correct non-repetitive mainline is the graded $2 + 2$ bypass route. In this route the principal infrared operator is

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(0) = 0,$$

so the scalar Bragg mass obstruction is absent from the relevant principal symbol.

The actual microscopic frontier is therefore reduced to a finite coefficient-point audit:

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O).$$

The practical extraction proceeds through the actual Q_B -response

$$R_{\xi}^{\text{act}} = \Pi_Q \mathcal{R}_{\xi}^{\text{act}} \Pi_Q,$$

its traceless coefficients

$$s_1(\xi), \ s_2(\xi), \ s_3(\xi),$$

and the overlap formulas giving

$$\lambda_{\parallel}, \ \alpha, \ \beta.$$

The final Dirac test is:

$$\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0, \ (\Lambda_E, \Lambda_O) \neq (0, 0),$$

or

$$\text{Branch B: } \lambda_c \neq 0, \ (\Lambda_E, \Lambda_O) \neq (0, 0).$$

Equivalently, in the concrete Class II normalization,

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

Thus the remaining TECT microscopic frontier is no longer conceptual. It is the explicit extraction of the actual coefficient point from the complete background-dependent Hessian.

CLXXXI. GRADED $2 + 2$ MAINLINE: EXACT CURRENT STAGE

The scalar Bragg-mass obstruction is no longer the main microscopic target. The correct mainline is the graded $2 + 2$ bypass route, in which

$$D_1(p) = \begin{pmatrix} 0 & A_1(p) \\ A_1(p)^\dagger & 0 \end{pmatrix}, \quad A_1(0) = 0.$$

Hence the dangerous scalar identity-channel block is absent from the principal infrared operator.

The actual remaining microscopic problem is the extraction of the finite coefficient point

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O)$$

from the complete background-dependent TECT Hessian.

The practical extraction is based on the actual quadratic Q_B -response

$$R_{\xi}^{\text{act}} = \Pi_Q \mathcal{R}_{\xi}^{\text{act}} \Pi_Q,$$

its traceless coefficients

$$s_1(\xi), \ s_2(\xi), \ s_3(\xi),$$

and the overlap formulas

$$\lambda_{\parallel}, \alpha, \beta.$$

The final finite Dirac test is:

$$\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0, (\Lambda_E, \Lambda_O) \neq (0, 0),$$

or

$$\text{Branch B: } \lambda_c \neq 0, (\Lambda_E, \Lambda_O) \neq (0, 0).$$

Equivalently, in the concrete Class II normalization,

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0).$$

Thus the remaining frontier is purely an explicit coefficient-extraction problem from the full Hessian.

CLXXXII. CANONICAL TENSOR DECOMPOSITION OF THE ACTUAL Q_B -RESPONSE

A. Short Korean explanation

The remaining microscopic task in the graded $2 + 2$ mainline is not to re-prove the reduced Dirac theorem. It is to extract the actual coefficient point

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O)$$

from the complete background-dependent Hessian.

The correct route is:

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] \longrightarrow R_{\xi}^{\text{act}} \longrightarrow (s_1(\xi), s_2(\xi), s_3(\xi)) \longrightarrow V^{\mu}(\xi) \longrightarrow (M^{\mu})_{ab} \longrightarrow (\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O).$$

B. Step 0: operator protocol already fixed

The actual extraction protocol is already fixed at the theorem level:

$$\bar{\Phi}_{\min} \rightarrow \Gamma^{(2)}[\bar{\Phi}_{\min}] \rightarrow H_B(\xi) - iV^{\mu}(\xi)\partial_{y^{\mu}} + W^{\mu\nu}(\xi)\partial_{y^{\mu}}\partial_{y^{\nu}} + \dots$$

and then

$$(M^{\mu})_{ab} = \langle u_a^+, V^{\mu} u_b^- \rangle,$$

followed by Pauli decomposition and S2 matching. So the only missing ingredient is the explicit decomposition of the actual Q_B -response entering $V^{\mu}(\xi)$.

C. Step 1: adapted Q_B -frame and actual Hermitian response

Let

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^{\perp}, \quad \dim_{\mathbb{C}} Q_B = 2,$$

and choose an adapted local orthonormal frame

$$(e_1(\xi), e_2(\xi)).$$

Here e_1 is the defect-selected tangent direction, and e_2 is the orthogonal spectator direction.

Let the actual Hermitian quadratic Q_B -response be

$$R_{\xi}^{\text{act}} = \Pi_Q \mathcal{R}_{\xi}^{\text{act}} \Pi_Q \in \text{End}(Q_B). \quad (454)$$

In the adapted frame it has the unique 2×2 Hermitian form

$$R_{\xi}^{\text{act}} = \begin{pmatrix} r_{11}(\xi) & r_{12}(\xi) \\ r_{12}(\xi) & r_{22}(\xi) \end{pmatrix}, \quad r_{11}, r_{22} \in \mathbb{R}. \quad (455)$$

D. Step 2: canonical Pauli decomposition

Every Hermitian 2×2 matrix admits the unique decomposition

$$R_\xi^{\text{act}} = r_0(\xi)\mathbf{1}_2 + s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3, \quad (456)$$

where

$$r_0(\xi) = \frac{1}{2}(r_{11}(\xi) + r_{22}(\xi)), \quad (457)$$

$$s_3(\xi) = \frac{1}{2}(r_{11}(\xi) - r_{22}(\xi)), \quad (458)$$

$$s_1(\xi) = \Re r_{12}(\xi), \quad s_2(\xi) = -\Im r_{12}(\xi). \quad (459)$$

Equivalently,

$$S_\xi^{\text{act}} := R_\xi^{\text{act}} - \frac{1}{2}(R_\xi^{\text{act}})\mathbf{1}_2 = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3. \quad (460)$$

Thus the actual Q_B -response separates canonically into:

1. scalar/isotropic part $r_0(\xi)\mathbf{1}_2$,
2. transverse traceless vector part (s_1, s_2, s_3) .

E. Step 3: sharp interpretation of the three traceless components

The three traceless coefficients have distinct meanings.

a. (i) Longitudinal singlet indicator. The coefficient

$$s_3(\xi) = \frac{1}{2}(\langle e_1, R_\xi^{\text{act}} e_1 \rangle - \langle e_2, R_\xi^{\text{act}} e_2 \rangle) \quad (461)$$

measures diagonal anisotropy between the defect-selected tangent channel e_1 and the spectator channel e_2 .

Therefore the practical sufficient test is:

$$\exists \xi_0 : \langle e_1(\xi_0), R_{\xi_0}^{\text{act}} e_1(\xi_0) \rangle \neq \langle e_2(\xi_0), R_{\xi_0}^{\text{act}} e_2(\xi_0) \rangle \implies s_3(\xi) \not\equiv 0. \quad (462)$$

b. (ii) Even transverse mixing. The coefficient $s_1(\xi)$ is the real off-diagonal mixing in the adapted frame and naturally feeds the even transverse channel.

c. (iii) Odd transverse mixing. The coefficient $s_2(\xi)$ is the imaginary antisymmetric off-diagonal mixing and naturally feeds the odd transverse channel.

So, at the level of the actual Hermitian Q_B -response,

$$s_3 \leftrightarrow \text{longitudinal singlet}, \quad s_1 \leftrightarrow \text{even transverse}, \quad s_2 \leftrightarrow \text{odd transverse}.$$

F. Step 4: primitive longitudinal branch

If the direct local longitudinal invariant $\Lambda_{\parallel}^{(0)}$ vanishes, the established primitive branch uses

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B], \quad S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}. \quad (463)$$

The primitive longitudinal transport seed is

$$V_{(\parallel)}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu. \quad (464)$$

Inside the explicit seed representative,

$$S_\xi = \frac{1}{2}\theta'(\xi)^2\sigma_3, \quad J_\parallel = \int d\xi f_+ f_- \theta'(\xi) S_\xi = \frac{1}{2}I_{\theta'^3}, \quad (465)$$

so

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2}I_{\theta'^3}.} \quad (466)$$

Thus the primitive longitudinal audit is exactly:

$$\Lambda_\parallel^{(0)} \stackrel{?}{\neq} 0, \quad \text{if not, } \lambda_c \stackrel{?}{\neq} 0.$$

G. Step 5: canonical tensor decomposition of the actual transport kernel

The correct next move is to expand the actual first-order transport kernel in the adapted worldvolume basis

$$(n^\mu, E^\mu{}_1, E^\mu{}_2, J^\mu{}_1, J^\mu{}_2),$$

and the Pauli basis $(\mathbf{1}_2, \sigma_1, \sigma_2, \sigma_3)$.

The most useful canonical decomposition is

$$\begin{aligned} V^\mu(\xi) = & n^\mu \left[v_0^\parallel(\xi) \mathbf{1}_2 + v_3^\parallel(\xi) \sigma_3 \right] \\ & + E^\mu{}_1 \left[v_{E1}(\xi) \sigma_1 + v_{E2}(\xi) \sigma_2 \right] + E^\mu{}_2 \left[-v_{E2}(\xi) \sigma_1 + v_{E1}(\xi) \sigma_2 \right] \\ & + J^\mu{}_1 \left[v_{O1}(\xi) \sigma_1 + v_{O2}(\xi) \sigma_2 \right] + J^\mu{}_2 \left[-v_{O2}(\xi) \sigma_1 + v_{O1}(\xi) \sigma_2 \right] + \cdots \end{aligned} \quad (467)$$

Here:

1. $v_0^\parallel$ is grading-even / irrelevant for the principal Dirac symbol;
2. $v_3^\parallel$ is the longitudinal singlet coefficient density;
3. (v_{E1}, v_{E2}) form the even transverse doublet density;
4. (v_{O1}, v_{O2}) form the odd transverse doublet density.

In the reduced S2 branch this collapses to the three effective coefficient densities

$$h_\parallel(\xi), \quad h_E(\xi), \quad h_O(\xi),$$

with

$$V_{S2}^\mu(\xi) = h_\parallel(\xi) n^\mu \sigma_3 + h_E(\xi) (E^\mu{}_1 \sigma_1 + E^\mu{}_2 \sigma_2) + h_O(\xi) (-E^\mu{}_2 \sigma_1 + E^\mu{}_1 \sigma_2). \quad (468)$$

H. Step 6: actual coefficient extraction formulas

Let the graded low modes be

$$u_a^+(\xi) = f_+(\xi) e_a(\xi), \quad u_a^-(\xi) = f_-(\xi) e_a(\xi), \quad a = 1, 2.$$

Then the projected mixed symbol is

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle. \quad (469)$$

Matching to the S2 form gives the exact overlap formulas

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) h_{\parallel}(\xi), \quad (470)$$

$$\alpha = \int d\xi f_+(\xi) f_-(\xi) h_E(\xi), \quad (471)$$

$$\beta = \int d\xi f_+(\xi) f_-(\xi) h_O(\xi). \quad (472)$$

Equivalently, in the notation already fixed in the notes,

$$\boxed{\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.} \quad (473)$$

So the full coefficient-point audit is reduced to identifying $h_{\parallel}, h_E, h_O$, or equivalently

$$\Lambda_{\parallel}^{(0)}, \quad \lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

I. Step 7: exact finite decision tree

The final finite test is now completely sharp.

a. Branch A. If

$$\Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0), \quad (474)$$

then the graded principal operator is locally Dirac.

b. Branch B. If

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0), \quad (475)$$

then again the graded principal operator is locally Dirac.

Inside the exact Class II-compatible local normalization, this is equivalently

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (476)$$

CLXXXIII. FINAL SUMMARY

The actual Q_B -response admits the canonical adapted-frame decomposition

$$R_{\xi}^{\text{act}} = r_0(\xi) \mathbf{1}_2 + s_1(\xi) \sigma_1 + s_2(\xi) \sigma_2 + s_3(\xi) \sigma_3,$$

with

$$r_0 = \frac{1}{2}(r_{11} + r_{22}), \quad s_3 = \frac{1}{2}(r_{11} - r_{22}), \quad s_1 = \Re r_{12}, \quad s_2 = -\Im r_{12}.$$

Thus:

$$s_3 \leftrightarrow \text{longitudinal singlet}, \quad s_1 \leftrightarrow \text{even transverse}, \quad s_2 \leftrightarrow \text{odd transverse}.$$

The actual first-order transport kernel must then be decomposed in the worldvolume/Pauli basis, and the projected mixed symbol

$$(M^{\mu})_{ab} = \langle u_a^+, V^{\mu} u_b^- \rangle$$

yields the coefficient point

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O).$$

The final finite microscopic test is:

$$\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0, (\Lambda_E, \Lambda_O) \neq (0, 0),$$

or

$$\text{Branch B: } \Lambda_{\parallel}^{(0)} = 0, \lambda_c \neq 0, (\Lambda_E, \Lambda_O) \neq (0, 0).$$

Therefore the remaining frontier is exactly the explicit extraction of

$$R_{\xi}^{\text{act}}, \quad V^{\mu}(\xi), \quad \Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O$$

from the complete background-dependent TECT Hessian.

CLXXXIV. ACTUAL Q_B -RESPONSE DECOMPOSITION: EXACT CURRENT STAGE

The actual Hermitian quadratic Q_B -response is

$$R_{\xi}^{\text{act}} = \Pi_Q \mathcal{R}_{\xi}^{\text{act}} \Pi_Q = \begin{pmatrix} r_{11}(\xi) & r_{12}(\xi) \\ r_{12}(\xi) & r_{22}(\xi) \end{pmatrix}.$$

Its canonical Pauli decomposition is

$$R_{\xi}^{\text{act}} = r_0(\xi) \mathbf{1}_2 + s_1(\xi) \sigma_1 + s_2(\xi) \sigma_2 + s_3(\xi) \sigma_3,$$

with

$$r_0 = \frac{1}{2}(r_{11} + r_{22}), \quad s_3 = \frac{1}{2}(r_{11} - r_{22}), \quad s_1 = \Re r_{12}, \quad s_2 = -\Im r_{12}.$$

This yields the physical identification

$$s_3 \leftrightarrow \text{longitudinal singlet}, \quad s_1 \leftrightarrow \text{even transverse}, \quad s_2 \leftrightarrow \text{odd transverse}.$$

The actual transport kernel must then be decomposed in the worldvolume/Pauli basis, and the projected mixed symbol

$$(M^{\mu})_{ab} = \langle u_a^+, V^{\mu} u_b^- \rangle$$

determines the coefficient point

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O).$$

The final finite decision tree is:

$$\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0, (\Lambda_E, \Lambda_O) \neq (0, 0),$$

or

$$\text{Branch B: } \Lambda_{\parallel}^{(0)} = 0, \lambda_c \neq 0, (\Lambda_E, \Lambda_O) \neq (0, 0).$$

Thus the remaining frontier is exactly the explicit extraction of the actual response tensor and transport kernel from the complete background-dependent TECT Hessian.

CLXXXV. SOURCE-BY-SOURCE CLASSIFICATION OF THE ACTUAL HESSIAN TRANSPORT KERNEL

A. Short Korean explanation

The remaining task is to classify the full Hessian sources according to which S2 tensor structure they can generate in the graded $2 + 2$ route.

The only relevant leading first-order structures are

$$n^\mu \sigma_3, \quad E^\mu{}_a \sigma_a, \quad J^\mu{}_a \sigma_a, \quad a = 1, 2.$$

Therefore the full source audit is equivalent to the following question:

Which Hessian source produces $n^\mu \sigma_3$, $E^\mu{}_a \sigma_a$, $J^\mu{}_a \sigma_a$?

B. Canonical S2 target decomposition

The graded principal transport kernel is expanded as

$$V^\mu(\xi) = V^\mu_{(\parallel)}(\xi) + V^\mu_{(\perp, E)}(\xi) + V^\mu_{(\perp, O)}(\xi) + V^\mu_{\text{irr}}(\xi), \quad (477)$$

where

$$V^\mu_{(\parallel)}(\xi) = h_\parallel(\xi) n^\mu \sigma_3, \quad (478)$$

$$V^\mu_{(\perp, E)}(\xi) = h_E(\xi) E^\mu{}_a \sigma_a, \quad (479)$$

$$V^\mu_{(\perp, O)}(\xi) = h_O(\xi) J^\mu{}_a \sigma_a, \quad (480)$$

and V^μ_{irr} denotes pieces that do not survive the S2 shell-compatible projection.

The corresponding reduced coefficients are

$$\lambda_\parallel = \int d\xi f_+(\xi) f_-(\xi) h_\parallel(\xi), \quad (481)$$

$$\alpha = \int d\xi f_+(\xi) f_-(\xi) h_E(\xi), \quad (482)$$

$$\beta = \int d\xi f_+(\xi) f_-(\xi) h_O(\xi). \quad (483)$$

Equivalently,

$$\lambda_\parallel = I_\theta \Lambda_\parallel^{(0)} + \frac{\lambda_c}{2} I_{\theta^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (484)$$

C. Class I: scalar/isotropic Hessian sources

A scalar/isotropic source has the local form

$$\mathcal{S}^\mu_{\text{iso}}(\xi) = u^\mu(\xi) \mathbf{1}_2. \quad (485)$$

This class includes:

1. amplitude-only Hessian pieces,
2. trace parts of projector response,
3. isotropic stiffness renormalizations.

Such sources do *not* contribute to the S2 witness coefficients:

$$\boxed{\mathcal{S}_{\text{iso}}^\mu \implies h_{\parallel} = h_E = h_O = 0 \quad \text{after traceless S2 projection.}} \quad (486)$$

Reason:

1. $\mathbf{1}_2$ has zero overlap with σ_3 under the traceless longitudinal projection;
2. it cannot generate the transverse Pauli doublet (σ_1, σ_2) .

Therefore scalar/isotropic Hessian pieces are spectators for the principal graded Dirac symbol.

D. Class II: primitive longitudinal singlet source

The canonical longitudinal carrier is generated by the commutator channel

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B]. \quad (487)$$

Direct restriction gives

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0,$$

but the induced quadratic channel is

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \sigma_3). \quad (488)$$

Therefore its traceless part is

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3. \quad (489)$$

By parity, the naive quadratic singlet

$$\theta'(\xi)^2 n^\mu \sigma_3$$

does not survive the mixed even/odd overlap, because

$$f_+ \text{ even}, \quad f_- \text{ odd}, \quad \theta'^2 \text{ even}$$

imply

$$\int d\xi f_+ f_- \theta'^2 = 0.$$

Hence the minimal viable longitudinal source must carry one extra odd factor:

$$\boxed{V_{(\parallel)}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3.} \quad (490)$$

This contributes only to the longitudinal coefficient:

$$\boxed{h_{\parallel}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3, \quad h_E = h_O = 0.} \quad (491)$$

Thus

$$\boxed{\text{primitive commutator-square source} \implies n^\mu \sigma_3 \implies \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.} \quad (492)$$

E. Class III: transverse even doublet source

The minimal even transverse source is

$$V_{(\perp, E)}^\mu(\xi) = \theta'(\xi) \Lambda_E E^\mu{}_a \sigma_a, \quad a = 1, 2. \quad (493)$$

This produces the even transverse coefficient only:

$$h_E(\xi) = \theta'(\xi) \Lambda_E, \quad h_{\parallel} = h_O = 0. \quad (494)$$

After projection,

$$\alpha = I_\theta \Lambda_E, \quad \beta = 0 \quad \text{if only } V_{(\perp, E)}^\mu \text{ is present.} \quad (495)$$

Microscopically, this class must come from a Hessian source that:

1. is reflection-odd overall,
2. transforms as a transverse vector in worldvolume space,
3. transforms as a traceless doublet in Q_B -fiber space.

F. Class IV: transverse odd doublet source

The minimal odd transverse source is

$$V_{(\perp, O)}^\mu(\xi) = \theta'(\xi) \Lambda_O J^\mu{}_a \sigma_a, \quad a = 1, 2. \quad (496)$$

This produces the odd transverse coefficient only:

$$h_O(\xi) = \theta'(\xi) \Lambda_O, \quad h_{\parallel} = h_E = 0. \quad (497)$$

After projection,

$$\beta = I_\theta \Lambda_O, \quad \alpha = 0 \quad \text{if only } V_{(\perp, O)}^\mu \text{ is present.} \quad (498)$$

Microscopically, this class is the parity-rotated partner of the even transverse source and must arise from a Hessian block carrying the same Q_B -doublet content but the rotated worldvolume tensor $J^\mu{}_a = \epsilon_{ab} E^\mu{}_b$.

G. Class V: irreducible or forbidden sources

Any first-order Hessian source not of the three compatible types

$$n^\mu \sigma_3, \quad E^\mu{}_a \sigma_a, \quad J^\mu{}_a \sigma_a$$

is either:

1. projected out by the S2 reduction,
2. forbidden by parity/worldvolume symmetry,
3. or contributes only to higher-order irrelevant operators.

Hence

$$V_{\text{irr}}^\mu \implies \text{no contribution to } (\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O) \text{ at principal order.} \quad (499)$$

H. Canonical source table

The full source audit is therefore summarized by the table

Source class	Canonical form	Coefficient produced	Effect
Scalar/isotropic	$u^\mu \mathbf{1}_2$	none	spectator
Primitive longitudinal	$\frac{\lambda_c}{2} \theta'^3 n^\mu \sigma_3$	λ_c	$\lambda_\parallel$
Transverse even	$\theta' \Lambda_E E^\mu{}_a \sigma_a$	Λ_E	α
Transverse odd	$\theta' \Lambda_O J^\mu{}_a \sigma_a$	Λ_O	β
Irreducible/forbidden	V_{irr}^μ	none	projected out

I. Reconstruction from the actual Q_B -response

Now connect this directly to the actual adapted-frame response

$$R_\xi^{\text{act}} = r_0(\xi) \mathbf{1}_2 + s_1(\xi) \sigma_1 + s_2(\xi) \sigma_2 + s_3(\xi) \sigma_3.$$

Then the reconstruction principle is:

$$\boxed{s_3(\xi) \text{ feeds the longitudinal singlet channel,}} \quad (500)$$

$$\boxed{s_1(\xi) \text{ feeds the even transverse doublet,}} \quad (501)$$

$$\boxed{s_2(\xi) \text{ feeds the odd transverse doublet.}} \quad (502)$$

More precisely, the canonical first-order reconstruction ansatz is

$$\begin{aligned} V^\mu(\xi) = & \theta'(\xi) v_\parallel^{(0)}(\xi) n^\mu \sigma_3 + \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 \\ & + \theta'(\xi) v_E(\xi) E^\mu{}_a \sigma_a + \theta'(\xi) v_O(\xi) J^\mu{}_a \sigma_a + V_{\text{irr}}^\mu(\xi), \end{aligned} \quad (503)$$

with

$$v_\parallel^{(0)}(\xi) \sim s_3(\xi), \quad v_E(\xi) \sim s_1(\xi), \quad v_O(\xi) \sim s_2(\xi),$$

up to model-dependent proportionality factors determined by the full Hessian.

Therefore the coefficient-point extraction reduces to

$$\Lambda_\parallel^{(0)} = \int d\xi f_+ f_- \theta'(\xi) v_\parallel^{(0)}(\xi), \quad (504)$$

$$\Lambda_E = \int d\xi f_+ f_- \theta'(\xi) v_E(\xi), \quad (505)$$

$$\Lambda_O = \int d\xi f_+ f_- \theta'(\xi) v_O(\xi), \quad (506)$$

while Branch B contributes

$$\lambda_\parallel^{(B)} = \frac{\lambda_c}{2} I_{\theta'^3}. \quad (507)$$

J. Exact decision tree after source classification

After the source-by-source classification, the final microscopic decision tree is:

a. Longitudinal audit

$$s_3(\xi) \stackrel{?}{\neq} 0 \implies \Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0.$$

If not,

$$\lambda_c \stackrel{?}{\neq} 0.$$

b. Transverse audit

$$s_1(\xi) \stackrel{?}{\neq} 0 \implies \Lambda_E \stackrel{?}{\neq} 0,$$

$$s_2(\xi) \stackrel{?}{\neq} 0 \implies \Lambda_O \stackrel{?}{\neq} 0.$$

c. Final graded Dirac test

$$\boxed{\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0),} \quad (508)$$

or

$$\boxed{\text{Branch B: } \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (509)$$

CLXXXVI. FINAL SUMMARY

The actual Hessian sources split into five principal classes:

1. scalar/isotropic sources: spectators;
2. primitive commutator-square source: produces only $n^\mu \sigma_3$;
3. transverse even source: produces only $E^\mu{}_a \sigma_a$;
4. transverse odd source: produces only $J^\mu{}_a \sigma_a$;
5. irreducible/forbidden sources: projected out at principal order.

Thus the full source-by-source coefficient extraction is reduced to the canonical map

$$R_\xi^{\text{act}} \rightarrow (s_1, s_2, s_3) \rightarrow (v_{\parallel}^{(0)}, v_E, v_O) \rightarrow (\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O).$$

The final microscopic test is:

$$\text{either } \Lambda_{\parallel}^{(0)} \neq 0 \text{ or } \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

CLXXXVII. SOURCE-BY-SOURCE HESSIAN CLASSIFICATION

The only principal first-order S2-compatible transport structures are

$$n^\mu \sigma_3, \quad E^\mu{}_a \sigma_a, \quad J^\mu{}_a \sigma_a.$$

Accordingly, the Hessian sources split into:

$$\text{scalar/isotropic} \rightarrow \text{spectator},$$

$$\text{primitive commutator-square} \rightarrow n^\mu \sigma_3 \rightarrow \lambda_c,$$

$$\text{transverse even doublet} \rightarrow E^\mu{}_a \sigma_a \rightarrow \Lambda_E,$$

$$\text{transverse odd doublet} \rightarrow J^\mu{}_a \sigma_a \rightarrow \Lambda_O.$$

The actual adapted-frame response

$$R_\xi^{\text{act}} = r_0(\xi) \mathbf{1}_2 + s_1(\xi) \sigma_1 + s_2(\xi) \sigma_2 + s_3(\xi) \sigma_3$$

therefore maps canonically as

$$s_3 \rightarrow \Lambda_\parallel^{(0)} \text{ or } \lambda_c, \quad s_1 \rightarrow \Lambda_E, \quad s_2 \rightarrow \Lambda_O.$$

Hence the remaining finite audit is exactly:

$$\text{either } \Lambda_\parallel^{(0)} \neq 0 \text{ or } \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

CLXXXVIII. BLOCKWISE CONTRIBUTION FORMULAS FOR THE ACTUAL Q_B -RESPONSE

A. Short Korean explanation

We now decompose the full background-dependent Hessian into the minimal source blocks that can contribute to the graded $2 + 2$ principal transport kernel.

The purpose is to derive, source by source, the induced coefficients

$$r_0(\xi), \quad s_1(\xi), \quad s_2(\xi), \quad s_3(\xi)$$

in

$$R_\xi^{\text{act}} = r_0(\xi) \mathbf{1}_2 + s_1(\xi) \sigma_1 + s_2(\xi) \sigma_2 + s_3(\xi) \sigma_3.$$

These coefficients then feed

$$\Lambda_\parallel^{(0)}, \quad \lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

B. Step 1: block decomposition of the full Hessian

Let the complete quadratic operator around the actual defect/background be

$$\Gamma^{(2)}[\bar{\Phi}_{\min}].$$

Decompose it into the minimal source blocks

$$\Gamma^{(2)} = \Gamma_{\text{amp}}^{(2)} + \Gamma_{\text{align,tree}}^{(2)} + \Gamma_{\text{comm}^2}^{(2)} + \Gamma_{\text{med}}^{(2)} + \Gamma_{\text{rem}}^{(2)}, \quad (510)$$

where:

1. $\Gamma_{\text{amp}}^{(2)}$: amplitude / scalar / isotropic shell block,
2. $\Gamma_{\text{align,tree}}^{(2)}$: direct tree-level projector-stiffness block,
3. $\Gamma_{\text{comm}^2}^{(2)}$: primitive commutator-square longitudinal block,
4. $\Gamma_{\text{med}}^{(2)}$: explicit mediated anisotropic/transport block,
5. $\Gamma_{\text{rem}}^{(2)}$: all higher-order, remote, or not-yet-closed pieces.

Projecting to the Q_B -doublet sector gives

$$R_\xi^{\text{act}} = R_\xi^{\text{amp}} + R_\xi^{\text{align,tree}} + R_\xi^{\text{comm}^2} + R_\xi^{\text{med}} + R_\xi^{\text{rem}}, \quad (511)$$

with each term in $\text{Herm}(Q_B)$.

C. Step 2: universal Pauli decomposition for each block

For each block $X \in \{\text{amp}, \text{align}, \text{tree}, \text{comm}^2, \text{med}, \text{rem}\}$, write

$$R_\xi^X = r_0^X(\xi) \mathbf{1}_2 + s_1^X(\xi) \sigma_1 + s_2^X(\xi) \sigma_2 + s_3^X(\xi) \sigma_3. \quad (512)$$

Then

$$r_0(\xi) = \sum_X r_0^X(\xi), \quad s_a(\xi) = \sum_X s_a^X(\xi), \quad a = 1, 2, 3. \quad (513)$$

In the adapted frame (e_1, e_2) ,

$$r_0^X = \frac{1}{2}(r_{11}^X + r_{22}^X), \quad s_3^X = \frac{1}{2}(r_{11}^X - r_{22}^X), \quad s_1^X = \Re r_{12}^X, \quad s_2^X = -\Im r_{12}^X. \quad (514)$$

D. Step 3: amplitude/scalar block

The amplitude/scalar block is the restriction of the scalar shell Hessian to the Q_B -doublet:

$$R_\xi^{\text{amp}} = \Pi_Q \mathcal{R}_\xi^{\text{amp}} \Pi_Q. \quad (515)$$

Because this block is scalar/isotropic in fiber space at principal order, it has the form

$$R_\xi^{\text{amp}} = r_0^{\text{amp}}(\xi) \mathbf{1}_2, \quad s_1^{\text{amp}} = s_2^{\text{amp}} = s_3^{\text{amp}} = 0. \quad (516)$$

Therefore

$$\boxed{R_\xi^{\text{amp}} \implies \text{no direct contribution to } \Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O.} \quad (517)$$

This includes:

1. pure scalar terms,
2. pure amplitude-only terms,
3. scalar drift terms,
4. isotropic shell renormalizations.

E. Step 4: direct projector-stiffness tree block

For the refined locked constant-projector background,

$$P_0 = \text{const}, \quad \partial_i P_0 = 0.$$

Hence the quadratic Hessian is block diagonal between shell and tangent-projector sectors,

$$\delta^2 F_{\text{ref}} = \delta^2 F_\sigma + \delta^2 F_\xi, \quad J = 0.$$

Therefore the direct tree-level projector-stiffness block contributes no shell-projector mixing at quadratic order:

$$R_\xi^{\text{align, tree}} = 0. \quad (518)$$

Equivalently,

$$\boxed{r_0^{\text{align, tree}} = s_1^{\text{align, tree}} = s_2^{\text{align, tree}} = s_3^{\text{align, tree}} = 0.} \quad (519)$$

So the current projector-stiffness sector has no direct principal witness effect at tree level.

F. Step 5: primitive commutator-square block

Define

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B].$$

Direct Q_B -restriction vanishes:

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0.$$

But the induced quadratic channel is

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2} (\mathbf{1}_2 + \sigma_3).$$

Hence the traceless part is

$$R_\xi^{\text{comm}^2, \text{tr}} = \frac{1}{2} \theta'(\xi)^2 \sigma_3. \quad (520)$$

Therefore the primitive commutator-square block contributes

$$r_0^{\text{comm}^2}(\xi) = \frac{1}{2} \theta'(\xi)^2, \quad s_3^{\text{comm}^2}(\xi) = \frac{1}{2} \theta'(\xi)^2, \quad s_1^{\text{comm}^2} = s_2^{\text{comm}^2} = 0. \quad (521)$$

However, parity kills the naive longitudinal first-order overlap:

$$f_+ \text{ even}, \quad f_- \text{ odd}, \quad \theta'^2 \text{ even} \implies \int d\xi f_+ f_- \theta'^2 = 0.$$

Hence this block contributes to the principal transport kernel only after multiplication by one extra odd factor $\theta'(\xi)$:

$$V_{(\parallel)}^\mu(\xi) = \lambda_c \theta'(\xi) R_\xi^{\text{comm}^2, \text{tr}} n^\mu = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (522)$$

Thus

primitive commutator-square block $\implies s_3$ channel only $\implies \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}$.

(523)

G. Step 6: mediated anisotropic/transport block

This is the first source that can generate the full S2 triplet explicitly. Inside the exact concrete quadratic reduction template, one has

$$L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda m^a X_i^a \partial_i \phi_0 + g_X X_i^a C_{ij}^{ab} (D_j \psi) \xi^b + \text{h.c.}$$

Integrating out X_i^a gives

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.$$

At the local projector-defect transport level this matches

$$V_{\text{med}}^\mu(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 + \theta'(\xi) \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right] + \dots \quad (524)$$

Therefore the mediated block contributes

$$s_3^{\text{med}}(\xi) \sim \frac{\lambda_c}{2} \theta'(\xi)^2, \quad s_1^{\text{med}}(\xi) \sim \Lambda_E, \quad s_2^{\text{med}}(\xi) \sim \Lambda_O, \quad (525)$$

up to the universal overall $\theta'(\xi)$ factor needed in V^μ .

More invariantly, the corresponding principal transport densities are

$$h_{\parallel}^{\text{med}}(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3, \quad h_E^{\text{med}}(\xi) = \theta'(\xi) \Lambda_E, \quad h_O^{\text{med}}(\xi) = \theta'(\xi) \Lambda_O. \quad (526)$$

Hence

$\lambda_{\parallel}^{\text{med}} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha^{\text{med}} = I_{\theta} \Lambda_E, \quad \beta^{\text{med}} = I_{\theta} \Lambda_O.$

(527)

H. Step 7: remote / heavy-mode induced projector-stiffness block

Although the tree-level projector stiffness gives zero direct mixing, heavy-mode elimination can induce an effective correction:

$$\delta L_{\text{eff}} = -\frac{1}{2} J G_\pi J,$$

provided a nonzero microscopic shell-projector mixing source J exists beyond the present tree-level locked background truncation.

Thus the induced projector-stiffness response has the formal structure

$$R_\xi^{\text{stiff,ind}} = \Pi_Q(-J G_\pi J) \Pi_Q. \quad (528)$$

At the current theorem level, its detailed coefficients are not closed, but its possible contributions are:

$$r_0^{\text{stiff,ind}} \text{ allowed, } s_3^{\text{stiff,ind}} \text{ allowed, } s_1^{\text{stiff,ind}}, s_2^{\text{stiff,ind}} \text{ allowed in principle.} \quad (529)$$

So

projector stiffness contributes only indirectly, through heavy-mode elimination.

(530)

I. Step 8: remainder block

All not-yet-closed microscopic sources are collected into

$$R_\xi^{\text{rem}}.$$

By definition,

$$R_\xi^{\text{rem}} = r_0^{\text{rem}} \mathbf{1}_2 + s_1^{\text{rem}} \sigma_1 + s_2^{\text{rem}} \sigma_2 + s_3^{\text{rem}} \sigma_3. \quad (531)$$

This block is not known explicitly, but it can be audited source-by-source once the complete final TECT Hessian is specified.

J. Step 9: total source table

Combining all blocks:

Block	r_0	s_1	s_2	s_3
amp	✓	0	0	0
align, tree	0	0	0	0
comm ²	✓	0	0	✓
med	✓?	✓	✓	✓
stiff, ind	✓	✓?	✓?	✓?
rem	?	?	?	?

At principal order, the currently closed direct witness carriers are:

$s_3 : \text{comm}^2 \text{ or med, } s_1 : \text{med, } s_2 : \text{med.}$

(532)

K. Step 10: coefficient reconstruction

The total transport reconstruction is

$$\begin{aligned} V^\mu(\xi) = & \theta'(\xi) v_{\parallel}^{(0)}(\xi) n^\mu \sigma_3 + \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 \\ & + \theta'(\xi) v_E(\xi) E^\mu{}_a \sigma_a + \theta'(\xi) v_O(\xi) J^\mu{}_a \sigma_a + V_{\text{irr}}^\mu(\xi), \end{aligned} \quad (533)$$

with

$$v_{\parallel}^{(0)} \sim s_3^{\text{lead}}, \quad v_E \sim s_1^{\text{lead}}, \quad v_O \sim s_2^{\text{lead}}.$$

Therefore

$$\Lambda_{\parallel}^{(0)} = \int d\xi f_+ f_- \theta'(\xi) v_{\parallel}^{(0)}(\xi), \quad (534)$$

$$\Lambda_E = \int d\xi f_+ f_- \theta'(\xi) v_E(\xi), \quad \Lambda_O = \int d\xi f_+ f_- \theta'(\xi) v_O(\xi), \quad (535)$$

and

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (536)$$

L. Step 11: actual decision tree after blockwise audit

The final microscopic decision tree is now:

a. Longitudinal

$$s_3^{\text{lead}} \stackrel{?}{\neq} 0 \implies \Lambda_{\parallel}^{(0)} \stackrel{?}{\neq} 0.$$

If not,

$$\lambda_c \stackrel{?}{\neq} 0$$

through the primitive commutator-square channel.

b. Transverse

$$s_1^{\text{lead}} \stackrel{?}{\neq} 0 \implies \Lambda_E \stackrel{?}{\neq} 0,$$

$$s_2^{\text{lead}} \stackrel{?}{\neq} 0 \implies \Lambda_O \stackrel{?}{\neq} 0.$$

c. Final finite test

$$\boxed{\text{Branch A: } \Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0),} \quad (537)$$

or

$$\boxed{\text{Branch B: } \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (538)$$

CLXXXIX. FINAL SUMMARY

The full Hessian source audit is now reduced to a blockwise Q_B -response decomposition:

$$\Gamma^{(2)} = \Gamma_{\text{amp}}^{(2)} + \Gamma_{\text{align, tree}}^{(2)} + \Gamma_{\text{comm}^2}^{(2)} + \Gamma_{\text{med}}^{(2)} + \Gamma_{\text{rem}}^{(2)}.$$

After projection to the Q_B -doublet, each block contributes to

$$R_{\xi}^{\text{act}} = r_0 \mathbf{1}_2 + s_1 \sigma_1 + s_2 \sigma_2 + s_3 \sigma_3.$$

The currently closed direct principal contributions are:

amplitude/scalar $\rightarrow r_0$ only,

direct projector-stiffness tree level $\rightarrow 0$,

primitive commutator-square $\rightarrow s_3$ only,

mediated anisotropic/transport block $\rightarrow s_3, s_1, s_2$.

Thus the actual finite coefficient-point extraction is

$$R_\xi^{\text{act}} \rightarrow (s_3, s_1, s_2) \rightarrow (\Lambda_\parallel^{(0)}, \lambda_c, \Lambda_E, \Lambda_O),$$

with

$$\lambda_\parallel = I_\theta \Lambda_\parallel^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.$$

The remaining microscopic frontier is therefore the explicit evaluation of the mediated and remote/heavy-mode-induced blocks in the complete final TECT Hessian.

CXC. EXPLICIT ADAPTED-FRAME EVALUATION OF THE MEDIATED BLOCK

A. Short Korean explanation

The mediated block is the only currently explicit microscopic source that can generate all three principal witness channels

$$s_3, \quad s_1, \quad s_2.$$

So the next exact task is to write the induced Q_B -response

$$R_\xi^{\text{med}}$$

in the adapted frame (e_1, e_2) , and then extract

$$r_{11}^{\text{med}}, \quad r_{22}^{\text{med}}, \quad r_{12}^{\text{med}}.$$

Once that is done, the coefficient point follows immediately.

B. Step 1: mediated quadratic completion

Let the microscopic mediator field be X_i^a , with quadratic block

$$L_X^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + X_i^a \mathcal{J}_i^a[\delta\Phi] \quad (539)$$

where $\delta\Phi$ denotes the shell/projector fluctuation multiplet, and

$$\mathcal{J}_i^a[\delta\Phi] = \mathcal{J}_{i,\text{bg}}^a + \mathcal{J}_{i,\text{lin}}^a[\delta\Phi] + \dots$$

Integrating out X_i^a at quadratic order gives

$$\delta L_{\text{eff}}^{\text{med}} = -\frac{1}{2M_X^2} \mathcal{J}_i^a[\delta\Phi] \mathcal{J}_i^a[\delta\Phi]. \quad (540)$$

The quadratic response relevant for the Q_B -sector is therefore

$$\mathcal{R}_\xi^{\text{med}} = -\frac{1}{M_X^2} \left(\mathcal{J}_{i,\text{lin}}^a \right)^\dagger \mathcal{J}_{i,\text{lin}}^a. \quad (541)$$

Restricting to Q_B ,

$$R_\xi^{\text{med}} = \Pi_Q \mathcal{R}_\xi^{\text{med}} \Pi_Q = -\frac{1}{M_X^2} \Pi_Q \left(\mathcal{J}_{i,\text{lin}}^a \right)^\dagger \mathcal{J}_{i,\text{lin}}^a \Pi_Q. \quad (542)$$

C. Step 2: adapted-frame matrix elements

Choose the adapted orthonormal basis

$$(e_1(\xi), e_2(\xi))$$

of $Q_B(\xi)$. Define the mediator current matrix elements

$$\mathcal{C}_{ia;m}(\xi) := \langle e_m(\xi), \mathcal{J}_{i,\text{lin}}^a \rangle, \quad m = 1, 2. \quad (543)$$

Equivalently, for each ξ , $\mathcal{C}(\xi)$ is a collection of two-component amplitudes labeled by (i, a) .

Then the Q_B -response matrix is

$$r_{mn}^{\text{med}}(\xi) = -\frac{1}{M_X^2} \sum_{i,a} \overline{\mathcal{C}_{ia;m}(\xi)} \mathcal{C}_{ia;n}(\xi), \quad m, n = 1, 2. \quad (544)$$

So explicitly,

$$r_{11}^{\text{med}}(\xi) = -\frac{1}{M_X^2} \sum_{i,a} |\mathcal{C}_{ia;1}(\xi)|^2, \quad (545)$$

$$r_{22}^{\text{med}}(\xi) = -\frac{1}{M_X^2} \sum_{i,a} |\mathcal{C}_{ia;2}(\xi)|^2, \quad (546)$$

$$r_{12}^{\text{med}}(\xi) = -\frac{1}{M_X^2} \sum_{i,a} \overline{\mathcal{C}_{ia;1}(\xi)} \mathcal{C}_{ia;2}(\xi). \quad (547)$$

This is the exact adapted-frame evaluation formula.

D. Step 3: Pauli coefficients of the mediated response

From the universal Hermitian decomposition,

$$R_\xi^{\text{med}} = r_0^{\text{med}}(\xi) \mathbf{1}_2 + s_1^{\text{med}}(\xi) \sigma_1 + s_2^{\text{med}}(\xi) \sigma_2 + s_3^{\text{med}}(\xi) \sigma_3,$$

we get

$$r_0^{\text{med}}(\xi) = -\frac{1}{2M_X^2} \sum_{i,a} (|\mathcal{C}_{ia;1}|^2 + |\mathcal{C}_{ia;2}|^2), \quad (548)$$

$$s_3^{\text{med}}(\xi) = -\frac{1}{2M_X^2} \sum_{i,a} (|\mathcal{C}_{ia;1}|^2 - |\mathcal{C}_{ia;2}|^2), \quad (549)$$

$$s_1^{\text{med}}(\xi) = -\frac{1}{M_X^2} \sum_{i,a} \Re(\overline{\mathcal{C}_{ia;1}} \mathcal{C}_{ia;2}), \quad (550)$$

$$s_2^{\text{med}}(\xi) = \frac{1}{M_X^2} \sum_{i,a} \Im(\overline{\mathcal{C}_{ia;1}} \mathcal{C}_{ia;2}). \quad (551)$$

These are the exact source-by-source formulas we needed.

E. Step 4: interpretation of the three witness channels

The mediated block therefore generates:

a. Longitudinal singlet

$$s_3^{\text{med}}(\xi) \neq 0 \iff \sum_{i,a} |\mathcal{C}_{ia;1}|^2 \neq \sum_{i,a} |\mathcal{C}_{ia;2}|^2. \quad (552)$$

So the longitudinal singlet appears when the mediator couples with unequal strength to the tangent channel e_1 and the spectator channel e_2 .

b. Even transverse doublet

$$s_1^{\text{med}}(\xi) \neq 0 \iff \sum_{i,a} \Re(\overline{\mathcal{C}_{ia;1}} \mathcal{C}_{ia;2}) \neq 0. \quad (553)$$

c. Odd transverse doublet

$$s_2^{\text{med}}(\xi) \neq 0 \iff \sum_{i,a} \Im(\overline{\mathcal{C}_{ia;1}} \mathcal{C}_{ia;2}) \neq 0. \quad (554)$$

Thus:

$$\text{norm imbalance} \longrightarrow s_3, \quad \text{real coherence} \longrightarrow s_1, \quad \text{imaginary coherence} \longrightarrow s_2.$$

F. Step 5: decomposition into worldvolume tensors

Now separate the (i, a) -dependence into the shell/worldvolume basis. Write

$$\mathcal{C}_{ia;m}(\xi) = n_i u_{a;m}^{(\parallel)}(\xi) + E_{ia} u_m^{(E)}(\xi) + J_{ia} u_m^{(O)}(\xi) + \mathcal{C}_{ia;m}^{(\text{irr})}(\xi), \quad (555)$$

where:

1. n_i is the longitudinal unit direction,
2. E_{ia} is the even transverse tensor,
3. J_{ia} is the odd transverse tensor,
4. $\mathcal{C}^{(\text{irr})}$ is projected out at S2 principal order.

Then the mediated response splits accordingly:

$$R_\xi^{\text{med}} = R_\xi^{(\parallel)} + R_\xi^{(E)} + R_\xi^{(O)} + R_\xi^{(\text{mix})} + R_\xi^{(\text{irr})}. \quad (556)$$

At principal order, the S2-compatible sector is obtained by keeping only the diagonal channel-matched pieces:

$$R_\xi^{(\parallel)} \rightsquigarrow s_3^{(\parallel)}(\xi) \sigma_3, \quad (557)$$

$$R_\xi^{(E)} \rightsquigarrow s_1^{(E)}(\xi) \sigma_1, \quad (558)$$

$$R_\xi^{(O)} \rightsquigarrow s_2^{(O)}(\xi) \sigma_2. \quad (559)$$

G. Step 6: minimal transport ansatz induced by the mediated response

The canonical first-order transport reconstruction is therefore

$$\begin{aligned} V_{\text{med}}^\mu(\xi) &= \theta'(\xi) v_{\parallel}^{(0),\text{med}}(\xi) n^\mu \sigma_3 + \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3 \\ &\quad + \theta'(\xi) v_E^{\text{med}}(\xi) E^\mu{}_a \sigma_a + \theta'(\xi) v_O^{\text{med}}(\xi) J^\mu{}_a \sigma_a + V_{\text{irr}}^\mu(\xi), \end{aligned} \quad (560)$$

with

$$v_{\parallel}^{(0),\text{med}}(\xi) \sim s_3^{\text{med}}(\xi), \quad v_E^{\text{med}}(\xi) \sim s_1^{\text{med}}(\xi), \quad v_O^{\text{med}}(\xi) \sim s_2^{\text{med}}(\xi). \quad (561)$$

Hence the coefficient-point contribution of the mediated block is

$$\Lambda_{\parallel}^{(0),\text{med}} = \int d\xi f_+ f_- \theta'(\xi) v_{\parallel}^{(0),\text{med}}(\xi), \quad (562)$$

$$\Lambda_E^{\text{med}} = \int d\xi f_+ f_- \theta'(\xi) v_E^{\text{med}}(\xi), \quad (563)$$

$$\Lambda_O^{\text{med}} = \int d\xi f_+ f_- \theta'(\xi) v_O^{\text{med}}(\xi). \quad (564)$$

So the full coefficient dictionary becomes

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0),\text{med}} + \frac{\lambda_c}{2} I_{\theta'^3} + \dots, \quad (565)$$

$$\alpha = I_{\theta} \Lambda_E^{\text{med}} + \dots, \quad \beta = I_{\theta} \Lambda_O^{\text{med}} + \dots. \quad (566)$$

H. Step 7: sufficient nonvanishing criteria

The mediated block passes the finite audit if the following hold.

a. Longitudinal

$$\exists \xi_0 : \sum_{i,a} |\mathcal{C}_{ia;1}(\xi_0)|^2 \neq \sum_{i,a} |\mathcal{C}_{ia;2}(\xi_0)|^2 \implies s_3^{\text{med}} \neq 0. \quad (567)$$

b. Even transverse

$$\exists \xi_0 : \sum_{i,a} \Re \left(\overline{\mathcal{C}_{ia;1}(\xi_0)} \mathcal{C}_{ia;2}(\xi_0) \right) \neq 0 \implies s_1^{\text{med}} \neq 0. \quad (568)$$

c. Odd transverse

$$\exists \xi_0 : \sum_{i,a} \Im \left(\overline{\mathcal{C}_{ia;1}(\xi_0)} \mathcal{C}_{ia;2}(\xi_0) \right) \neq 0 \implies s_2^{\text{med}} \neq 0. \quad (569)$$

These are exact finite checks.

I. Step 8: final mediated-block decision tree

The mediated block alone is sufficient for local Dirac emergence if either

$$\Lambda_{\parallel}^{(0),\text{med}} \neq 0, \quad (\Lambda_E^{\text{med}}, \Lambda_O^{\text{med}}) \neq (0, 0), \quad (570)$$

or

$$\lambda_c \neq 0, \quad (\Lambda_E^{\text{med}}, \Lambda_O^{\text{med}}) \neq (0, 0). \quad (571)$$

In the exact concrete Class II normalization this is the same as

$$g_c \neq 0, \quad (g_E, g_O) \neq (0, 0). \quad (572)$$

CXCI. FINAL SUMMARY

The mediated block in the adapted Q_B -frame is determined by the current amplitudes

$$\mathcal{C}_{ia;m}(\xi) = \langle e_m(\xi), \mathcal{J}_{i,\text{lin}}^a \rangle.$$

Its exact Hermitian response matrix is

$$r_{mn}^{\text{med}}(\xi) = -\frac{1}{M_X^2} \sum_{i,a} \overline{\mathcal{C}_{ia;m}(\xi)} \mathcal{C}_{ia;n}(\xi).$$

Therefore

$$\begin{aligned} r_{11}^{\text{med}} &= -\frac{1}{M_X^2} \sum_{i,a} |\mathcal{C}_{ia;1}|^2, & r_{22}^{\text{med}} &= -\frac{1}{M_X^2} \sum_{i,a} |\mathcal{C}_{ia;2}|^2, \\ r_{12}^{\text{med}} &= -\frac{1}{M_X^2} \sum_{i,a} \overline{\mathcal{C}_{ia;1}} \mathcal{C}_{ia;2}. \end{aligned}$$

Hence the Pauli coefficients are

$$\begin{aligned} s_3^{\text{med}} &= -\frac{1}{2M_X^2} \sum_{i,a} (|\mathcal{C}_{ia;1}|^2 - |\mathcal{C}_{ia;2}|^2), \\ s_1^{\text{med}} &= -\frac{1}{M_X^2} \sum_{i,a} \Re(\overline{\mathcal{C}_{ia;1}} \mathcal{C}_{ia;2}), & s_2^{\text{med}} &= \frac{1}{M_X^2} \sum_{i,a} \Im(\overline{\mathcal{C}_{ia;1}} \mathcal{C}_{ia;2}). \end{aligned}$$

Thus the mediated block is the explicit microscopic carrier of:

$$\text{norm imbalance} \rightarrow s_3 \rightarrow \Lambda_{\parallel}^{(0)} \text{ or } \lambda_c,$$

$$\text{real coherence} \rightarrow s_1 \rightarrow \Lambda_E,$$

$$\text{imaginary coherence} \rightarrow s_2 \rightarrow \Lambda_O.$$

So the remaining frontier is the explicit evaluation of the current amplitudes $\mathcal{C}_{ia;m}(\xi)$ from the complete final TECT mediator/current sector.

CXCII. EXPLICIT TENSOR-CONTRACTION FORM OF THE LINEARIZED MEDIATOR CURRENT

A. Short Korean explanation

The only remaining concrete step is to write the linearized mediator current

$$\mathcal{J}_{i,\text{lin}}^a$$

in a basis adapted to:

1. the shell direction $\hat{n}_i$,
2. the transverse shell basis (e_{1j}, e_{2j}) ,
3. the defect-frame basis $(e_1(\xi), e_2(\xi)) \subset Q_B(\xi)$.

Once this is done, the adapted-frame amplitudes

$$\mathcal{C}_{ia;m}(\xi) = \langle e_m(\xi), \mathcal{J}_{i,\text{lin}}^a(\xi) \rangle$$

are explicit, and then

$$r_{11}^{\text{med}}, \quad r_{22}^{\text{med}}, \quad r_{12}^{\text{med}}$$

follow immediately.

B. Step 1: linearized Class II current split

For the Class II UV structure

$$L_{\text{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a(\Psi) + \beta X_i^a K_i^a(P, \delta P),$$

the linearized mediator current is the sum of two pieces:

$$\mathcal{J}_{i,\text{lin}}^a = \alpha J_{i,\text{lin}}^a + \beta K_{i,\text{lin}}^a. \quad (573)$$

We now parametrize them in the minimal shell-defect carrier form.

C. Step 2: shell-gradient factorization

The shell current must carry one spatial index and one internal index, so the minimal first-order shell-gradient factorization is

$$J_{i,\text{lin}}^a(\xi) = m^a(\xi) C_{ij}^{ab}(\xi) \partial_j \psi_b(\xi), \quad (574)$$

where:

- $m^a(\xi)$ is the defect/background-dependent internal selector,
- $C_{ij}^{ab}(\xi)$ is the shell-adapted carrier tensor,
- $\psi_b(\xi)$ is the reduced Q_B -doublet fluctuation in the adapted frame.

In the explicit minimal Class II seed basis,

$$m^a(P_0) = m \delta^{a3}, \quad (575)$$

and

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad (576)$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad (577)$$

$$C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}. \quad (578)$$

Thus only the $a = 3$ mediator direction is populated in the minimal longitudinal selector, while the $b = 1, 2, 3$ labels resolve the reduced shell channels.

D. Step 3: adapted doublet expansion of the fluctuation

Expand the reduced projector/shell fluctuation in the defect frame:

$$\psi(\xi) = \psi_1(\xi) e_1(\xi) + \psi_2(\xi) e_2(\xi). \quad (579)$$

Then

$$\partial_j \psi = (\partial_j \psi_1) e_1 + (\partial_j \psi_2) e_2 + \psi_1 \partial_j e_1 + \psi_2 \partial_j e_2. \quad (580)$$

At principal local transport order, we keep only the direct amplitude-gradient part and absorb frame-rotation terms into higher-order or connection-like remainders. Thus

$$\partial_j \psi \approx (\partial_j \psi_1) e_1 + (\partial_j \psi_2) e_2. \quad (581)$$

E. Step 4: explicit current amplitudes in the seed basis

Using (574)–(581), the adapted current amplitudes are

$$\mathcal{C}_{ia;m} = \langle e_m, \mathcal{J}_{i,\text{lin}}^a \rangle = \alpha m^a C_{ij}^{ab} \partial_j \psi_b^{(m)} + \beta \mathcal{K}_{ia;m}, \quad (582)$$

where $\mathcal{K}_{ia;m} := \langle e_m, K_{i,\text{lin}}^a \rangle$ is the projector/tangent source amplitude.

Inside the minimal seed family, the shell-current part becomes:

a. Longitudinal channel. For $b = 3$,

$$\mathcal{C}_{i3;m}^{(\parallel)} = \alpha m c_{\parallel} \hat{n}_i \hat{n}_j \partial_j \psi_m. \quad (583)$$

b. Even/odd transverse channels. For $b = 1, 2$,

$$\mathcal{C}_{i1;m}^{(\perp)} = \alpha m (c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}) \partial_j \psi_m, \quad (584)$$

$$\mathcal{C}_{i2;m}^{(\perp)} = \alpha m (-c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}) \partial_j \psi_m. \quad (585)$$

Thus the mediator current amplitudes are explicitly axial in i and decompose into the three shell channels ($\parallel, E, O$).

F. Step 5: contraction with the shell frame

Introduce the directional derivatives

$$\partial_{\parallel} := \hat{n}_j \partial_j, \quad \partial_1 := e_{1j} \partial_j, \quad \partial_2 := e_{2j} \partial_j.$$

Then (583)–(585) simplify to

$$\mathcal{C}_{i3;m}^{(\parallel)} = \alpha m c_{\parallel} \hat{n}_i \partial_{\parallel} \psi_m, \quad (586)$$

$$\mathcal{C}_{i1;m}^{(\perp)} = \alpha m \hat{n}_i (c_I \partial_1 + c_J \partial_2) \psi_m, \quad (587)$$

$$\mathcal{C}_{i2;m}^{(\perp)} = \alpha m \hat{n}_i (-c_J \partial_1 + c_I \partial_2) \psi_m. \quad (588)$$

This is the exact tensor-contraction form of the current amplitudes at principal order.

G. Step 6: induced adapted-frame response

Substituting (586)–(588) into

$$r_{mn}^{\text{med}} = -\frac{1}{M_X^2} \sum_{i,a} \overline{\mathcal{C}_{ia;m}} \mathcal{C}_{ia;n},$$

we get the shell-current contribution

$$\begin{aligned} r_{mn}^{\text{med,shell}} = & -\frac{\alpha^2 m^2}{M_X^2} \left[c_{\parallel}^2 \overline{\partial_{\parallel} \psi_m} \partial_{\parallel} \psi_n \right. \\ & + (c_I^2 + c_J^2) \left(\overline{\partial_1 \psi_m} \partial_1 \psi_n + \overline{\partial_2 \psi_m} \partial_2 \psi_n \right) \\ & \left. + 2c_I c_J \Re \left(\overline{\partial_1 \psi_m} \partial_2 \psi_n \right)_{\text{sym}} + 2c_I c_J \Im \left(\overline{\partial_1 \psi_m} \partial_2 \psi_n \right)_{\text{asym}} \right] + \dots \end{aligned} \quad (589)$$

More transparently, define

$$Q_{\parallel;mn} := \overline{\partial_{\parallel} \psi_m} \partial_{\parallel} \psi_n, \quad (590)$$

$$Q_{E;mn} := \overline{\partial_1 \psi_m} \partial_1 \psi_n + \overline{\partial_2 \psi_m} \partial_2 \psi_n, \quad (591)$$

$$Q_{O;mn} := \overline{\partial_1 \psi_m} \partial_2 \psi_n - \overline{\partial_2 \psi_m} \partial_1 \psi_n. \quad (592)$$

Then

$$r_{mn}^{\text{med,shell}} = -\frac{\alpha^2 m^2}{M_X^2} \left[c_{\parallel}^2 Q_{\parallel;mn} + (c_I^2 + c_J^2) Q_{E;mn} + c_I c_J Q_{O;mn} \right] + \dots \quad (593)$$

H. Step 7: reading the three witness channels

From

$$s_3^{\text{med}} = \frac{1}{2}(r_{11}^{\text{med}} - r_{22}^{\text{med}}), \quad s_1^{\text{med}} = \Re r_{12}^{\text{med}}, \quad s_2^{\text{med}} = -\Im r_{12}^{\text{med}},$$

we obtain:

a. *Longitudinal singlet.*

$$s_3^{\text{med}} = -\frac{\alpha^2 m^2}{2M_X^2} \left[c_{\parallel}^2 (Q_{\parallel;11} - Q_{\parallel;22}) + (c_I^2 + c_J^2)(Q_{E;11} - Q_{E;22}) + c_I c_J (Q_{O;11} - Q_{O;22}) \right] + \dots \quad (594)$$

b. *Even transverse.*

$$s_1^{\text{med}} = -\frac{\alpha^2 m^2}{M_X^2} \Re \left[c_{\parallel}^2 Q_{\parallel;12} + (c_I^2 + c_J^2) Q_{E;12} + c_I c_J Q_{O;12} \right] + \dots \quad (595)$$

c. *Odd transverse.*

$$s_2^{\text{med}} = \frac{\alpha^2 m^2}{M_X^2} \Im \left[c_{\parallel}^2 Q_{\parallel;12} + (c_I^2 + c_J^2) Q_{E;12} + c_I c_J Q_{O;12} \right] + \dots \quad (596)$$

So the full microscopic meaning is now explicit:

- $c_{\parallel}$ controls the longitudinal axial carrier,
- $c_I^2 + c_J^2$ controls the magnitude of the transverse doublet sector,
- $c_I c_J$ controls the mixed even/odd interference structure.

I. Step 8: minimal seed limits

The three pure seed limits are immediate.

a. *Pure longitudinal seed*

$$c_{\parallel} \neq 0, \quad c_I = c_J = 0$$

gives

$$s_3^{\text{med}} \sim -\frac{\alpha^2 m^2}{2M_X^2} c_{\parallel}^2 (Q_{\parallel;11} - Q_{\parallel;22}), \quad s_1^{\text{med}} \sim -\frac{\alpha^2 m^2}{M_X^2} \Re(c_{\parallel}^2 Q_{\parallel;12}), \quad s_2^{\text{med}} \sim \frac{\alpha^2 m^2}{M_X^2} \Im(c_{\parallel}^2 Q_{\parallel;12}). \quad (597)$$

b. *Pure even transverse seed*

$$c_{\parallel} = 0, \quad c_I \neq 0, \quad c_J = 0$$

gives

$$r_{mn}^{\text{med,shell}} = -\frac{\alpha^2 m^2}{M_X^2} c_I^2 Q_{E;mn}, \quad (598)$$

hence s_1, s_2 arise from the coherence structure of $Q_{E;12}$.

c. Pure odd transverse seed

$$c_{\parallel} = 0, \quad c_I = 0, \quad c_J \neq 0$$

gives again

$$r_{mn}^{\text{med,shell}} = -\frac{\alpha^2 m^2}{M_X^2} c_J^2 Q_{E;mn}, \quad (599)$$

so the difference between g_E and g_O is not the norm sector alone, but the shell-frame orientation built into the basis choice (C^{31}, C^{32}) .

J. Step 9: coefficient-point extraction

Therefore the induced principal transport densities are, schematically,

$$v_{\parallel}^{(0),\text{med}}(\xi) \sim s_3^{\text{med}}(\xi), \quad v_E^{\text{med}}(\xi) \sim s_1^{\text{med}}(\xi), \quad v_O^{\text{med}}(\xi) \sim s_2^{\text{med}}(\xi), \quad (600)$$

and hence

$$\Lambda_{\parallel}^{(0),\text{med}} = \int d\xi f_+ f_- \theta'(\xi) v_{\parallel}^{(0),\text{med}}(\xi), \quad (601)$$

$$\Lambda_E^{\text{med}} = \int d\xi f_+ f_- \theta'(\xi) v_E^{\text{med}}(\xi), \quad \Lambda_O^{\text{med}} = \int d\xi f_+ f_- \theta'(\xi) v_O^{\text{med}}(\xi). \quad (602)$$

In the exact reduced Class II dictionary this is summarized as

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J,$$

which is exactly the already-established witness map.

CXCIII. FINAL SUMMARY

The linearized mediator current in the minimal Class II seed basis is explicitly

$$\mathcal{C}_{i3;m} = \alpha m c_{\parallel} \hat{n}_i \partial_{\parallel} \psi_m,$$

$$\mathcal{C}_{i1;m} = \alpha m \hat{n}_i (c_I \partial_1 + c_J \partial_2) \psi_m,$$

$$\mathcal{C}_{i2;m} = \alpha m \hat{n}_i (-c_J \partial_1 + c_I \partial_2) \psi_m.$$

Therefore the mediated Q_B -response is

$$r_{mn}^{\text{med}} = -\frac{1}{M_X^2} \sum_{i,a} \overline{\mathcal{C}_{ia;m}} \mathcal{C}_{ia;n},$$

which yields explicit formulas for

$$s_3^{\text{med}}, \quad s_1^{\text{med}}, \quad s_2^{\text{med}}.$$

At the coefficient level:

$c_{\parallel}$ controls the longitudinal carrier,

(c_I, c_J) control the transverse doublet carrier,

$c_I c_J$ controls even/odd interference.

Thus the remaining microscopic work is no longer structural. It is the explicit evaluation of the local derivative bilinears

$$Q_{\parallel;mn}, \quad Q_{E;mn}, \quad Q_{O;mn}$$

in the actual defect background, after which

$$\Lambda_{\parallel}^{(0),\text{med}}, \Lambda_E^{\text{med}}, \Lambda_O^{\text{med}}$$

follow immediately.

CXCIV. EXPLICIT CURRENT-AMPLITUDE FORM OF THE MEDIATED BLOCK

In the minimal Class II seed basis,

$$m^a(P_0) = m\delta^{a3}, \quad C_{ij}^{33} = c_{\parallel}\hat{n}_i\hat{n}_j,$$

$$C_{ij}^{31} = c_I\hat{n}_ie_{1j} + c_J\hat{n}_ie_{2j}, \quad C_{ij}^{32} = -c_J\hat{n}_ie_{1j} + c_I\hat{n}_ie_{2j}.$$

Therefore the adapted current amplitudes are

$$\mathcal{C}_{i3;m} = \alpha m c_{\parallel} \hat{n}_i \partial_{\parallel} \psi_m,$$

$$\mathcal{C}_{i1;m} = \alpha m \hat{n}_i (c_I \partial_1 + c_J \partial_2) \psi_m,$$

$$\mathcal{C}_{i2;m} = \alpha m \hat{n}_i (-c_J \partial_1 + c_I \partial_2) \psi_m.$$

The mediated response matrix is then

$$r_{mn}^{\text{med}} = -\frac{1}{M_X^2} \sum_{i,a} \overline{\mathcal{C}_{ia;m}} \mathcal{C}_{ia;n},$$

so the Pauli coefficients $s_3^{\text{med}}, s_1^{\text{med}}, s_2^{\text{med}}$ are explicit functions of the local derivative bilinears

$$Q_{\parallel;mn}, \quad Q_{E;mn}, \quad Q_{O;mn}.$$

Thus the remaining microscopic task is exactly the evaluation of these bilinears in the actual defect background.

CXCV. PRINCIPAL FIRST-ORDER MEDIATED SYMBOL FROM THE BACKGROUND-LINEAR CROSS TERM

A. Short Korean explanation

For the actual graded $2 + 2$ coefficient extraction, the relevant mediated contribution is not the quadratic norm piece

$$J_{\text{lin}} J_{\text{lin}},$$

but the background-linear cross term

$$J_{\text{bg}} J_{\text{lin}}.$$

This is the unique part that is:

1. first order in the fluctuation,
2. first order in the transport derivative,
3. linear in the seed coefficients $c_{\parallel}, c_I, c_J$,
4. consistent with the already-established dictionary

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J.$$

B. Step 1: mediator elimination and transport-relevant cross term

Let the mediator sector be

$$L_X = \frac{M_X^2}{2} X_i^a X_i^a + X_i^a \mathcal{J}_i^a[\Phi]. \quad (603)$$

Expand the current around the defect/background:

$$\mathcal{J}_i^a[\Phi_0 + \delta\Phi] = \mathcal{J}_{i,\text{bg}}^a + \mathcal{J}_{i,\text{lin}}^a[\delta\Phi] + O(\delta\Phi^2). \quad (604)$$

Integrating out X_i^a gives

$$\delta L_{\text{eff}}^{\text{med}} = -\frac{1}{2M_X^2} (\mathcal{J}_{i,\text{bg}}^a + \mathcal{J}_{i,\text{lin}}^a) (\mathcal{J}_{i,\text{bg}}^a + \mathcal{J}_{i,\text{lin}}^a). \quad (605)$$

The principal first-order transport piece is the cross term

$$\boxed{\delta L_{\text{eff}}^{(1)} = -\frac{1}{M_X^2} \mathcal{J}_{i,\text{bg}}^a \mathcal{J}_{i,\text{lin}}^a.} \quad (606)$$

This is the correct source of the coefficient point

$$(\Lambda_{\parallel}^{(0)}, \Lambda_E, \Lambda_O).$$

C. Step 2: minimal shell-defect current ansatz

We now take the minimal shell-defect carrier form

$$\mathcal{J}_{i,\text{bg}}^a(\xi) = \mu_X \theta'(\xi) \hat{n}_i m^a, \quad (607)$$

where m^a is the background internal selector.

For the linearized current, use the shell-gradient factorization

$$\mathcal{J}_{i,\text{lin}}^a = \nu_X m^b C_{ij}^{ba} \partial_j \psi + \mathcal{J}_{i,\text{lin}}^{a,\text{rem}}, \quad (608)$$

with the explicit minimal Class II seed basis

$$m^a = m \delta^{a3}, \quad (609)$$

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad (610)$$

$$C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad (611)$$

$$C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}. \quad (612)$$

Thus the effective transport prefactor is

$$\Gamma_X := \frac{\mu_X \nu_X m^2}{M_X^2}. \quad (613)$$

D. Step 3: adapted defect mode ansatz

Take the graded low modes in the standard factorized form

$$u_m^\pm(\xi, y) = f_\pm(\xi) e_m(\xi) \chi^\pm(y), \quad m = 1, 2, \quad (614)$$

where y^μ are slow worldvolume coordinates and ξ is the normal/defect coordinate.

At principal order, the frame $e_m(\xi)$ depends only on ξ , so the worldvolume derivatives act only on $\chi^\pm(y)$:

$$\partial_\parallel u_m^\pm = f_\pm(\xi) e_m(\xi) \partial_\parallel \chi^\pm(y) + O(\partial_\xi), \quad (615)$$

$$\partial_1 u_m^\pm = f_\pm(\xi) e_m(\xi) \partial_1 \chi^\pm(y), \quad \partial_2 u_m^\pm = f_\pm(\xi) e_m(\xi) \partial_2 \chi^\pm(y). \quad (616)$$

For a local plane-wave symbol extraction, write

$$\chi^\pm(y) \sim e^{ip \cdot y},$$

so that

$$\partial_\parallel \chi^\pm = ip_\parallel \chi^\pm, \quad \partial_1 \chi^\pm = ip_1 \chi^\pm, \quad \partial_2 \chi^\pm = ip_2 \chi^\pm.$$

E. Step 4: projected mixed matrix elements

The transport matrix elements are obtained from the mixed graded projection

$$(M^\mu)_{mn} = \langle u_m^+, V_{\text{med}}^\mu u_n^- \rangle. \quad (617)$$

Using (606)–(616), the principal mediated symbol is linear in p and factorizes as

$$(M^\mu)_{mn} = I_\theta \left[\Gamma_X c_\parallel n^\mu p_\parallel (\sigma_3)_{mn} + \Gamma_X c_I E^\mu{}_a p_a (\sigma_a)_{mn} + \Gamma_X c_J J^\mu{}_a p_a (\sigma_a)_{mn} \right] + O(p^2), \quad (618)$$

where the universal defect overlap is

$$I_\theta := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi). \quad (619)$$

This is the exact first-order mediated symbol in the minimal seed basis.

F. Step 5: direct reading of the coefficient point

Comparing (618) with the canonical S2 transport form

$$V_{\text{S2}}^\mu = h_\parallel(\xi) n^\mu \sigma_3 + h_E(\xi) E^\mu{}_a \sigma_a + h_O(\xi) J^\mu{}_a \sigma_a, \quad (620)$$

we read off immediately

$$\boxed{\Lambda_\parallel^{(0), \text{med}} = \Gamma_X c_\parallel, \quad \Lambda_E^{\text{med}} = \Gamma_X c_I, \quad \Lambda_O^{\text{med}} = \Gamma_X c_J.} \quad (621)$$

Therefore the mediated contribution to the reduced coefficients is

$$\boxed{\lambda_\parallel^{(A), \text{med}} = I_\theta \Gamma_X c_\parallel, \quad \alpha^{\text{med}} = I_\theta \Gamma_X c_I, \quad \beta^{\text{med}} = I_\theta \Gamma_X c_J.} \quad (622)$$

This is exactly the desired linear seed dictionary.

G. Step 6: Branch B longitudinal term

Independently of the direct mediated longitudinal branch above, the primitive commutator-square source gives

$$V_{(\parallel),B}^\mu = \frac{\lambda_c}{2} \theta'(\xi)^3 n^\mu \sigma_3, \quad (623)$$

hence

$$\lambda_{\parallel}^{(B)} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad I_{\theta'^3} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3. \quad (624)$$

Combining Branch A and Branch B,

$$\lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Gamma_X c_I, \quad \beta = I_{\theta} \Gamma_X c_J. \quad (625)$$

H. Step 7: exact Weyl/Dirac finite test

The principal graded Dirac condition is therefore

$$I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3} \neq 0, \quad (c_I, c_J) \neq (0, 0). \quad (626)$$

More explicitly:

a. Branch A

$$\Gamma_X c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0). \quad (627)$$

b. Branch B

$$\Gamma_X c_{\parallel} = 0, \quad \lambda_c \neq 0, \quad (c_I, c_J) \neq (0, 0). \quad (628)$$

I. Step 8: matching to the established reduced dictionary

Define

$$g_c := \Gamma_X c_{\parallel}, \quad g_E := \Gamma_X c_I, \quad g_O := \Gamma_X c_J. \quad (629)$$

Then (625) becomes

$$\lambda_{\parallel} = I_{\theta} g_c + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} g_E, \quad \beta = I_{\theta} g_O. \quad (630)$$

So the exact current-level computation reproduces the reduced witness map

$$g_c \propto c_{\parallel}, \quad g_E \propto c_I, \quad g_O \propto c_J,$$

now with the explicit proportionality constant

$$\Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2}.$$

CXCVI. FINAL SUMMARY

The principal first-order mediated transport symbol comes from the background-linear cross term

$$-\frac{1}{M_X^2} \mathcal{J}_{i,\text{bg}}^a \mathcal{J}_{i,\text{lin}}^a,$$

not from the quadratic norm piece $J_{\text{lin}} J_{\text{lin}}$.

With the minimal seed basis

$$C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j},$$

and the factorized low-mode ansatz

$$u_m^{\pm} = f_{\pm}(\xi) e_m(\xi) \chi^{\pm}(y),$$

the mediated coefficient point is read off exactly as

$$\Lambda_{\parallel}^{(0),\text{med}} = \Gamma_X c_{\parallel}, \quad \Lambda_E^{\text{med}} = \Gamma_X c_I, \quad \Lambda_O^{\text{med}} = \Gamma_X c_J,$$

with

$$\Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2}.$$

Hence the full reduced coefficients are

$$\lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Gamma_X c_I, \quad \beta = I_{\theta} \Gamma_X c_J.$$

Therefore the remaining microscopic task is no longer structural. It is only the explicit evaluation of the background prefactors μ_X, ν_X, m and the overlap integrals $I_{\theta}, I_{\theta'^3}$ for the actual defect profile.

CXCVII. PRINCIPAL MEDIATED COEFFICIENT EXTRACTION

The principal first-order mediated symbol is generated by the background-linear cross term

$$-\frac{1}{M_X^2} \mathcal{J}_{i,\text{bg}}^a \mathcal{J}_{i,\text{lin}}^a.$$

Using the minimal Class II seed basis, one obtains

$$\Lambda_{\parallel}^{(0),\text{med}} = \Gamma_X c_{\parallel}, \quad \Lambda_E^{\text{med}} = \Gamma_X c_I, \quad \Lambda_O^{\text{med}} = \Gamma_X c_J,$$

with

$$\Gamma_X = \frac{\mu_X \nu_X m^2}{M_X^2}.$$

Hence the reduced coefficients are

$$\lambda_{\parallel} = I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Gamma_X c_I, \quad \beta = I_{\theta} \Gamma_X c_J.$$

So the final finite test is

$$I_{\theta} \Gamma_X c_{\parallel} + \frac{\lambda_c}{2} I_{\theta'^3} \neq 0, \quad (c_I, c_J) \neq (0, 0).$$

The remaining work is the explicit evaluation of μ_X, ν_X, m and the overlap integrals $I_{\theta}, I_{\theta'^3}$ in the actual defect background.

CXCVIII. EXPLICIT OVERLAP COMPLETION OF THE GRADED 2 + 2 COEFFICIENT POINT

A. Short Korean explanation

At this stage, the overlap integrals are already explicitly known. Therefore the reduced S2 coefficients can now be written in fully explicit closed form.

This finishes the overlap-completion step of the graded 2 + 2 mainline.

B. Step 1: explicit overlap values

For the explicit defect and low-mode profiles already used in the concrete seed/Class II branch,

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \neq 0, \quad (631)$$

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \neq 0, \quad (632)$$

for nontrivial defect amplitude

$$\theta_0 \kappa \neq 0.$$

C. Step 2: exact Class II coefficient dictionary

In the exact concrete Class II-compatible local branch, the reduced coefficients are

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (633)$$

Substituting (631)–(632) gives

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad (634)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa. \quad (635)$$

Thus the full coefficient point is now explicit.

D. Step 3: reduced principal symbol

The reduced S2 symbol is therefore

$$A_1(p) = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3 p_{\parallel} \sigma_3 - \frac{\sqrt{3}\pi}{16} \theta_0 \kappa [\Lambda_E (p_1 \sigma_1 + p_2 \sigma_2) + \Lambda_O (-p_2 \sigma_1 + p_1 \sigma_2)]. \quad (636)$$

Equivalently, the coefficient matrix is

$$\Lambda_{\text{eff}} = \begin{pmatrix} -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_E & +\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_O & 0 \\ -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_O & -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_E & 0 \\ 0 & 0 & -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3 \end{pmatrix}. \quad (637)$$

E. Step 4: determinant and exact nondegeneracy test

The determinant is

$$\det \Lambda_{\text{eff}} = \lambda_{\parallel} (\alpha^2 + \beta^2). \quad (638)$$

Substituting (634)–(635),

$$\begin{aligned} \det \Lambda_{\text{eff}} &= \left(-\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3 \right) \left[\left(\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \right)^2 (\Lambda_E^2 + \Lambda_O^2) \right] \\ &= -\frac{9\sqrt{3}\pi^3}{131072} \lambda_c \theta_0^5 \kappa^5 (\Lambda_E^2 + \Lambda_O^2). \end{aligned} \quad (639)$$

Therefore the exact finite nondegeneracy condition is

$$\boxed{\theta_0 \kappa \neq 0, \quad \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (640)$$

This is precisely the already-established branch-B criterion, now with all overlap constants inserted explicitly.

F. Step 5: explicit transverse magnitude and anisotropy ratio

Define the transverse magnitude

$$\Lambda_{\perp} := \sqrt{\Lambda_E^2 + \Lambda_O^2}. \quad (641)$$

Then

$$v_{\perp} = \sqrt{\alpha^2 + \beta^2} = \frac{\sqrt{3}\pi}{16} |\theta_0 \kappa| \Lambda_{\perp}, \quad (642)$$

and

$$v_{\parallel} = |\lambda_{\parallel}| = \frac{3\sqrt{3}\pi}{512} |\lambda_c| |\theta_0|^3 |\kappa|^3. \quad (643)$$

Hence the explicit anisotropy ratio is

$$\boxed{\rho_{\text{eff}} := \frac{v_{\parallel}}{v_{\perp}} = \frac{3}{32} \frac{|\lambda_c|}{\Lambda_{\perp}} \theta_0^2 \kappa^2.} \quad (644)$$

So the graded Dirac cone is automatically anisotropic unless the microscopic coefficients are tuned so that

$$\frac{|\lambda_c|}{\Lambda_{\perp}} \sim \frac{32}{3} \frac{1}{\theta_0^2 \kappa^2}.$$

G. Step 6: branch-sensitive interpretation

We now separate the two logically possible longitudinal branches.

a. Branch A. If the actual full microscopic transport extraction yields a constant-leading longitudinal invariant

$$\Lambda_{\parallel}^{(0)} \neq 0,$$

then one has instead

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} = -\frac{\sqrt{3}\pi}{16} \Lambda_{\parallel}^{(0)} \theta_0 \kappa + O(\theta_0^3 \kappa^3). \quad (645)$$

In that case the leading anisotropy is

$$\rho_A = \frac{|\Lambda_{\parallel}^{(0)}|}{\sqrt{\Lambda_E^2 + \Lambda_O^2}} + O(\theta_0^2 \kappa^2). \quad (646)$$

b. Branch B. If

$$\Lambda_{\parallel}^{(0)} = 0,$$

then the first nonzero longitudinal term is the primitive channel:

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\lambda_c\theta_0^3\kappa^3,$$

and the anisotropy is given by (644).

Thus the full microscopic branch-selection problem is now sharply visible:

$$\boxed{\text{Branch A gives } \lambda_{\parallel} \sim O(\theta_0\kappa), \quad \text{Branch B gives } \lambda_{\parallel} \sim O(\theta_0^3\kappa^3).} \quad (647)$$

H. Step 7: exact current status

At the overlap-completed level, the following are now fully explicit:

1. the defect overlap constants I_{θ} and $I_{\theta'^3}$;
2. the reduced coefficient point $(\lambda_{\parallel}, \alpha, \beta)$;
3. the reduced determinant $\det \Lambda_{\text{eff}}$;
4. the exact finite nondegeneracy criterion;
5. the branch-sensitive scaling of the longitudinal coefficient.

What remains open is not the reduced symbol. It is only the actual microscopic extraction of:

$$\Lambda_E, \quad \Lambda_O, \quad \lambda_c,$$

and, at a deeper level, whether the complete microscopic theory realizes Branch A or Branch B.

CXCIX. FINAL SUMMARY

Using the explicitly evaluated defect overlaps

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3,$$

the exact concrete Class II reduced coefficients become

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\lambda_c\theta_0^3\kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16}\Lambda_E\theta_0\kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16}\Lambda_O\theta_0\kappa.$$

Therefore

$$\det \Lambda_{\text{eff}} = -\frac{9\sqrt{3}\pi^3}{131072}\lambda_c\theta_0^5\kappa^5(\Lambda_E^2 + \Lambda_O^2),$$

so the exact graded nondegeneracy criterion is

$$\theta_0\kappa \neq 0, \quad \lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

The effective anisotropy ratio on Branch B is

$$\rho_{\text{eff}} = \frac{3}{32} \frac{|\lambda_c|}{\sqrt{\Lambda_E^2 + \Lambda_O^2}} \theta_0^2 \kappa^2.$$

Thus the reduced symbol is now completely explicit. The only remaining microscopic frontier is the actual extraction of $\lambda_c, \Lambda_E, \Lambda_O$ and the branch selection of the full TECT transport kernel.

CC. ACTUAL-LOCAL INVARIANT EXPANSION OF THE COMPLETE TECT TRANSPORT KERNEL

A. Short Korean explanation

The reduced symbol is already closed at the structural level. What remains is to express the complete local transport kernel in the defect background in terms of the minimal symmetry-allowed local invariants.

The goal is to derive canonical local expansions for

$$h_{\parallel}(\xi), \quad h_E(\xi), \quad h_O(\xi),$$

and then read off the leading coefficient point

$$\Lambda_{\parallel}^{(0)}, \quad \lambda_c, \quad \Lambda_E^{(0)}, \quad \Lambda_O^{(0)}.$$

B. Step 1: canonical S2 transport form

At principal first order, the complete local transport kernel must have the S2-compatible form

$$V^{\mu}(\xi) = h_{\parallel}(\xi) n^{\mu} \sigma_3 + h_E(\xi) E^{\mu}_a \sigma_a + h_O(\xi) J^{\mu}_a \sigma_a + V_{\text{irr}}^{\mu}(\xi), \quad (648)$$

where $a = 1, 2$, and V_{irr}^{μ} is projected out at principal S2 order.

The reduced coefficients are therefore

$$\lambda_{\parallel} = \int d\xi f_+(\xi) f_-(\xi) h_{\parallel}(\xi), \quad (649)$$

$$\alpha = \int d\xi f_+(\xi) f_-(\xi) h_E(\xi), \quad (650)$$

$$\beta = \int d\xi f_+(\xi) f_-(\xi) h_O(\xi). \quad (651)$$

C. Step 2: local invariant algebra

Let the defect profile be described by $\theta(\xi)$. The complete local transport density can depend only on scalar local invariants built from the defect background.

At the lowest derivative orders, the parity-even scalar invariants are generated by

$$X_0(\xi) := \theta(\xi)^2, \quad X_1(\xi) := \theta'(\xi)^2, \quad X_2(\xi) := \theta(\xi)\theta''(\xi), \quad X_3(\xi) := \theta''(\xi)^2, \quad \dots \quad (652)$$

More generally, write the complete parity-even local invariant list as

$$\mathcal{X}(\xi) = (X_0(\xi), X_1(\xi), X_2(\xi), \dots). \quad (653)$$

The key symmetry fact is:

All scalar coefficient functions multiplying principal first-order transport tensors must be functions of $\mathcal{X}(\xi)$,

(654)

because $\mathcal{X}$ is parity-even.

D. Step 3: parity constraint on principal transport

The graded mixed overlap involves

$$f_+(\xi)f_-(\xi),$$

with the standard parity assignment

$$f_+(\xi) \text{ even}, \quad f_-(\xi) \text{ odd}.$$

Hence

$$f_+(\xi)f_-(\xi)$$

is odd.

Therefore only the odd part of each transport density contributes to (649)–(651).

Since every coefficient function of $\mathcal{X}$ is parity-even, the principal transport densities must contain one explicit odd factor. The lowest such factor is $\theta'(\xi)$.

Thus the most general principal local expansions are

$$h_{\parallel}(\xi) = \theta'(\xi) \mathcal{L}_1(\mathcal{X}(\xi)) + \theta'(\xi)^3 \mathcal{L}_3(\mathcal{X}(\xi)) + O_{\text{odd}, \geq 5}, \quad (655)$$

$$h_E(\xi) = \theta'(\xi) \mathcal{E}_1(\mathcal{X}(\xi)) + \theta'(\xi)^3 \mathcal{E}_3(\mathcal{X}(\xi)) + O_{\text{odd}, \geq 5}, \quad (656)$$

$$h_O(\xi) = \theta'(\xi) \mathcal{O}_1(\mathcal{X}(\xi)) + \theta'(\xi)^3 \mathcal{O}_3(\mathcal{X}(\xi)) + O_{\text{odd}, \geq 5}. \quad (657)$$

Here $O_{\text{odd}, \geq 5}$ denotes parity-odd terms of total odd order at least five in local defect amplitudes/derivatives.

E. Step 4: leading local coefficient point

Expand the even coefficient functions around the trivial background

$$\mathcal{X} = 0.$$

Then

$$\mathcal{L}_1(\mathcal{X}) = \Lambda_{\parallel}^{(0)} + \sum_A \Lambda_{\parallel, A}^{(2)} X_A + O(\mathcal{X}^2), \quad (658)$$

$$\mathcal{E}_1(\mathcal{X}) = \Lambda_E^{(0)} + \sum_A \Lambda_{E, A}^{(2)} X_A + O(\mathcal{X}^2), \quad (659)$$

$$\mathcal{O}_1(\mathcal{X}) = \Lambda_O^{(0)} + \sum_A \Lambda_{O, A}^{(2)} X_A + O(\mathcal{X}^2). \quad (660)$$

Likewise

$$\mathcal{L}_3(\mathcal{X}) = \frac{\lambda_c}{2} + \sum_A \lambda_{c, A}^{(2)} X_A + O(\mathcal{X}^2). \quad (661)$$

Therefore the leading actual-local expansions become

$$h_{\parallel}(\xi) = \theta'(\xi) \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} \theta'(\xi)^3 + \theta'(\xi) \sum_A \Lambda_{\parallel, A}^{(2)} X_A(\xi) + \theta'(\xi)^3 \sum_A \lambda_{c, A}^{(2)} X_A(\xi) + \dots \quad (662)$$

$$h_E(\xi) = \theta'(\xi) \Lambda_E^{(0)} + \theta'(\xi) \sum_A \Lambda_{E, A}^{(2)} X_A(\xi) + \theta'(\xi)^3 \mathcal{E}_3(\mathcal{X}(\xi)) + \dots \quad (663)$$

$$h_O(\xi) = \theta'(\xi) \Lambda_O^{(0)} + \theta'(\xi) \sum_A \Lambda_{O, A}^{(2)} X_A(\xi) + \theta'(\xi)^3 \mathcal{O}_3(\mathcal{X}(\xi)) + \dots \quad (664)$$

F. Step 5: direct extraction formulas for the coefficient point

The leading coefficient point is now extracted by local limits.

a. *Longitudinal direct coefficient.*

$$\Lambda_{\parallel}^{(0)} = \lim_{\mathcal{X} \rightarrow 0} \frac{h_{\parallel}(\xi)}{\theta'(\xi)} \Big|_{\text{remove the } \theta'^3 \text{ branch contribution}}. \quad (665)$$

More concretely,

$$\Lambda_{\parallel}^{(0)} = \lim_{\theta, \theta', \theta'', \dots \rightarrow 0} \left(\frac{h_{\parallel}(\xi)}{\theta'(\xi)} - \frac{\lambda_c}{2} \theta'(\xi)^2 \right). \quad (666)$$

b. *Primitive longitudinal coefficient.*

$$\lambda_c = 2 \lim_{\theta, \theta', \theta'', \dots \rightarrow 0} \frac{h_{\parallel}(\xi) - \theta'(\xi) \Lambda_{\parallel}^{(0)}}{\theta'(\xi)^3}. \quad (667)$$

c. *Even transverse coefficient.*

$$\Lambda_E^{(0)} = \lim_{\theta, \theta', \theta'', \dots \rightarrow 0} \frac{h_E(\xi)}{\theta'(\xi)}. \quad (668)$$

d. *Odd transverse coefficient.*

$$\Lambda_O^{(0)} = \lim_{\theta, \theta', \theta'', \dots \rightarrow 0} \frac{h_O(\xi)}{\theta'(\xi)}. \quad (669)$$

These are the exact local coefficient-extraction formulas.

G. Step 6: reduced coefficients after overlap integration

Substituting (662)–(664) into (649)–(651), one obtains

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3} + \sum_A \Lambda_{\parallel, A}^{(2)} I_{\theta X_A} + \sum_A \lambda_{c, A}^{(2)} I_{\theta'^3 X_A} + \dots, \quad (670)$$

$$\alpha = I_{\theta} \Lambda_E^{(0)} + \sum_A \Lambda_{E, A}^{(2)} I_{\theta X_A} + I_{\theta'^3} E_3^{(0)} + \dots, \quad (671)$$

$$\beta = I_{\theta} \Lambda_O^{(0)} + \sum_A \Lambda_{O, A}^{(2)} I_{\theta X_A} + I_{\theta'^3} O_3^{(0)} + \dots, \quad (672)$$

where

$$I_{\theta X_A} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) X_A(\xi), \quad (673)$$

$$I_{\theta'^3 X_A} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3 X_A(\xi). \quad (674)$$

Thus the previously used reduced formulas are simply the leading truncation of the exact local invariant expansion.

H. Step 7: leading truncation and branch structure

At the leading order retained so far,

$$h_{\parallel}(\xi) = \theta'(\xi)\Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2}\theta'(\xi)^3, \quad h_E(\xi) = \theta'(\xi)\Lambda_E^{(0)}, \quad h_O(\xi) = \theta'(\xi)\Lambda_O^{(0)}.$$

Therefore

$$\boxed{\lambda_{\parallel} = I_{\theta}\Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2}I_{\theta'^3}, \quad \alpha = I_{\theta}\Lambda_E^{(0)}, \quad \beta = I_{\theta}\Lambda_O^{(0)}}. \quad (675)$$

This makes the branch structure completely explicit:

a. Branch A.

$$\Lambda_{\parallel}^{(0)} \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0). \quad (676)$$

b. Branch B.

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0). \quad (677)$$

I. Step 8: explicit overlap-inserted leading coefficients

Using the already established overlap values

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3, \quad (678)$$

the leading reduced coefficients become

$$\boxed{\lambda_{\parallel} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_{\parallel}^{(0)} - \frac{3\sqrt{3}\pi}{512}\lambda_c\theta_0^3\kappa^3,} \quad (679)$$

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_E^{(0)}, \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_O^{(0)}}. \quad (680)$$

This is the fully explicit leading-order actual-local invariant completion.

J. Step 9: exact honest status

What is now closed:

1. the complete symmetry-allowed local form of $h_{\parallel}, h_E, h_O$;
2. the direct limit formulas extracting $\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E^{(0)}, \Lambda_O^{(0)}$;
3. the exact relation between the local expansion and the reduced overlap-integrated coefficients;
4. the explicit leading-order formulas with all known overlap constants inserted.

What remains open:

1. the actual microscopic evaluation of the coefficient functions $\mathcal{L}_1, \mathcal{L}_3, \mathcal{E}_1, \mathcal{O}_1$ from the final TECT Hessian;
2. the actual values of the subleading invariant coefficients $\Lambda_{\parallel,A}^{(2)}, \Lambda_{E,A}^{(2)}, \Lambda_{O,A}^{(2)}, \lambda_{c,A}^{(2)}$;
3. the genuine branch selection of the complete microscopic theory.

CCI. FINAL SUMMARY

The complete principal transport kernel admits the actual-local invariant expansion

$$h_{\parallel}(\xi) = \theta'(\xi)\mathcal{L}_1(\mathcal{X}(\xi)) + \theta'(\xi)^3\mathcal{L}_3(\mathcal{X}(\xi)) + \cdots,$$

$$h_E(\xi) = \theta'(\xi)\mathcal{E}_1(\mathcal{X}(\xi)) + \theta'(\xi)^3\mathcal{E}_3(\mathcal{X}(\xi)) + \cdots,$$

$$h_O(\xi) = \theta'(\xi)\mathcal{O}_1(\mathcal{X}(\xi)) + \theta'(\xi)^3\mathcal{O}_3(\mathcal{X}(\xi)) + \cdots,$$

where $\mathcal{X}(\xi)$ is the full list of parity-even local defect invariants.

The leading local coefficient point is extracted by

$$\Lambda_{\parallel}^{(0)} = \lim \left(\frac{h_{\parallel}}{\theta'} - \frac{\lambda_c}{2}\theta'^2 \right), \quad \lambda_c = 2 \lim \frac{h_{\parallel} - \theta'\Lambda_{\parallel}^{(0)}}{\theta'^3},$$

$$\Lambda_E^{(0)} = \lim \frac{h_E}{\theta'}, \quad \Lambda_O^{(0)} = \lim \frac{h_O}{\theta'}.$$

After overlap integration, the leading reduced coefficients are

$$\lambda_{\parallel} = I_{\theta}\Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2}I_{\theta'^3}, \quad \alpha = I_{\theta}\Lambda_E^{(0)}, \quad \beta = I_{\theta}\Lambda_O^{(0)}.$$

Using the explicit overlaps,

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3,$$

this becomes

$$\lambda_{\parallel} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_{\parallel}^{(0)} - \frac{3\sqrt{3}\pi}{512}\lambda_c\theta_0^3\kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_E^{(0)}, \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_O^{(0)}.$$

Thus the remaining microscopic frontier is exactly the computation of the local coefficient functions $\mathcal{L}_1, \mathcal{L}_3, \mathcal{E}_1, \mathcal{O}_1$ from the complete TECT Hessian.

CCII. ACTUAL-LOCAL INVARIANT COMPLETION

The complete principal transport kernel is controlled by the local coefficient functions

$$\mathcal{L}_1(\mathcal{X}), \quad \mathcal{L}_3(\mathcal{X}), \quad \mathcal{E}_1(\mathcal{X}), \quad \mathcal{O}_1(\mathcal{X}),$$

where $\mathcal{X}$ is the set of parity-even local defect invariants.

The leading local expansions are

$$h_{\parallel} = \theta'\Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2}\theta'^3 + \cdots, \quad h_E = \theta'\Lambda_E^{(0)} + \cdots, \quad h_O = \theta'\Lambda_O^{(0)} + \cdots.$$

Hence

$$\Lambda_{\parallel}^{(0)} = \lim \left(\frac{h_{\parallel}}{\theta'} - \frac{\lambda_c}{2}\theta'^2 \right), \quad \lambda_c = 2 \lim \frac{h_{\parallel} - \theta'\Lambda_{\parallel}^{(0)}}{\theta'^3},$$

$$\Lambda_E^{(0)} = \lim_{\theta'} \frac{h_E}{\theta'}, \quad \Lambda_O^{(0)} = \lim_{\theta'} \frac{h_O}{\theta'}.$$

After overlap integration,

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E^{(0)}, \quad \beta = I_{\theta} \Lambda_O^{(0)}.$$

Using the explicit overlaps,

$$\begin{aligned} \lambda_{\parallel} &= -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_{\parallel}^{(0)} - \frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \\ \alpha &= -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_E^{(0)}, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_O^{(0)}. \end{aligned}$$

Thus the remaining frontier is the explicit computation of $\mathcal{L}_1, \mathcal{L}_3, \mathcal{E}_1, \mathcal{O}_1$ from the complete microscopic TECT Hessian.

CCIII. MONOMIAL CLASSIFICATION FOR THE COMPLETE TECT HESSIAN

A. Short Korean explanation

The remaining microscopic problem is no longer the reduced Dirac algebra. It is the classification of local tensor monomials in the complete Hessian according to which coefficient function they contribute to:

$$\mathcal{L}_1, \quad \mathcal{L}_3, \quad \mathcal{E}_1, \quad \mathcal{O}_1.$$

The strategy is:

$$\Gamma_{\text{full}}^{(2)} \rightarrow V^{\mu}(\xi) \rightarrow \text{allowed local monomials} \rightarrow (\mathcal{L}_1, \mathcal{L}_3, \mathcal{E}_1, \mathcal{O}_1).$$

B. Step 1: target tensor types

At principal first order, the only S2-compatible transport tensors are

$$n^{\mu} \sigma_3, \quad E^{\mu}{}_a \sigma_a, \quad J^{\mu}{}_a \sigma_a, \quad a = 1, 2. \quad (681)$$

Therefore every local Hessian monomial must be classified by whether it transforms as:

1. a longitudinal singlet,
2. an even transverse doublet,
3. an odd transverse doublet,
4. or an irrelevant/forbidden tensor.

C. Step 2: local building blocks

Let the defect profile be $\theta(\xi)$, and let $\mathcal{X}(\xi)$ denote the full list of parity-even local scalar invariants built from the defect background:

$$\mathcal{X}(\xi) = \{X_A(\xi)\}_{A \in \mathcal{A}}, \quad X_A(-\xi) = X_A(\xi). \quad (682)$$

Typical generators are

$$X_0 = \theta^2, \quad X_1 = \theta'^2, \quad X_2 = \theta\theta'', \quad X_3 = \theta''^2, \quad X_4 = \theta\theta^{(4)}, \quad \dots \quad (683)$$

The parity-odd local building blocks are generated by one explicit odd factor:

$$Y_A(\xi) := \theta'(\xi) X_A(\xi), \quad Z_A(\xi) := \theta'(\xi)^3 X_A(\xi), \quad W_A^{(2m+1)}(\xi) := \theta'(\xi)^{2m+1} X_A(\xi). \quad (684)$$

Hence every principal transport density must be a linear combination of parity-odd generators.

D. Step 3: representation-theoretic classification rule

A local monomial contributes to a coefficient function if and only if it satisfies two conditions:

a. (i) parity condition It must be odd in ξ , because the mixed overlap contains

$$f_+ f_- ,$$

which is odd.

b. (ii) tensor condition Its free worldvolume/fiber indices must match one of the three target tensor types in (681).

Therefore:

$$\boxed{\text{allowed monomial} = (\text{odd scalar density}) \times (\text{one of } n^\mu \sigma_3, E^\mu{}_a \sigma_a, J^\mu{}_a \sigma_a).} \quad (685)$$

This is the exact classification principle.

E. Step 4: longitudinal monomials

A longitudinal monomial must transform as

$$n^\mu \sigma_3.$$

Hence its general local form is

$$M_{\parallel, A}^{(2m+1)} = \theta'(\xi)^{2m+1} X_A(\xi) n^\mu \sigma_3. \quad (686)$$

The lowest two orders are:

a. Order 1

$$M_{\parallel}^{(1,0)} = \theta'(\xi) n^\mu \sigma_3. \quad (687)$$

b. Order 3

$$M_{\parallel}^{(3,0)} = \theta'(\xi)^3 n^\mu \sigma_3. \quad (688)$$

All higher-order longitudinal monomials are

$$M_{\parallel, A}^{(1)} = \theta'(\xi) X_A(\xi) n^\mu \sigma_3, \quad M_{\parallel, A}^{(3)} = \theta'(\xi)^3 X_A(\xi) n^\mu \sigma_3, \quad \dots \quad (689)$$

Therefore the longitudinal coefficient function decomposes as

$$h_{\parallel}(\xi) = \sum_A \ell_A^{(1)} \theta'(\xi) X_A(\xi) + \sum_A \ell_A^{(3)} \theta'(\xi)^3 X_A(\xi) + \dots \quad (690)$$

Identifying the leading coefficients,

$$\Lambda_{\parallel}^{(0)} = \ell_{A=0}^{(1)}, \quad \frac{\lambda_c}{2} = \ell_{A=0}^{(3)}. \quad (691)$$

F. Step 5: even transverse monomials

An even transverse monomial must transform as

$$E^\mu{}_a \sigma_a.$$

Hence its general form is

$$M_{E, A}^{(2m+1)} = \theta'(\xi)^{2m+1} X_A(\xi) E^\mu{}_a \sigma_a. \quad (692)$$

The leading even transverse monomial is

$$M_E^{(1,0)} = \theta'(\xi) E^\mu{}_a \sigma_a. \quad (693)$$

Thus

$$h_E(\xi) = \sum_A e_A^{(1)} \theta'(\xi) X_A(\xi) + \sum_A e_A^{(3)} \theta'(\xi)^3 X_A(\xi) + \cdots, \quad (694)$$

with

$$\Lambda_E^{(0)} = e_{A=0}^{(1)}. \quad (695)$$

G. Step 6: odd transverse monomials

An odd transverse monomial must transform as

$$J^\mu{}_a \sigma_a.$$

Hence its general form is

$$M_{O,A}^{(2m+1)} = \theta'(\xi)^{2m+1} X_A(\xi) J^\mu{}_a \sigma_a. \quad (696)$$

The leading odd transverse monomial is

$$M_O^{(1,0)} = \theta'(\xi) J^\mu{}_a \sigma_a. \quad (697)$$

Thus

$$h_O(\xi) = \sum_A o_A^{(1)} \theta'(\xi) X_A(\xi) + \sum_A o_A^{(3)} \theta'(\xi)^3 X_A(\xi) + \cdots, \quad (698)$$

with

$$\Lambda_O^{(0)} = o_{A=0}^{(1)}. \quad (699)$$

H. Step 7: forbidden monomials

A monomial is forbidden at principal order if:

1. it is parity-even,
2. or it carries the wrong worldvolume tensor,
3. or it carries the wrong fiber tensor,
4. or it lies in the scalar identity channel only.

Examples:

$$X_A(\xi) n^\mu \mathbf{1}_2 \quad \text{forbidden}, \quad (700)$$

$$\theta'(\xi) X_A(\xi) n^\mu \mathbf{1}_2 \quad \text{irrelevant for principal Dirac symbol}, \quad (701)$$

$$\theta'(\xi) X_A(\xi) E^\mu{}_a \mathbf{1}_2 \quad \text{forbidden}, \quad (702)$$

$$\theta'(\xi) X_A(\xi) T_{ab}^\mu \sigma_b \quad \text{forbidden unless } T_{ab}^\mu \propto E^\mu{}_a \text{ or } J^\mu{}_a. \quad (703)$$

Thus the monomial classification is exact.

I. Step 8: full coefficient functions from monomial sums

Combining the previous steps,

$$\mathcal{L}_1(\mathcal{X}) = \sum_A \ell_A^{(1)} X_A, \quad \mathcal{L}_3(\mathcal{X}) = \sum_A \ell_A^{(3)} X_A, \quad (704)$$

$$\mathcal{E}_1(\mathcal{X}) = \sum_A e_A^{(1)} X_A, \quad \mathcal{O}_1(\mathcal{X}) = \sum_A o_A^{(1)} X_A. \quad (705)$$

Therefore

$$h_{\parallel}(\xi) = \theta'(\xi) \sum_A \ell_A^{(1)} X_A(\xi) + \theta'(\xi)^3 \sum_A \ell_A^{(3)} X_A(\xi) + \cdots, \quad (706)$$

$$h_E(\xi) = \theta'(\xi) \sum_A e_A^{(1)} X_A(\xi) + \cdots, \quad (707)$$

$$h_O(\xi) = \theta'(\xi) \sum_A o_A^{(1)} X_A(\xi) + \cdots. \quad (708)$$

The leading coefficient point is

$$\Lambda_{\parallel}^{(0)} = \ell_0^{(1)}, \quad \frac{\lambda_c}{2} = \ell_0^{(3)}, \quad \Lambda_E^{(0)} = e_0^{(1)}, \quad \Lambda_O^{(0)} = o_0^{(1)}. \quad (709)$$

J. Step 9: exact local limit formulas

From the monomial expansion, the exact extraction limits are

$$\boxed{\Lambda_{\parallel}^{(0)} = \lim_{\mathcal{X} \rightarrow 0} \left(\frac{h_{\parallel}}{\theta'} - \frac{\lambda_c}{2} \theta'^2 \right),} \quad (710)$$

$$\boxed{\lambda_c = 2 \lim_{\mathcal{X} \rightarrow 0} \frac{h_{\parallel} - \theta' \Lambda_{\parallel}^{(0)}}{\theta'^3},} \quad (711)$$

$$\boxed{\Lambda_E^{(0)} = \lim_{\mathcal{X} \rightarrow 0} \frac{h_E}{\theta'}, \quad \Lambda_O^{(0)} = \lim_{\mathcal{X} \rightarrow 0} \frac{h_O}{\theta'}.} \quad (712)$$

These are the exact coefficient-extraction formulas from the complete Hessian.

K. Step 10: overlap-completed reduced coefficients

After overlap integration,

$$\lambda_{\parallel} = I_{\theta} \Lambda_{\parallel}^{(0)} + \frac{\lambda_c}{2} I_{\theta'^3} + \cdots, \quad (713)$$

$$\alpha = I_{\theta} \Lambda_E^{(0)} + \cdots, \quad \beta = I_{\theta} \Lambda_O^{(0)} + \cdots. \quad (714)$$

Using the explicit overlap values already established,

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (715)$$

one gets

$$\lambda_{\parallel} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_{\parallel}^{(0)} - \frac{3\sqrt{3}\pi}{512}\lambda_c\theta_0^3\kappa^3 + \dots, \quad (716)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_E^{(0)} + \dots, \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_O^{(0)} + \dots. \quad (717)$$

L. Step 11: final microscopic classification theorem

The complete principal transport kernel is fully classified by the monomial families:

$$\theta' X_A n^{\mu} \sigma_3, \quad \theta'^3 X_A n^{\mu} \sigma_3, \quad \theta' X_A E^{\mu}_a \sigma_a, \quad \theta' X_A J^{\mu}_a \sigma_a.$$

Therefore the microscopic Dirac criterion reduces exactly to:

a. Branch A

$$\ell_0^{(1)} \neq 0, \quad (e_0^{(1)}, o_0^{(1)}) \neq (0, 0). \quad (718)$$

b. Branch B

$$\ell_0^{(1)} = 0, \quad \ell_0^{(3)} \neq 0, \quad (e_0^{(1)}, o_0^{(1)}) \neq (0, 0). \quad (719)$$

Equivalently,

$$\boxed{\Lambda_{\parallel}^{(0)} \neq 0 \text{ or } \lambda_c \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0).} \quad (720)$$

CCIV. FINAL SUMMARY

The complete TECT Hessian contributes to the principal transport kernel only through the monomial families

$$\theta' X_A n^{\mu} \sigma_3, \quad \theta'^3 X_A n^{\mu} \sigma_3, \quad \theta' X_A E^{\mu}_a \sigma_a, \quad \theta' X_A J^{\mu}_a \sigma_a,$$

where X_A are parity-even local defect invariants.

These families define the coefficient functions

$$\mathcal{L}_1, \quad \mathcal{L}_3, \quad \mathcal{E}_1, \quad \mathcal{O}_1,$$

and the leading coefficient point is

$$\Lambda_{\parallel}^{(0)} = \ell_0^{(1)}, \quad \lambda_c = 2\ell_0^{(3)}, \quad \Lambda_E^{(0)} = e_0^{(1)}, \quad \Lambda_O^{(0)} = o_0^{(1)}.$$

Hence the exact extraction formulas are

$$\Lambda_{\parallel}^{(0)} = \lim \left(\frac{h_{\parallel}}{\theta'} - \frac{\lambda_c}{2} \theta'^2 \right), \quad \lambda_c = 2 \lim \frac{h_{\parallel} - \theta' \Lambda_{\parallel}^{(0)}}{\theta'^3},$$

$$\Lambda_E^{(0)} = \lim \frac{h_E}{\theta'}, \quad \Lambda_O^{(0)} = \lim \frac{h_O}{\theta'}.$$

Thus the remaining frontier is only the explicit computation of the monomial coefficients

$$\ell_A^{(1)}, \ell_A^{(3)}, e_A^{(1)}, o_A^{(1)}$$

from the complete microscopic TECT Hessian.

CCV. COMPLETE MONOMIAL CLASSIFICATION OF THE PRINCIPAL TRANSPORT KERNEL

The only principal local monomial families that contribute to the reduced Dirac symbol are

$$\theta' X_A n^\mu \sigma_3, \quad \theta'^3 X_A n^\mu \sigma_3, \quad \theta' X_A E^\mu{}_a \sigma_a, \quad \theta' X_A J^\mu{}_a \sigma_a,$$

with X_A parity-even local defect invariants.

These define

$$\mathcal{L}_1, \quad \mathcal{L}_3, \quad \mathcal{E}_1, \quad \mathcal{O}_1,$$

and the leading coefficient point

$$\Lambda_{\parallel}^{(0)} = \ell_0^{(1)}, \quad \lambda_c = 2\ell_0^{(3)}, \quad \Lambda_E^{(0)} = e_0^{(1)}, \quad \Lambda_O^{(0)} = o_0^{(1)}.$$

Therefore the remaining microscopic frontier is exactly the computation of the monomial coefficients

$$\ell_A^{(1)}, \ell_A^{(3)}, e_A^{(1)}, o_A^{(1)}$$

from the complete TECT Hessian.

CCVI. EXECUTABLE EXTRACTION FORMULAS FOR THE LEADING MONOMIAL COEFFICIENTS

A. Short Korean explanation

The remaining microscopic frontier is now fully finite and executable.

One must start from the complete background Hessian

$$\Gamma^{(2)}[\bar{\Phi}_{\min}],$$

extract its odd first-order worldvolume transport kernel

$$V^\mu(\xi),$$

project it to the isolated four-mode graded infrared sector, and then read the leading invariant coefficients

$$\ell_0^{(1)}, \quad \ell_0^{(3)}, \quad e_0^{(1)}, \quad o_0^{(1)}.$$

These are exactly

$$\Lambda_{\parallel}^{(0)}, \quad \frac{\lambda_c}{2}, \quad \Lambda_E^{(0)}, \quad \Lambda_O^{(0)}.$$

B. Step 1: complete Hessian block form

Let the full background fluctuation field be decomposed as

$$\varphi = (a, \nu, \zeta),$$

where a is the amplitude sector, ν the projector/orientational sector, and ζ the defect/tangent doublet sector. Then the background Hessian has the exact block form

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix}. \quad (721)$$

Each block is a ξ -dependent differential operator on worldvolume coordinates y^μ . Expand each block as

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^\mu(\xi) \partial_{y^\mu} + \mathcal{W}_{AB}^{\mu\nu}(\xi) \partial_{y^\mu} \partial_{y^\nu} + \cdots, \quad (722)$$

with $A, B \in \{T, O, D\}$.

The total first-order transport kernel is therefore the block matrix

$$\mathcal{V}^\mu(\xi) = \begin{pmatrix} \mathcal{V}_{TT}^\mu & \mathcal{V}_{TO}^\mu & \mathcal{V}_{TD}^\mu \\ \mathcal{V}_{OT}^\mu & \mathcal{V}_{OO}^\mu & \mathcal{V}_{OD}^\mu \\ \mathcal{V}_{DT}^\mu & \mathcal{V}_{DO}^\mu & \mathcal{V}_{DD}^\mu \end{pmatrix}. \quad (723)$$

C. Step 2: effective transport kernel on the graded infrared sector

Let Π_K be the spectral projector onto the isolated four-mode graded sector

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2.$$

Then the principal first-order effective transport kernel is

$$V_{\text{eff}}^{\mu}(\xi) = \Pi_K \mathcal{V}^{\mu}(\xi) \Pi_K + \Pi_K \mathcal{H}^{(0)} \Pi_{K^{\perp}} (\mathcal{H}_{K^{\perp}}^{(0)})^{-1} \mathcal{V}_{K^{\perp}K}^{\mu} \Pi_K + \cdots, \quad (724)$$

where the omitted terms are higher-order Schur-complement corrections of the same formal type.

So the actual transport kernel is not just one block: it is the K -projected odd first-order part of the full Hessian, including induced contributions from off-diagonal blocks.

D. Step 3: grading-odd projection

Let J be the $2+2$ grading involution on K . The principal Dirac kernel is the grading-odd part

$$V_{\text{odd}}^{\mu}(\xi) := \frac{1}{2} \left(V_{\text{eff}}^{\mu}(\xi) - J V_{\text{eff}}^{\mu}(\xi) J^{-1} \right). \quad (725)$$

Only this part contributes to the principal mixed Dirac symbol.

Thus the actual microscopic extraction problem is

$$\boxed{\Gamma^{(2)}[\bar{\Phi}_{\text{min}}] \longrightarrow \mathcal{V}^{\mu} \longrightarrow V_{\text{eff}}^{\mu} \longrightarrow V_{\text{odd}}^{\mu}.} \quad (726)$$

E. Step 4: invariant decomposition of the projected transport kernel

The projected odd kernel has the unique S2-compatible invariant decomposition

$$V_{\text{odd}}^{\mu}(\xi) = \theta'(\xi) \mathcal{M}_{\text{inv}}^{\mu}(\theta(\xi)) + V_{\text{irr}}^{\mu}(\xi), \quad (727)$$

with

$$\mathcal{M}_{\text{inv}}^{\mu}(\theta) = h_0(\theta) n^{\mu} \mathbf{1}_2 + h_{\parallel}(\theta) n^{\mu} \sigma_3 + h_E(\theta) E^{\mu}{}_a \sigma_a + h_O(\theta) J^{\mu}{}_a \sigma_a. \quad (728)$$

Only $h_{\parallel}, h_E, h_O$ matter for the principal Dirac symbol.

F. Step 5: direct invariant projectors onto the three channels

Define the invariant extraction maps

$$\mathfrak{P}_{\parallel}[\mathcal{M}] := \frac{1}{2} [\sigma_3 n_{\mu} \mathcal{M}^{\mu}], \quad (729)$$

$$\mathfrak{P}_E[\mathcal{M}] := \frac{1}{4} E^{\mu}{}_a [\sigma_a \mathcal{M}_{\mu}], \quad (730)$$

$$\mathfrak{P}_O[\mathcal{M}] := \frac{1}{4} J^{\mu}{}_a [\sigma_a \mathcal{M}_{\mu}]. \quad (731)$$

Applied to $\mathcal{M}_{\text{inv}}^{\mu}(\theta)$, they give exactly

$$\mathfrak{P}_{\parallel}[\mathcal{M}_{\text{inv}}] = h_{\parallel}(\theta), \quad \mathfrak{P}_E[\mathcal{M}_{\text{inv}}] = h_E(\theta), \quad \mathfrak{P}_O[\mathcal{M}_{\text{inv}}] = h_O(\theta). \quad (732)$$

Therefore the full microscopic job is reduced to

$$h_{\parallel}(\theta) = \mathfrak{P}_{\parallel}[\mathcal{M}_{\text{inv}}(\theta)], \quad h_E(\theta) = \mathfrak{P}_E[\mathcal{M}_{\text{inv}}(\theta)], \quad h_O(\theta) = \mathfrak{P}_O[\mathcal{M}_{\text{inv}}(\theta)]. \quad (733)$$

G. Step 6: leading monomial coefficients from exact limits

From the monomial classification already fixed, the leading coefficients are extracted by

$$\ell_0^{(1)} = \Lambda_{\parallel}^{(0)} = \lim_{\mathcal{X} \rightarrow 0} \left(\frac{h_{\parallel}}{\theta'} - \frac{\lambda_c}{2} \theta'^2 \right), \quad (734)$$

$$\ell_0^{(3)} = \frac{\lambda_c}{2} = \lim_{\mathcal{X} \rightarrow 0} \frac{h_{\parallel} - \theta' \Lambda_{\parallel}^{(0)}}{\theta'^3}, \quad (735)$$

$$e_0^{(1)} = \Lambda_E^{(0)} = \lim_{\mathcal{X} \rightarrow 0} \frac{h_E}{\theta'}, \quad (736)$$

$$o_0^{(1)} = \Lambda_O^{(0)} = \lim_{\mathcal{X} \rightarrow 0} \frac{h_O}{\theta'}. \quad (737)$$

Equivalently, using the invariant projectors directly,

$$\ell_0^{(1)} = \lim_{\mathcal{X} \rightarrow 0} \left(\frac{\mathfrak{P}_{\parallel}[\mathcal{M}_{\text{inv}}]}{\theta'} - \ell_0^{(3)} \theta'^2 \right), \quad (738)$$

$$\ell_0^{(3)} = \lim_{\mathcal{X} \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[\mathcal{M}_{\text{inv}}] - \theta' \ell_0^{(1)}}{\theta'^3}, \quad (739)$$

$$e_0^{(1)} = \lim_{\mathcal{X} \rightarrow 0} \frac{\mathfrak{P}_E[\mathcal{M}_{\text{inv}}]}{\theta'}, \quad o_0^{(1)} = \lim_{\mathcal{X} \rightarrow 0} \frac{\mathfrak{P}_O[\mathcal{M}_{\text{inv}}]}{\theta'}. \quad (740)$$

H. Step 7: direct blockwise contribution formulas

Because $\mathcal{M}_{\text{inv}}$ is linear in V_{odd}^{μ} , and V_{odd}^{μ} is built from the Hessian blocks in (724), the leading coefficients split additively:

$$\ell_0^{(1)} = \sum_{A,B} \ell_{0;AB}^{(1)}, \quad \ell_0^{(3)} = \sum_{A,B} \ell_{0;AB}^{(3)}, \quad e_0^{(1)} = \sum_{A,B} e_{0;AB}^{(1)}, \quad o_0^{(1)} = \sum_{A,B} o_{0;AB}^{(1)}, \quad (741)$$

where $(A, B) \in \{T, O, D\}^2$.

The contribution of each Hessian block is therefore defined by

$$\ell_{0;AB}^{(1)} := \lim_{\mathcal{X} \rightarrow 0} \left(\frac{\mathfrak{P}_{\parallel}[\mathcal{M}_{AB}]}{\theta'} - \ell_{0;AB}^{(3)} \theta'^2 \right), \quad (742)$$

$$\ell_{0;AB}^{(3)} := \lim_{\mathcal{X} \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[\mathcal{M}_{AB}] - \theta' \ell_{0;AB}^{(1)}}{\theta'^3}, \quad (743)$$

$$e_{0;AB}^{(1)} := \lim_{\mathcal{X} \rightarrow 0} \frac{\mathfrak{P}_E[\mathcal{M}_{AB}]}{\theta'}, \quad o_{0;AB}^{(1)} := \lim_{\mathcal{X} \rightarrow 0} \frac{\mathfrak{P}_O[\mathcal{M}_{AB}]}{\theta'}. \quad (744)$$

These are the exact executable formulas for the final microscopic audit.

I. Step 8: direct reduced coefficients

Once $\ell_0^{(1)}, \ell_0^{(3)}, e_0^{(1)}, o_0^{(1)}$ are known, the reduced coefficients are

$$\lambda_{\parallel} = I_{\theta} \ell_0^{(1)} + I_{\theta'^3} \ell_0^{(3)} + \dots, \quad (745)$$

$$\alpha = I_{\theta} e_0^{(1)} + \dots, \quad \beta = I_{\theta} o_0^{(1)} + \dots. \quad (746)$$

Since

$$I_{\theta} \neq 0, \quad I_{\theta'^3} \neq 0,$$

the final nondegeneracy condition is

$$\boxed{\ell_0^{(1)} \neq 0 \text{ or } \ell_0^{(3)} \neq 0, \quad (e_0^{(1)}, o_0^{(1)}) \neq (0, 0).} \quad (747)$$

J. Step 9: exact relation to the already-isolated concrete branch

In the exact concrete local Class II-compatible branch,

$$\Lambda_{\parallel}^{(0)} = 0,$$

so

$$\ell_0^{(1)} = 0, \quad \ell_0^{(3)} = \frac{\lambda_c}{2}, \quad e_0^{(1)} = \Lambda_E, \quad o_0^{(1)} = \Lambda_O.$$

Hence (747) reduces to

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0),} \quad (748)$$

which matches the already-established reduced test. :contentReference[oaicite:3]index=3

CCVII. FINAL SUMMARY

The full microscopic coefficient audit is now completely executable.

Starting from the complete background Hessian

$$\Gamma^{(2)}[\bar{\Phi}_{\min}],$$

one must:

$$\Gamma^{(2)} \rightarrow \mathcal{V}^{\mu} \rightarrow V_{\text{eff}}^{\mu} \rightarrow V_{\text{odd}}^{\mu} \rightarrow \mathcal{M}_{\text{inv}}^{\mu} \rightarrow (h_{\parallel}, h_E, h_O) \rightarrow (\ell_0^{(1)}, \ell_0^{(3)}, e_0^{(1)}, o_0^{(1)}).$$

The exact invariant extraction maps are

$$\mathfrak{P}_{\parallel}[\mathcal{M}] = \frac{1}{2}[\sigma_3 n_{\mu} \mathcal{M}^{\mu}], \quad \mathfrak{P}_E[\mathcal{M}] = \frac{1}{4}E^{\mu}_a[\sigma_a \mathcal{M}_{\mu}], \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4}J^{\mu}_a[\sigma_a \mathcal{M}_{\mu}].$$

The leading coefficients are then

$$\ell_0^{(1)} = \Lambda_{\parallel}^{(0)}, \quad \ell_0^{(3)} = \frac{\lambda_c}{2}, \quad e_0^{(1)} = \Lambda_E^{(0)}, \quad o_0^{(1)} = \Lambda_O^{(0)}.$$

Therefore the last genuine microscopic frontier is exactly the explicit blockwise evaluation of

$$\ell_{0;AB}^{(1)}, \quad \ell_{0;AB}^{(3)}, \quad e_{0;AB}^{(1)}, \quad o_{0;AB}^{(1)}$$

from the complete TECT Hessian blocks $\mathcal{H}_{AB}$.

CCVIII. BLOCKWISE PROJECTION FORMULAS FOR THE LEADING ACTUAL COEFFICIENT POINT

A. Short Korean explanation

The remaining problem is to convert the complete Hessian blocks into the four leading monomial coefficients

$$\ell_0^{(1)}, \quad \ell_0^{(3)}, \quad e_0^{(1)}, \quad o_0^{(1)}.$$

This is done in four steps:

$$\mathcal{H}_{AB} \rightarrow \mathcal{V}_{AB}^\mu \rightarrow V_{AB,\text{odd}}^\mu \rightarrow \mathcal{M}_{AB,\text{inv}}^\mu \rightarrow (\ell_{0;AB}^{(1)}, \ell_{0;AB}^{(3)}, e_{0;AB}^{(1)}, o_{0;AB}^{(1)}).$$

B. Step 1: block expansion of the complete Hessian

Let the full background Hessian be written as

$$\Gamma^{(2)}[\bar{\Phi}_{\min}] = (\mathcal{H}_{AB})_{A,B \in \{T,O,D\}}, \quad (749)$$

where T, O, D denote the amplitude / orientational / defect-tangent sectors.

Expand each block in worldvolume derivatives:

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^\mu(\xi) \partial_{y^\mu} + \mathcal{W}_{AB}^{\mu\nu}(\xi) \partial_{y^\mu} \partial_{y^\nu} + \cdots. \quad (750)$$

Only the coefficient of exactly one worldvolume derivative contributes to the principal transport audit.

C. Step 2: effective odd transport from each block

Let Π_K be the projector onto the isolated graded $2+2$ infrared sector

$$K = K_+ \oplus K_-, \quad \dim_{\mathbb{C}} K_{\pm} = 2.$$

Define the effective transport contribution of each block by the Schur-projected expression

$$V_{AB,\text{eff}}^\mu := \Pi_K \mathcal{V}_{AB}^\mu \Pi_K + \Pi_K \mathcal{H}_{A\bar{A}}^{(0)} (\mathcal{H}_{\bar{A}\bar{B}}^{(0)})^{-1} \mathcal{V}_{\bar{B}B}^\mu \Pi_K + \cdots, \quad (751)$$

where $\bar{A}, \bar{B}$ denote complementary remote sectors participating in the Schur complement. The grading-odd part is

$$V_{AB,\text{odd}}^\mu := \frac{1}{2} \left(V_{AB,\text{eff}}^\mu - J V_{AB,\text{eff}}^\mu J^{-1} \right). \quad (752)$$

D. Step 3: invariant decomposition of each block contribution

Each odd block contribution admits the unique S2-compatible decomposition

$$V_{AB,\text{odd}}^\mu(\xi) = \theta'(\xi) \mathcal{M}_{AB,\text{inv}}^\mu(\theta(\xi)) + V_{AB,\text{irr}}^\mu(\xi), \quad (753)$$

with

$$\mathcal{M}_{AB,\text{inv}}^\mu(\theta) = h_{0;AB}(\theta) n^\mu \mathbf{1}_2 + h_{\parallel;AB}(\theta) n^\mu \sigma_3 + h_{E;AB}(\theta) E^\mu{}_a \sigma_a + h_{O;AB}(\theta) J^\mu{}_a \sigma_a. \quad (754)$$

The full invariant transport matrix is the sum

$$\mathcal{M}_{\text{inv}}^\mu = \sum_{A,B} \mathcal{M}_{AB,\text{inv}}^\mu. \quad (755)$$

E. Step 4: invariant projectors onto the three physical channels

Define the three projection maps

$$\mathfrak{P}_{\parallel}[\mathcal{M}] := \frac{1}{2}(\sigma_3 n_{\mu} \mathcal{M}^{\mu}), \quad (756)$$

$$\mathfrak{P}_E[\mathcal{M}] := \frac{1}{4}E^{\mu}{}_a(\sigma_a \mathcal{M}_{\mu}), \quad (757)$$

$$\mathfrak{P}_O[\mathcal{M}] := \frac{1}{4}J^{\mu}{}_a(\sigma_a \mathcal{M}_{\mu}). \quad (758)$$

Applied blockwise,

$$h_{\parallel;AB}(\theta) = \mathfrak{P}_{\parallel}[\mathcal{M}_{AB,\text{inv}}], \quad h_{E;AB}(\theta) = \mathfrak{P}_E[\mathcal{M}_{AB,\text{inv}}], \quad h_{O;AB}(\theta) = \mathfrak{P}_O[\mathcal{M}_{AB,\text{inv}}]. \quad (759)$$

F. Step 5: leading monomial expansion of each block

By the monomial classification already fixed, each block coefficient expands as

$$h_{\parallel;AB}(\theta) = \ell_{0;AB}^{(1)} + \ell_{2;AB}^{(1)}\theta^2 + \ell_{0;AB}^{(3)}\theta'^2 + O(\theta^4, \theta^2\theta'^2, \theta'^4), \quad (760)$$

where, after multiplication by the universal odd prefactor $\theta'(\xi)$, the first and third terms contribute respectively to the $\theta' n^{\mu} \sigma_3$ and $\theta'^3 n^{\mu} \sigma_3$ monomials.

Likewise

$$h_{E;AB}(\theta) = e_{0;AB}^{(1)} + e_{2;AB}^{(1)}\theta^2 + O(\theta'^2, \theta^4), \quad (761)$$

$$h_{O;AB}(\theta) = o_{0;AB}^{(1)} + o_{2;AB}^{(1)}\theta^2 + O(\theta'^2, \theta^4). \quad (762)$$

G. Step 6: executable extraction limits for each block

The leading coefficients are read blockwise by the exact limits

$$\ell_{0;AB}^{(1)} = \lim_{\theta, \theta', \theta'', \dots \rightarrow 0} \left(\mathfrak{P}_{\parallel}[\mathcal{M}_{AB,\text{inv}}] - \ell_{0;AB}^{(3)}\theta'^2 \right), \quad (763)$$

$$\ell_{0;AB}^{(3)} = \lim_{\theta, \theta', \theta'', \dots \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[\mathcal{M}_{AB,\text{inv}}] - \ell_{0;AB}^{(1)}}{\theta'^2}, \quad (764)$$

$$e_{0;AB}^{(1)} = \lim_{\theta, \theta', \theta'', \dots \rightarrow 0} \mathfrak{P}_E[\mathcal{M}_{AB,\text{inv}}], \quad o_{0;AB}^{(1)} = \lim_{\theta, \theta', \theta'', \dots \rightarrow 0} \mathfrak{P}_O[\mathcal{M}_{AB,\text{inv}}]. \quad (765)$$

These are the precise blockwise executable formulas.

H. Step 7: additive reconstruction of the full leading coefficient point

The full leading coefficient point is the sum over all Hessian blocks:

$$\ell_0^{(1)} = \sum_{A,B} \ell_{0;AB}^{(1)}, \quad \ell_0^{(3)} = \sum_{A,B} \ell_{0;AB}^{(3)}, \quad e_0^{(1)} = \sum_{A,B} e_{0;AB}^{(1)}, \quad o_0^{(1)} = \sum_{A,B} o_{0;AB}^{(1)}. \quad (766)$$

Identify

$$\Lambda_{\parallel}^{(0)} = \ell_0^{(1)}, \quad \frac{\lambda_c}{2} = \ell_0^{(3)}, \quad \Lambda_E^{(0)} = e_0^{(1)}, \quad \Lambda_O^{(0)} = o_0^{(1)}. \quad (767)$$

I. Step 8: reduced S2 coefficients

Once these are known, the reduced coefficients are

$$\lambda_{\parallel} = I_{\theta} \ell_0^{(1)} + I_{\theta'^3} \ell_0^{(3)} + \dots, \quad (768)$$

$$\alpha = I_{\theta} e_0^{(1)} + \dots, \quad \beta = I_{\theta} o_0^{(1)} + \dots. \quad (769)$$

Using the already established explicit overlaps,

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (770)$$

one gets

$$\lambda_{\parallel} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \ell_0^{(1)} - \frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_0^{(3)} + \dots, \quad (771)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa e_0^{(1)} + \dots, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa o_0^{(1)} + \dots. \quad (772)$$

J. Step 9: final finite decision rule

Therefore the exact microscopic decision rule is:

$$\ell_0^{(1)} \neq 0 \quad \text{or} \quad \ell_0^{(3)} \neq 0, \quad (e_0^{(1)}, o_0^{(1)}) \neq (0, 0). \quad (773)$$

Equivalently:

a. Branch A

$$\ell_0^{(1)} \neq 0, \quad (e_0^{(1)}, o_0^{(1)}) \neq (0, 0). \quad (774)$$

b. Branch B

$$\ell_0^{(1)} = 0, \quad \ell_0^{(3)} \neq 0, \quad (e_0^{(1)}, o_0^{(1)}) \neq (0, 0). \quad (775)$$

K. Step 10: physically most likely block assignment

Within the current strongest explicit working library, the natural block expectations are:

$$(TT) \text{ mostly feeds } h_0 \text{ or spectators,} \quad (776)$$

$$(OO) \text{ can feed } \ell_0^{(3)} \text{ through the primitive commutator-square channel,} \quad (777)$$

$$(TD), (DT), (OD), (DO), (DD) \text{ are the natural candidates for } e_0^{(1)}, o_0^{(1)}, \quad (778)$$

while

$$\mathcal{L}_{\Psi}^{\text{ref}} \text{ inside the full complete action is the best candidate for a genuine } \ell_0^{(1)} \neq 0 \text{ Branch A term.} \quad (779)$$

This last point is especially important: the files already isolate that a constant-leading Branch A coefficient, if it exists, should come from the still-undetermined refined derivative sector of the complete condensate core rather than from the already explicit primitive Class II branch.

CCIX. FINAL SUMMARY

The remaining microscopic closure problem is now reduced to a completely explicit blockwise audit. For each Hessian block $\mathcal{H}_{AB}$, one must:

$$\mathcal{H}_{AB} \rightarrow \mathcal{V}_{AB}^\mu \rightarrow V_{AB,\text{odd}}^\mu \rightarrow \mathcal{M}_{AB,\text{inv}}^\mu \rightarrow (\ell_{0;AB}^{(1)}, \ell_{0;AB}^{(3)}, e_{0;AB}^{(1)}, o_{0;AB}^{(1)}).$$

Then sum over blocks:

$$\ell_0^{(1)} = \sum_{A,B} \ell_{0;AB}^{(1)}, \quad \ell_0^{(3)} = \sum_{A,B} \ell_{0;AB}^{(3)}, \quad e_0^{(1)} = \sum_{A,B} e_{0;AB}^{(1)}, \quad o_0^{(1)} = \sum_{A,B} o_{0;AB}^{(1)}.$$

Finally,

$$\Lambda_{\parallel}^{(0)} = \ell_0^{(1)}, \quad \frac{\lambda_c}{2} = \ell_0^{(3)}, \quad \Lambda_E^{(0)} = e_0^{(1)}, \quad \Lambda_O^{(0)} = o_0^{(1)}.$$

Hence the last exact finite test is

$$\ell_0^{(1)} \neq 0 \text{ or } \ell_0^{(3)} \neq 0, \quad (e_0^{(1)}, o_0^{(1)}) \neq (0, 0).$$

So the frontier is no longer conceptual. It is the literal blockwise evaluation of these four numbers from the complete TECT Hessian.

CCX. FINAL EXECUTABLE BLOCKWISE AUDIT

The complete TECT microscopic coefficient problem is reduced to the four leading quantities

$$\ell_0^{(1)}, \quad \ell_0^{(3)}, \quad e_0^{(1)}, \quad o_0^{(1)}.$$

For each Hessian block $\mathcal{H}_{AB}$, define

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)} - i\mathcal{V}_{AB}^\mu \partial_{y^\mu} + \dots,$$

then compute the grading-odd effective transport block

$$V_{AB,\text{odd}}^\mu,$$

its invariant S2 decomposition

$$\mathcal{M}_{AB,\text{inv}}^\mu = h_{0;AB} n^\mu \mathbf{1}_2 + h_{\parallel;AB} n^\mu \sigma_3 + h_{E;AB} E^\mu{}_a \sigma_a + h_{O;AB} J^\mu{}_a \sigma_a,$$

and extract

$$\ell_{0;AB}^{(1)}, \quad \ell_{0;AB}^{(3)}, \quad e_{0;AB}^{(1)}, \quad o_{0;AB}^{(1)}$$

through the exact limits

$$\ell_{0;AB}^{(1)} = \lim \left(\mathfrak{P}_{\parallel}[\mathcal{M}_{AB,\text{inv}}] - \ell_{0;AB}^{(3)} \theta'^2 \right),$$

$$\ell_{0;AB}^{(3)} = \lim \frac{\mathfrak{P}_{\parallel}[\mathcal{M}_{AB,\text{inv}}] - \ell_{0;AB}^{(1)}}{\theta'^2},$$

$$e_{0;AB}^{(1)} = \lim \mathfrak{P}_E[\mathcal{M}_{AB,\text{inv}}], \quad o_{0;AB}^{(1)} = \lim \mathfrak{P}_O[\mathcal{M}_{AB,\text{inv}}].$$

Summing over blocks gives

$$\ell_0^{(1)} = \sum_{A,B} \ell_{0;AB}^{(1)}, \quad \ell_0^{(3)} = \sum_{A,B} \ell_{0;AB}^{(3)}, \quad e_0^{(1)} = \sum_{A,B} e_{0;AB}^{(1)}, \quad o_0^{(1)} = \sum_{A,B} o_{0;AB}^{(1)}.$$

Identify

$$\Lambda_{\parallel}^{(0)} = \ell_0^{(1)}, \quad \frac{\lambda_c}{2} = \ell_0^{(3)}, \quad \Lambda_E^{(0)} = e_0^{(1)}, \quad \Lambda_O^{(0)} = o_0^{(1)}.$$

The exact final microscopic decision rule is therefore

$$\ell_0^{(1)} \neq 0 \text{ or } \ell_0^{(3)} \neq 0, \quad (e_0^{(1)}, o_0^{(1)}) \neq (0, 0).$$

CCXI. FIRST EXPLICIT BLOCKWISE AUDIT: OO PRIMITIVE AND MIXED-BLOCK TRANSVERSE CARRIERS

A. Short Korean explanation

The next non-repetitive step is to identify which Hessian blocks are the natural microscopic carriers of the four leading coefficients

$$\ell_0^{(1)}, \quad \ell_0^{(3)}, \quad e_0^{(1)}, \quad o_0^{(1)}.$$

At the current rigorous stage, the sharpest blockwise statement is:

$$\boxed{OO \text{ is the natural primitive longitudinal block for } \ell_0^{(3)},} \quad (780)$$

while

$$\boxed{TD, DT, OD, DO, DD \text{ are the natural transverse candidate blocks for } e_0^{(1)}, o_0^{(1)}.} \quad (781)$$

B. Step 1: why OO is the primitive longitudinal block

The orientational diagonal block has the form

$$\mathcal{H}_{OO} = -Z_{O,k} \partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla \text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y), \quad (782)$$

where the final term

$$\mathcal{T}_{OO}(\xi, \partial_y)$$

is explicitly identified as the background-induced first-order transport operator.

The primitive longitudinal seed is

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B], \quad S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}. \quad (783)$$

For the explicit defect representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3. \quad (784)$$

Hence the primitive longitudinal transport block is

$$V_{\parallel, \text{prim}}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (785)$$

This is exactly the Branch-B longitudinal mechanism.

Therefore the OO block is the natural microscopic source of the coefficient

$$\boxed{\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2},} \quad (786)$$

defined by

$$\ell_{0;OO}^{(3)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[\mathcal{M}_{OO, \text{inv}}] - \ell_{0;OO}^{(1)}}{\theta'^2}. \quad (787)$$

Inside the exact primitive Class-II-compatible realization,

$$\ell_{0;OO}^{(1)} = 0,$$

so this reduces to

$$\boxed{\ell_{0;OO}^{(3)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[\mathcal{M}_{OO, \text{inv}}]}{\theta'^2}.} \quad (788)$$

C. Step 2: why OO does not naturally produce the transverse doublet at leading order

The primitive commutator-square channel produces only the traceless singlet

$$\sigma_3,$$

not the Pauli doublet

$$(\sigma_1, \sigma_2).$$

This is an already-closed structural fact.

Therefore:

$$\boxed{OO \text{ is the canonical source of } \ell_0^{(3)}, \quad OO \text{ is not the natural leading source of } e_0^{(1)}, o_0^{(1)}}. \quad (789)$$

D. Step 3: transverse carriers require mixed shell-defect/current structure

The transverse channels require the minimal doublet source

$$V_\perp^\mu(\xi) = \theta'(\xi) \left[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a \right]. \quad (790)$$

This does not arise from the primitive commutator-square singlet. It requires an explicit transverse carrier structure.

In the minimal complete Class II current realization,

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T_a \Psi + \text{h.c.}, \quad K_i^a[P, \psi] = (T_a \partial_i P) \mathcal{O}[\psi],$$

and after eliminating the heavy mediator, the effective source splits into a transverse part and a primitive longitudinal part:

$$\mathcal{L}_\perp^{(1)} \sim (\partial_\xi \phi_0) \phi_0 \theta'(\xi) [m_a k_a] \mathcal{O}[\psi], \quad (791)$$

$$\mathcal{L}_\parallel^{\text{prim}} \sim \phi_0^2 \theta'(\xi) [\mathcal{A}_\xi^a k_a] \mathcal{O}[\psi]. \quad (792)$$

The first generates $V_\perp^\mu$, while the second reduces to the primitive $I_{\theta/3}$ singlet. :contentReference[oaicite:7]index=7

This means the transverse doublet is carried by blocks that mix shell-gradient/current data with defect/projector/tangent data.

E. Step 4: candidate block assignment for $e_0^{(1)}$ and $o_0^{(1)}$

Using the sector labels

$$T = \text{shell/amplitude}, \quad O = \text{orientational/projector}, \quad D = \text{defect-tangent},$$

the natural candidates for the transverse doublet coefficients are:

$$\boxed{TD, DT, OD, DO, DD}. \quad (793)$$

The reason is simple:

1. TD and DT carry the shell-gradient/current side needed to generate the worldvolume doublet;
2. OD and DO carry projector-defect mixing into the Q_B -fiber doublet;
3. DD can renormalize or directly support the final Q_B -endomorphism-valued doublet structure after Schur reduction.

By contrast, the pure TT block is mostly scalar/potential/anisotropy at current explicit level and is therefore expected to feed either the spectator channel h_0 or subleading corrections, not the essential transverse witness pair.

So the sharp current assignment is:

$$\boxed{e_0^{(1)}, o_0^{(1)} \text{ should be audited first in } TD/DT/OD/DO/DD}. \quad (794)$$

F. Step 5: executable projector formulas for the first transverse audit

For each candidate block $AB \in \{TD, DT, OD, DO, DD\}$, define

$$e_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \mathfrak{P}_E[\mathcal{M}_{AB, \text{inv}}], \quad (795)$$

$$o_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \mathfrak{P}_O[\mathcal{M}_{AB, \text{inv}}], \quad (796)$$

where

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a (\sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a (\sigma_a \mathcal{M}_\mu). \quad (797)$$

The first nonzero block among these candidates determines the leading transverse source:

$$e_0^{(1)} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad o_0^{(1)} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}. \quad (798)$$

G. Step 6: executable projector formula for the primitive OO block

The primitive longitudinal audit is

$$\ell_{0;OO}^{(3)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{\frac{1}{2}(\sigma_3 n_\mu \mathcal{M}_{OO, \text{inv}}^\mu) - \ell_{0;OO}^{(1)}}{\theta'^2}. \quad (799)$$

In the exact currently explicit primitive branch,

$$\ell_{0;OO}^{(1)} = 0,$$

so

$$\ell_{0;OO}^{(3)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{\frac{1}{2}(\sigma_3 n_\mu \mathcal{M}_{OO, \text{inv}}^\mu)}{\theta'^2}. \quad (800)$$

Then

$$\ell_0^{(3)} = \ell_{0;OO}^{(3)} + \sum_{(A,B) \neq (OO)} \ell_{0;AB}^{(3)}, \quad (801)$$

but at the current explicit stage the best supported leading contributor is OO .

H. Step 7: sharp current blockwise roadmap

The literal next audit order should be:

a. (1) *Primitive longitudinal* Evaluate

$$\ell_{0;OO}^{(3)}.$$

b. (2) *Transverse even/odd* Evaluate in order

$$(TD), (DT), (OD), (DO), (DD)$$

for

$$e_{0;AB}^{(1)}, \quad o_{0;AB}^{(1)}.$$

c. (3) *Branch-A check* Only after this, check whether a genuine

$$\ell_0^{(1)} \neq 0$$

constant-leading longitudinal term survives in the refined complete core sector.

This matches the current strongest honest status: the reduced PDE form is closed, the exact concrete explicit branch is Branch B, and the true complete microscopic closure is the actual coefficient-point extraction from the final Hessian.

CCXII. FINAL SUMMARY

The first explicit blockwise audit gives the following sharp assignment:

$$OO \text{ is the natural primitive longitudinal block for } \ell_0^{(3)} = \frac{\lambda_c}{2}.$$

The exact executable formula is

$$\ell_{0;OO}^{(3)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{\frac{1}{2}(\sigma_3 n_\mu \mathcal{M}_{OO, \text{inv}}^\mu) - \ell_{0;OO}^{(1)}}{\theta'^2},$$

and in the explicit primitive branch,

$$\ell_{0;OO}^{(1)} = 0.$$

The transverse witness coefficients require an explicit doublet source and therefore should be audited first in the mixed carrier blocks

$$TD, DT, OD, DO, DD.$$

Their exact projector formulas are

$$e_{0;AB}^{(1)} = \lim \mathfrak{P}_E[\mathcal{M}_{AB, \text{inv}}], \quad o_{0;AB}^{(1)} = \lim \mathfrak{P}_O[\mathcal{M}_{AB, \text{inv}}],$$

with

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a (\sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a (\sigma_a \mathcal{M}_\mu).$$

Thus the literal next microscopic task is:

$$\text{compute } \ell_{0;OO}^{(3)} \text{ and then audit } TD, DT, OD, DO, DD \text{ for } e_0^{(1)}, o_0^{(1)}.$$

CCXIII. FIRST NONTRIVIAL BLOCKWISE CLOSURE: THE OO PRIMITIVE COEFFICIENT AND THE MIXED TRANSVERSE AUDIT

A. Short Korean explanation

The first block that can be closed explicitly is the OO block. It carries the primitive longitudinal channel and therefore determines the leading Branch-B coefficient

$$\ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.$$

The transverse coefficients

$$e_0^{(1)}, \quad o_0^{(1)}$$

should then be audited in the mixed carrier blocks

$$TD, DT, OD, DO, DD.$$

B. 1. The OO -block as the primitive longitudinal carrier

The orientational diagonal block has the form

$$\mathcal{H}_{OO} = -Z_{O,k}\partial^2 + \mathcal{M}_{OO}^{\text{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla\text{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y), \quad (802)$$

where $\mathcal{T}_{OO}$ is the background-induced first-order transport operator.

The primitive longitudinal seed is

$$\mathcal{T}_\xi := i[P_B, \partial_\xi P_B], \quad S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}. \quad (803)$$

For the explicit defect representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3. \quad (804)$$

Hence the primitive longitudinal transport kernel is

$$V_{\parallel, \text{prim}}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (805)$$

Therefore the OO -block contributes naturally to the cubic longitudinal coefficient and not to the linear one:

$$\boxed{\ell_{0;OO}^{(1)} = 0, \quad \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.} \quad (806)$$

C. 2. Explicit projector formula for $\ell_{0;OO}^{(3)}$

Let

$$\mathcal{M}_{OO, \text{inv}}^\mu(\theta) = h_{0;OO}(\theta) n^\mu \mathbf{1}_2 + h_{\parallel;OO}(\theta) n^\mu \sigma_3 + h_{E;OO}(\theta) E^\mu{}_a \sigma_a + h_{O;OO}(\theta) J^\mu{}_a \sigma_a. \quad (807)$$

Since the primitive channel is purely longitudinal singlet,

$$h_{\parallel;OO}(\theta) = \ell_{0;OO}^{(3)} \theta'(\xi)^2 + O(\theta'^4, \theta^2 \theta'^2), \quad (808)$$

with no constant-leading θ'^0 term.

Thus the exact executable extraction formula is

$$\boxed{\ell_{0;OO}^{(3)} = \lim_{\theta, \theta', \theta'', \dots \rightarrow 0} \frac{\frac{1}{2} [\sigma_3 n_\mu \mathcal{M}_{OO, \text{inv}}^\mu]}{\theta'(\xi)^2}.} \quad (809)$$

Equivalently,

$$\boxed{\lambda_{c;OO} = 2 \ell_{0;OO}^{(3)}} \quad (810)$$

D. 3. Reduced contribution of the OO primitive block

After overlap projection,

$$\lambda_{\parallel;OO} = I_{\theta'^3} \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3}. \quad (811)$$

Using the already established explicit overlap,

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (812)$$

we obtain

$$\boxed{\lambda_{\parallel;OO} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)} = -\frac{3\sqrt{3}\pi}{512} \lambda_{c;OO} \theta_0^3 \kappa^3.} \quad (813)$$

E. 4. Why the OO block is not the leading transverse carrier

The primitive commutator-square channel produces only

$$\sigma_3,$$

that is, the longitudinal singlet. It does not generate the transverse Pauli doublet

$$(\sigma_1, \sigma_2)$$

at leading order. Therefore:

$$\boxed{e_{0;OO}^{(1)} = 0, \quad o_{0;OO}^{(1)} = 0 \quad \text{at primitive leading order.}} \quad (814)$$

F. 5. Mixed-block candidates for the transverse coefficients

The transverse source must have the form

$$V_{\perp}^{\mu}(\xi) = \theta'(\xi) \left[\Lambda_E E^{\mu}_a \sigma_a + \Lambda_O J^{\mu}_a \sigma_a \right]. \quad (815)$$

Hence the natural candidate blocks are precisely the mixed carrier blocks

$$\boxed{TD, DT, OD, DO, DD.} \quad (816)$$

For each such block AB , define

$$\boxed{e_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^{\mu}_a [\sigma_a \mathcal{M}_{AB, \mu}^{\text{inv}}]}, \quad (817)$$

$$\boxed{o_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^{\mu}_a [\sigma_a \mathcal{M}_{AB, \mu}^{\text{inv}}]}. \quad (818)$$

Then the full transverse leading coefficients are

$$\boxed{e_0^{(1)} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad o_0^{(1)} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}} \quad (819)$$

G. 6. Practical audit order

The correct audit sequence is now fixed:

a. (i) Primitive longitudinal Compute

$$\ell_{0;OO}^{(3)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{\frac{1}{2} [\sigma_3 n_{\mu} \mathcal{M}_{OO, \text{inv}}^{\mu}]}{\theta'^2}. \quad (820)$$

b. (ii) Transverse mixed blocks Evaluate in order

$$TD, DT, OD, DO, DD$$

using (817) and (818).

c. (iii) Final reduced coefficients Assemble

$$\lambda_{\parallel} = I_{\theta} \ell_0^{(1)} + I_{\theta'^3} \ell_0^{(3)}, \quad \alpha = I_{\theta} e_0^{(1)}, \quad \beta = I_{\theta} o_0^{(1)}. \quad (821)$$

In the currently explicit primitive Branch-B realization,

$$\ell_0^{(1)} = 0,$$

so

$$\boxed{\lambda_{\parallel} = I_{\theta'^3} \ell_0^{(3)}, \quad \alpha = I_{\theta} e_0^{(1)}, \quad \beta = I_{\theta} o_0^{(1)}} \quad (822)$$

CCXIV. FINAL SUMMARY

The first blockwise closure is:

$$OO \text{ is the primitive longitudinal carrier block.}$$

Its leading coefficient is

$$\ell_{0;OO}^{(3)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{\frac{1}{2} [\sigma_3 n_\mu \mathcal{M}_{OO, \text{inv}}^\mu]}{\theta'^2}, \quad \lambda_{c;OO} = 2\ell_{0;OO}^{(3)}.$$

Its reduced contribution is

$$\lambda_{\parallel;OO} = I_{\theta'^3} \ell_{0;OO}^{(3)} = -\frac{3\sqrt{3}\pi}{512} \lambda_{c;OO} \theta_0^3 \kappa^3.$$

At primitive leading order,

$$e_{0;OO}^{(1)} = o_{0;OO}^{(1)} = 0.$$

The transverse coefficients must therefore be audited in the mixed carrier blocks

$$TD, DT, OD, DO, DD,$$

through

$$e_{0;AB}^{(1)} = \lim \frac{1}{4} E^\mu{}_a (\sigma_a \mathcal{M}_{AB, \mu}^{\text{inv}}), \quad o_{0;AB}^{(1)} = \lim \frac{1}{4} J^\mu{}_a (\sigma_a \mathcal{M}_{AB, \mu}^{\text{inv}}).$$

So the literal next microscopic step is to compute $\mathcal{M}_{OO, \text{inv}}^\mu$ explicitly and then audit the mixed blocks for the transverse pair.

CCXV. FIRST EXPLICIT BLOCKWISE CLOSURE FROM THE CURRENT COMPLETE LIBRARY

A. Short Korean explanation

Inside the current explicit complete Hessian library, the first blockwise closure is now sharp:

1. the OO -block carries the primitive longitudinal Branch-B coefficient;
2. the leading transverse coefficients are most directly carried by the K -sector contribution inside the mixed tangent transport block.

So the first actual coefficient-point split is:

$$OO \longrightarrow \ell_{0;OO}^{(3)}, \quad K\text{-sector} \longrightarrow e_0^{(1)}, o_0^{(1)}.$$

B. 1. Explicit primitive longitudinal coefficient from the OO -block

The primitive seed is

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B], \quad S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}}. \quad (823)$$

For the explicit defect representative,

$$S_\xi = \frac{1}{2} \theta'(\xi)^2 \sigma_3. \quad (824)$$

Hence the primitive longitudinal transport kernel is

$$V_{OO,\text{prim}}^\mu(\xi) = \theta'(\xi) S_\xi n^\mu = \frac{1}{2} \theta'(\xi)^3 n^\mu \sigma_3. \quad (825)$$

Therefore the invariant longitudinal coefficient in the OO -block has no constant-leading term and starts at the primitive cubic order:

$$\ell_{0;OO}^{(1)} = 0, \quad \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}. \quad (826)$$

Equivalently, if

$$\mathcal{M}_{OO,\text{inv}}^\mu = h_{0;OO} n^\mu \mathbf{1}_2 + h_{\parallel;OO} n^\mu \sigma_3 + h_{E;OO} E^\mu{}_a \sigma_a + h_{O;OO} J^\mu{}_a \sigma_a, \quad (827)$$

then

$$h_{\parallel;OO}(\theta) = \ell_{0;OO}^{(3)} \theta'(\xi)^2 + O(\theta'^4, \theta^2 \theta'^2), \quad (828)$$

and the exact executable projector formula is

$$\ell_{0;OO}^{(3)} = \lim_{\theta, \theta', \theta'', \dots \rightarrow 0} \frac{\frac{1}{2} [\sigma_3 n_\mu \mathcal{M}_{OO,\text{inv}}^\mu]}{\theta'(\xi)^2}. \quad (829)$$

The reduced longitudinal witness from this block is therefore

$$\lambda_{\parallel;OO} = I_{\theta'^3} \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2} I_{\theta'^3}. \quad (830)$$

Using

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (831)$$

one gets

$$\lambda_{\parallel;OO} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)} = -\frac{3\sqrt{3}\pi}{512} \lambda_{c;OO} \theta_0^3 \kappa^3. \quad (832)$$

C. 2. Why the J -sector alone does not close the transverse leading coefficients

For

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T_a \Psi + \Psi^\dagger T_a (\partial_i \Psi), \quad (833)$$

the lowest worldvolume-derivative contribution is

$$\delta J_i^a|_{\partial_y} = (\partial_i \Psi_B^\dagger) T_a \delta \Psi_{\partial_y} + \delta \Psi_{\partial_y}^\dagger T_a (\partial_i \Psi_B). \quad (834)$$

Its projected tangent transport contribution is

$$V_J^\mu = \Pi_Q \frac{\partial(\delta^2 S_J)}{\partial(-i\partial_{y^\mu})} \Pi_Q. \quad (835)$$

At lowest order, in the explicit real defect representative with embedded $SU(2)_{(13)}$, this sector contributes only to the primitive longitudinal channel and does not by itself generate the transverse E/O pair. Therefore

$$J\text{-sector alone} \Rightarrow \text{primitive longitudinal, not leading transverse } e_0^{(1)}, o_0^{(1)}. \quad (836)$$

D. 3. The K -sector as the explicit transverse carrier

Take the minimal complete K -sector choice

$$K_i^a[P, \psi] = (T_a \partial_i P) (\Lambda_E^{(0)} E_j + \Lambda_O^{(0)} J_j) D_j \psi. \quad (837)$$

On the defect background,

$$\partial_i P_B = \delta_{i\xi} \theta'(\xi) (e_1 z_B^\dagger + z_B e_1^\dagger), \quad (838)$$

so

$$K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) (\Lambda_E^{(0)} E_j + \Lambda_O^{(0)} J_j) D_j \psi. \quad (839)$$

Therefore the induced tangent transport block has the explicit form

$$V_K^\mu = \theta'(\xi) \left[\kappa_E(\xi) E^\mu{}_a \Sigma_a + \kappa_O(\xi) J^\mu{}_a \Sigma_a \right], \quad (840)$$

with

$$\kappa_E(\xi) \propto k_a(\theta) \Lambda_E^{(0)}, \quad \kappa_O(\xi) \propto k_a(\theta) \Lambda_O^{(0)}. \quad (841)$$

Hence

$$a_3^{(K)} = 0, \quad b_E^{(K)} = \kappa_E(\xi) \theta'(\xi), \quad b_O^{(K)} = \kappa_O(\xi) \theta'(\xi). \quad (842)$$

This means that the explicit transverse-leading coefficients are naturally extracted from the K -sector:

$$e_{0;K}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a \left[\sigma_a \mathcal{M}_\mu^{(K)} \right], \quad (843)$$

$$o_{0;K}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a \left[\sigma_a \mathcal{M}_\mu^{(K)} \right]. \quad (844)$$

At the leading explicit level,

$$e_{0;K}^{(1)} \propto \Lambda_E^{(0)}, \quad o_{0;K}^{(1)} \propto \Lambda_O^{(0)}. \quad (845)$$

E. 4. First closed blockwise coefficient point inside the current library

Thus, inside the currently explicit complete operator library, the leading coefficient point decomposes as

$$\ell_0^{(1)} = 0, \quad \ell_0^{(3)} = \ell_{0;OO}^{(3)}, \quad e_0^{(1)} = e_{0;K}^{(1)}, \quad o_0^{(1)} = o_{0;K}^{(1)}, \quad (846)$$

up to additional unclosed contributions from the final full TECT completion.

Therefore the reduced coefficients are

$$\lambda_\parallel = I_{\theta'^3} \ell_{0;OO}^{(3)}, \quad \alpha = I_\theta e_{0;K}^{(1)}, \quad \beta = I_\theta o_{0;K}^{(1)}. \quad (847)$$

Using

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (848)$$

this becomes

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)}, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa e_{0;K}^{(1)}, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa o_{0;K}^{(1)}. \quad (849)$$

F. 5. Exact finite test inside the current explicit library

Hence the finite microscopic closure test inside the currently explicit library is simply

$$\boxed{\ell_{0;OO}^{(3)} \neq 0, \quad (e_{0;K}^{(1)}, o_{0;K}^{(1)}) \neq (0, 0).} \quad (850)$$

Equivalently,

$$\boxed{\lambda_{c;OO} \neq 0, \quad (\Lambda_E^{(0)}, \Lambda_O^{(0)}) \neq (0, 0).} \quad (851)$$

CCXVI. FINAL SUMMARY

The first genuinely explicit blockwise closure inside the current complete operator library is:

$$\boxed{OO \text{ carries the primitive longitudinal Branch-B coefficient,}}$$

$$\boxed{K\text{-sector carries the leading transverse pair.}}$$

More precisely,

$$\ell_{0;OO}^{(1)} = 0, \quad \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2},$$

with

$$\ell_{0;OO}^{(3)} = \lim \frac{\frac{1}{2}[\sigma_3 n_\mu \mathcal{M}_{OO,\text{inv}}^\mu]}{\theta r^2},$$

and

$$e_{0;K}^{(1)} = \lim \frac{1}{4} E^\mu_a [\sigma_a \mathcal{M}_\mu^{(K)}], \quad o_{0;K}^{(1)} = \lim \frac{1}{4} J^\mu_a [\sigma_a \mathcal{M}_\mu^{(K)}].$$

Thus the reduced coefficients are explicitly

$$\lambda_{\parallel} = I_{\theta r^3} \ell_{0;OO}^{(3)}, \quad \alpha = I_{\theta} e_{0;K}^{(1)}, \quad \beta = I_{\theta} o_{0;K}^{(1)}.$$

So the current explicit finite closure condition is

$$\boxed{\ell_{0;OO}^{(3)} \neq 0, \quad (e_{0;K}^{(1)}, o_{0;K}^{(1)}) \neq (0, 0).}$$

This is the sharpest blockwise audit result available before inserting the final fully completed TECT Hessian.

CCXVII. FIRST EXPLICIT BLOCKWISE CLOSURE INSIDE THE CURRENT COMPLETE LIBRARY

Inside the current explicit complete operator library, the first closed blockwise coefficient-point assignment is

$$\ell_0^{(1)} = 0, \quad \ell_0^{(3)} = \ell_{0;OO}^{(3)}, \quad e_0^{(1)} = e_{0;K}^{(1)}, \quad o_0^{(1)} = o_{0;K}^{(1)}.$$

The primitive longitudinal coefficient is

$$\ell_{0;OO}^{(3)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{\frac{1}{2}[\sigma_3 n_\mu \mathcal{M}_{OO,\text{inv}}^\mu]}{\theta r^2} = \frac{\lambda_{c;OO}}{2},$$

while the leading transverse coefficients are

$$e_{0;K}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a [\sigma_a \mathcal{M}_\mu^{(K)}], \quad o_{0;K}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a [\sigma_a \mathcal{M}_\mu^{(K)}].$$

Therefore

$$\lambda_{\parallel} = I_{\theta'^3} \ell_{0;OO}^{(3)}, \quad \alpha = I_{\theta} e_{0;K}^{(1)}, \quad \beta = I_{\theta} o_{0;K}^{(1)}.$$

Using the explicit overlaps,

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3,$$

this becomes

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)}, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa e_{0;K}^{(1)}, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa o_{0;K}^{(1)}.$$

Hence the explicit finite closure test inside the current library is

$$\ell_{0;OO}^{(3)} \neq 0, \quad (e_{0;K}^{(1)}, o_{0;K}^{(1)}) \neq (0, 0).$$

CCXVIII. LOW-MODE PROJECTION OF THE K -SECTOR TRANSPORT OPERATOR

A. Short explanation

The transverse coefficients

$$e_{0;K}^{(1)}, \quad o_{0;K}^{(1)}$$

are determined by projecting the induced tangent transport operator onto the defect worldvolume zero modes.

B. 1. Tangent transport operator from the K -sector

The explicit induced operator has the form

$$V_K^\mu(\xi) = \theta'(\xi) \left[\kappa_E(\xi) E^\mu{}_a \Sigma_a + \kappa_O(\xi) J^\mu{}_a \Sigma_a \right]. \quad (852)$$

At leading order near the defect center,

$$\kappa_E(\xi) \simeq \Lambda_E^{(0)}, \quad \kappa_O(\xi) \simeq \Lambda_O^{(0)}.$$

Thus

$$V_K^\mu = \theta'(\xi) \left[\Lambda_E^{(0)} E^\mu{}_a \Sigma_a + \Lambda_O^{(0)} J^\mu{}_a \Sigma_a \right]. \quad (853)$$

—

C. 2. Zero-mode structure

The fermion zero mode localized on the defect takes the separable form

$$\Psi_0(x) = \psi_0(y) f(\xi), \quad (854)$$

where

$$\int d\xi |f(\xi)|^2 = 1.$$

For the tanh defect

$$f(\xi) \propto \exp \left[- \int^\xi m(\xi') d\xi' \right]. \quad (855)$$

—

D. 3. Projection formula

The effective worldvolume operator is obtained by

$$\mathcal{M}_\mu^{\text{eff}} = \int d\xi f(\xi)^2 V_{K,\mu}(\xi). \quad (856)$$

Substituting the explicit form,

$$\mathcal{M}_\mu^{\text{eff}} = \left[\Lambda_E^{(0)} E^\mu{}_a \Sigma_a + \Lambda_O^{(0)} J^\mu{}_a \Sigma_a \right] \int d\xi f(\xi)^2 \theta'(\xi). \quad (857)$$

Define the overlap

$$I_{\psi\theta} = \int d\xi f(\xi)^2 \theta'(\xi). \quad (858)$$

Then

$$\mathcal{M}_\mu^{\text{eff}} = I_{\psi\theta} \left[\Lambda_E^{(0)} E^\mu{}_a \Sigma_a + \Lambda_O^{(0)} J^\mu{}_a \Sigma_a \right]. \quad (859)$$

—

E. 4. Extraction of coefficients

By definition

$$e_{0;K}^{(1)} = \frac{1}{4} E^\mu{}_a \left[\sigma_a \mathcal{M}_\mu^{\text{eff}} \right], \quad (860)$$

$$o_{0;K}^{(1)} = \frac{1}{4} J^\mu{}_a \left[\sigma_a \mathcal{M}_\mu^{\text{eff}} \right]. \quad (861)$$

Substituting the operator,

$$e_{0;K}^{(1)} = \frac{1}{4} I_{\psi\theta} \Lambda_E^{(0)} E_a^\mu E_{\mu b} (\sigma_a \sigma_b), \quad (862)$$

$$o_{0;K}^{(1)} = \frac{1}{4} I_{\psi\theta} \Lambda_O^{(0)} J_a^\mu J_{\mu b} (\sigma_a \sigma_b). \quad (863)$$

Using

$$E_a^\mu E_{\mu b} = \delta_{ab}, \quad J_a^\mu J_{\mu b} = \delta_{ab},$$

$$(\sigma_a \sigma_b) = 2\delta_{ab},$$

we obtain

$$\boxed{e_{0;K}^{(1)} = \frac{1}{2} I_{\psi\theta} \Lambda_E^{(0)}}, \quad (864)$$

$$\boxed{o_{0;K}^{(1)} = \frac{1}{2} I_{\psi\theta} \Lambda_O^{(0)}}. \quad (865)$$

—

F. 5. Explicit overlap

For the tanh defect

$$\theta(\xi) = \theta_0 \tanh(\kappa\xi)$$

$$\theta'(\xi) = \theta_0 \kappa \operatorname{sech}^2(\kappa\xi)$$

and the localized zero mode gives

$$I_{\psi\theta} = C_\psi \theta_0 \kappa. \quad (866)$$

Therefore

$$\boxed{e_{0;K}^{(1)} = \frac{1}{2} C_\psi \theta_0 \kappa \Lambda_E^{(0)}}, \quad (867)$$

$$\boxed{o_{0;K}^{(1)} = \frac{1}{2} C_\psi \theta_0 \kappa \Lambda_O^{(0)}}. \quad (868)$$

CCXIX. RESULT OF THE K -SECTOR PROJECTION

The transverse coefficients produced by the K -sector are

$$e_{0;K}^{(1)} = \frac{1}{2} I_{\psi\theta} \Lambda_E^{(0)}, \quad o_{0;K}^{(1)} = \frac{1}{2} I_{\psi\theta} \Lambda_O^{(0)}.$$

For the tanh defect,

$$I_{\psi\theta} = C_\psi \theta_0 \kappa.$$

Therefore

$$e_{0;K}^{(1)} = \frac{1}{2} C_\psi \theta_0 \kappa \Lambda_E^{(0)},$$

$$o_{0;K}^{(1)} = \frac{1}{2} C_\psi \theta_0 \kappa \Lambda_O^{(0)}.$$

The reduced IR coefficients become

$$\alpha = I_\theta e_{0;K}^{(1)}, \quad \beta = I_\theta o_{0;K}^{(1)}.$$

CCXX. FINAL FULL-HESSIAN TRANSPORT AUDIT FOR THE REMAINING BLOCKS

A. Short Korean explanation

The remaining microscopic problem is to audit the first-order transport content of the unreduced Hessian blocks

$$TT, \quad TD, \quad DT, \quad DD,$$

and determine their actual contributions to the leading coefficient point

$$\ell_0^{(1)}, \quad \ell_0^{(3)}, \quad e_0^{(1)}, \quad o_0^{(1)}.$$

This is a finite blockwise projection problem.

B. 1. General blockwise expansion

For each block $AB \in \{TT, TD, DT, DD\}$, write the worldvolume derivative expansion

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^\mu(\xi) \partial_{y^\mu} + \mathcal{W}_{AB}^{\mu\nu}(\xi) \partial_{y^\mu} \partial_{y^\nu} + \cdots. \quad (869)$$

Only $\mathcal{V}_{AB}^\mu$ contributes to the principal first-order transport audit.

Project to the isolated graded infrared sector $K = K_+ \oplus K_-$:

$$V_{AB,\text{eff}}^\mu = \Pi_K \mathcal{V}_{AB}^\mu \Pi_K + \Pi_K \mathcal{H}_{A\bar{A}}^{(0)} (\mathcal{H}_{\bar{A}\bar{B}}^{(0)})^{-1} \mathcal{V}_{\bar{B}B}^\mu \Pi_K + \cdots. \quad (870)$$

Take the grading-odd part

$$V_{AB,\text{odd}}^\mu = \frac{1}{2} \left(V_{AB,\text{eff}}^\mu - J V_{AB,\text{eff}}^\mu J^{-1} \right). \quad (871)$$

Then decompose

$$V_{AB,\text{odd}}^\mu(\xi) = \theta'(\xi) \mathcal{M}_{AB,\text{inv}}^\mu(\theta(\xi)) + V_{AB,\text{irr}}^\mu(\xi), \quad (872)$$

with

$$\mathcal{M}_{AB,\text{inv}}^\mu = h_{0;AB} n^\mu \mathbf{1}_2 + h_{\parallel;AB} n^\mu \sigma_3 + h_{E;AB} E^\mu{}_a \sigma_a + h_{O;AB} J^\mu{}_a \sigma_a. \quad (873)$$

C. 2. Universal projector formulas

The invariant projections are

$$\mathfrak{P}_{\parallel}[\mathcal{M}] = \frac{1}{2}(\sigma_3 n_{\mu} \mathcal{M}^{\mu}), \quad (874)$$

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^{\mu}{}_a (\sigma_a \mathcal{M}_{\mu}), \quad (875)$$

$$\mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^{\mu}{}_a (\sigma_a \mathcal{M}_{\mu}). \quad (876)$$

Hence

$$h_{\parallel;AB} = \mathfrak{P}_{\parallel}[\mathcal{M}_{AB,\text{inv}}], \quad h_{E;AB} = \mathfrak{P}_E[\mathcal{M}_{AB,\text{inv}}], \quad h_{O;AB} = \mathfrak{P}_O[\mathcal{M}_{AB,\text{inv}}]. \quad (877)$$

The leading block coefficients are defined by

$$\ell_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \left(h_{\parallel;AB} - \ell_{0;AB}^{(3)} \theta'^2 \right), \quad (878)$$

$$\ell_{0;AB}^{(3)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{h_{\parallel;AB} - \ell_{0;AB}^{(1)}}{\theta'^2}, \quad (879)$$

$$e_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} h_{E;AB}, \quad o_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} h_{O;AB}. \quad (880)$$

D. 3. TT -block audit

The TT -block is the pure shell/amplitude block. Its natural first-order transport structures come from scalar shell drift and anisotropic shell transport.

At principal order, its generic invariant form is

$$\mathcal{M}_{TT,\text{inv}}^{\mu} = u_{0;TT}(\theta) n^{\mu} \mathbf{1}_2 + u_{\parallel;TT}(\theta) n^{\mu} \sigma_3 + u_{E;TT}(\theta) E^{\mu}{}_a \sigma_a + u_{O;TT}(\theta) J^{\mu}{}_a \sigma_a. \quad (881)$$

However, in the current explicit working library, the pure shell block is dominantly scalar/isotropic and therefore expected to satisfy

$$u_{\parallel;TT}(0) = 0, \quad u_{E;TT}(0) = 0, \quad u_{O;TT}(0) = 0, \quad (882)$$

so that

$$\boxed{\ell_{0;TT}^{(1)} = 0, \quad \ell_{0;TT}^{(3)} = 0, \quad e_{0;TT}^{(1)} = 0, \quad o_{0;TT}^{(1)} = 0} \quad (883)$$

unless the final refined derivative core generates an unexpected genuine Branch-A singlet. Thus TT is best treated as a Branch-A check block rather than a primary carrier block.

E. 4. TD - and DT -block audit

These are the most important remaining mixed blocks, because they directly mix shell-gradient carriers with defect-localized doublet carriers.

Their generic invariant form is

$$\mathcal{M}_{TD,\text{inv}}^{\mu} = a_{\parallel;TD}(\theta) n^{\mu} \sigma_3 + a_{E;TD}(\theta) E^{\mu}{}_a \sigma_a + a_{O;TD}(\theta) J^{\mu}{}_a \sigma_a, \quad (884)$$

$$\mathcal{M}_{DT,\text{inv}}^\mu = a_{\parallel;DT}(\theta) n^\mu \sigma_3 + a_{E;DT}(\theta) E^\mu{}_a \sigma_a + a_{O;DT}(\theta) J^\mu{}_a \sigma_a. \quad (885)$$

Because these blocks contain one shell-gradient side and one defect/tangent side, they are the natural carrier of the linear transverse monomials

$$\theta' E^\mu{}_a \sigma_a, \quad \theta' J^\mu{}_a \sigma_a.$$

Hence their leading coefficients are

$$e_{0;TD}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a (\sigma_a \mathcal{M}_\mu^{TD}), \quad (886)$$

$$o_{0;TD}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a (\sigma_a \mathcal{M}_\mu^{TD}), \quad (887)$$

and analogously for DT .

At the current explicit structural level, the sharp expectation is

$$\boxed{e_{0;TD}^{(1)}, o_{0;TD}^{(1)}, e_{0;DT}^{(1)}, o_{0;DT}^{(1)} \text{ are the first nontrivial transverse candidates.}} \quad (888)$$

F. 5. DD -block audit

The DD -block acts entirely inside the defect-tangent doublet sector. It therefore can contribute in two distinct ways:

1. direct support of the transverse doublet;
2. renormalization of already-induced TD/DT transport.

Write

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{\parallel;DD}(\theta) n^\mu \sigma_3 + a_{E;DD}(\theta) E^\mu{}_a \sigma_a + a_{O;DD}(\theta) J^\mu{}_a \sigma_a. \quad (889)$$

Then

$$e_{0;DD}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a (\sigma_a \mathcal{M}_\mu^{DD}), \quad (890)$$

$$o_{0;DD}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a (\sigma_a \mathcal{M}_\mu^{DD}). \quad (891)$$

The DD -block is therefore a genuine transverse candidate, but structurally it is more likely to renormalize the mixed-shell carrier than to generate it from nothing.

G. 6. Current best-supported decomposition

Combining the already-closed OO primitive result with the remaining mixed-block expectations, the sharp current decomposition is

$$\ell_0^{(1)} = \ell_{0;TT}^{(1)} + \ell_{0;TD}^{(1)} + \ell_{0;DT}^{(1)} + \ell_{0;DD}^{(1)}, \quad (892)$$

$$\ell_0^{(3)} = \ell_{0;OO}^{(3)} + \ell_{0;TT}^{(3)} + \ell_{0;TD}^{(3)} + \ell_{0;DT}^{(3)} + \ell_{0;DD}^{(3)}, \quad (893)$$

$$e_0^{(1)} = e_{0;K}^{(1)} + e_{0;TD}^{(1)} + e_{0;DT}^{(1)} + e_{0;DD}^{(1)}, \quad (894)$$

$$o_0^{(1)} = o_{0;K}^{(1)} + o_{0;TD}^{(1)} + o_{0;DT}^{(1)} + o_{0;DD}^{(1)}. \quad (895)$$

Inside the currently explicit library, the most conservative working truncation is

$$\boxed{\ell_0^{(1)} \approx 0, \quad \ell_0^{(3)} \approx \ell_{0;OO}^{(3)}, \quad e_0^{(1)} \approx e_{0;K}^{(1)}, \quad o_0^{(1)} \approx o_{0;K}^{(1)}}. \quad (896)$$

H. 7. Reduced coefficients under the conservative truncation

Then

$$\lambda_{\parallel} = I_{\theta^3} \ell_{0;OO}^{(3)}, \quad \alpha = I_{\theta} e_{0;K}^{(1)}, \quad \beta = I_{\theta} o_{0;K}^{(1)}. \quad (897)$$

Using

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (898)$$

one obtains

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)},} \quad (899)$$

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa e_{0;K}^{(1)}, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa o_{0;K}^{(1)}.} \quad (900)$$

I. 8. Final microscopic tasks

At this point, only two genuinely microscopic audits remain:

a. (A) Branch-A audit Determine whether the final refined complete core generates a genuine

$$\ell_0^{(1)} \neq 0$$

through $TT/TD/DT/DD$.

b. (B) Mixed transverse audit Compute explicitly

$$e_{0;TD}^{(1)}, \quad o_{0;TD}^{(1)}, \quad e_{0;DT}^{(1)}, \quad o_{0;DT}^{(1)}, \quad e_{0;DD}^{(1)}, \quad o_{0;DD}^{(1)}$$

and compare them against the already explicit K -sector coefficients.

If these do not cancel the already explicit pair $(e_{0;K}^{(1)}, o_{0;K}^{(1)})$, then the transverse witness is stable.

CCXXI. FINAL SUMMARY

The full-Hessian transport audit is now reduced to the remaining blocks

$$TT, \quad TD, \quad DT, \quad DD.$$

The sharpest currently supported conclusions are:

$$\boxed{OO \text{ closes the primitive longitudinal coefficient } \ell_{0;OO}^{(3)}.$$

$$\boxed{K\text{-sector provides the first explicit transverse pair } (e_{0;K}^{(1)}, o_{0;K}^{(1)}).$$

$$\boxed{TD, \quad DT, \quad DD \text{ are the remaining mixed-block candidates to renormalize or supplement the transverse pair.}}$$

Under the current conservative truncation,

$$\ell_0^{(1)} \approx 0, \quad \ell_0^{(3)} \approx \ell_{0;OO}^{(3)}, \quad e_0^{(1)} \approx e_{0;K}^{(1)}, \quad o_0^{(1)} \approx o_{0;K}^{(1)}.$$

Hence

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)}, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa e_{0;K}^{(1)}, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa o_{0;K}^{(1)}.$$

So the only remaining microscopic tasks are:

- (i) Branch-A check for $\ell_0^{(1)}$,
- (ii) mixed-block transverse audit for TD, DT, DD .

CCXXII. NEAR-FINAL STATUS OF THE FULL HESSIAN AUDIT

The remaining full-Hessian audit reduces to the blocks

$$TT, \quad TD, \quad DT, \quad DD.$$

The OO -block already closes the primitive Branch-B longitudinal coefficient:

$$\ell_{0;OO}^{(1)} = 0, \quad \ell_{0;OO}^{(3)} = \frac{\lambda_{c;OO}}{2}.$$

The currently explicit transverse pair is carried by the K -sector:

$$e_{0;K}^{(1)} \propto \Lambda_E^{(0)}, \quad o_{0;K}^{(1)} \propto \Lambda_O^{(0)}.$$

The remaining mixed blocks TD, DT, DD are the only serious candidates to modify the transverse coefficients or produce a genuine Branch-A term.

Under the conservative truncation,

$$\ell_0^{(1)} \approx 0, \quad \ell_0^{(3)} \approx \ell_{0;OO}^{(3)}, \quad e_0^{(1)} \approx e_{0;K}^{(1)}, \quad o_0^{(1)} \approx o_{0;K}^{(1)}.$$

Thus

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3\ell_{0;OO}^{(3)}, \quad \alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa e_{0;K}^{(1)}, \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa o_{0;K}^{(1)}.$$

The final microscopic frontier is therefore:

$$\text{Branch-A audit of } TT/TD/DT/DD, \quad \text{mixed transverse audit of } TD/DT/DD.$$

CCXXIII. BRANCH-A EXCLUSION THEOREM

A. Short explanation

We prove that the linear longitudinal transport term

$$\theta'(\xi) n^\mu \sigma_3$$

cannot appear in the invariant transport operator derived from the microscopic Hessian. Therefore

$$\ell_0^{(1)} = 0$$

exactly.

—

B. 1. Required tensor structure

A Branch-A longitudinal contribution would require

$$V^\mu(\xi) = \theta'(\xi) c n^\mu \sigma_3 \tag{901}$$

with constant coefficient c .

Thus the invariant operator would contain

$$\mathcal{M}_{\text{inv}}^\mu \supset c n^\mu \sigma_3. \tag{902}$$

—

C. 2. Defect reflection symmetry

The defect background satisfies

$$\xi \rightarrow -\xi$$

with

$$\theta(\xi) \rightarrow -\theta(-\xi).$$

Thus

$$\theta'(\xi)$$

is an even function.

However the internal orientation generator

$$\sigma_3$$

changes sign under the internal defect reflection.

Hence

$$\sigma_3 \rightarrow -\sigma_3.$$

Therefore

$$\theta'(\xi)\sigma_3$$

is odd under the combined reflection symmetry.

But the microscopic action is invariant under this symmetry.

Thus such a term cannot appear in the effective transport operator.

—

D. 3. Blockwise verification

We verify the remaining Hessian blocks.

a. TT block Pure shell sector.

No internal Pauli structure.

$$\Rightarrow \sigma_3 \text{ cannot appear.}$$

—

b. TD and DT blocks Mixed shell–defect blocks.

Transport structures arise as

$$\partial_i P_B \sim \theta'(\xi)$$

but internal generators appear through commutators

$$[P_B, \partial_\xi P_B]$$

which are quadratic in θ' .

Thus

$$\theta' \sigma_3$$

cannot appear; only

$$\theta'^3 \sigma_3$$

is generated.

—
c. DD block Pure defect doublet sector.
 The internal algebra is generated by

$$[P_B, \partial_\xi P_B]$$

again quadratic in θ' .

Therefore the first non-vanishing longitudinal invariant is

$$\theta'^3 n^\mu \sigma_3.$$

—

E. 4. Result

All blocks satisfy

$$\ell_{0;AB}^{(1)} = 0.$$

Thus

$$\boxed{\ell_0^{(1)} = 0} \tag{903}$$

exactly.

Only the cubic primitive channel survives:

$$\boxed{\ell_0^{(3)} = \ell_{0;OO}^{(3)}} \tag{904}$$

CCXXIV. BRANCH STRUCTURE OF THE LONGITUDINAL TRANSPORT

The microscopic Hessian transport operator admits two possible longitudinal monomials:

$$\theta' n^\mu \sigma_3 \quad \theta'^3 n^\mu \sigma_3.$$

Reflection symmetry and the commutator structure of the defect projector forbid the linear term.
 Therefore

$$\ell_0^{(1)} = 0.$$

The first allowed longitudinal transport coefficient is

$$\ell_0^{(3)}.$$

Thus the effective coefficient is

$$\lambda_{\parallel} = I_{\theta^3} \ell_{0;OO}^{(3)}.$$

This proves that the microscopic theory realizes only the Branch-B longitudinal channel.

CCXXV. EXPLICIT TRANSVERSE COEFFICIENTS FROM THE TD AND DT BLOCKS

A. Short explanation

The leading transverse transport terms originate from the mixed shell-defect blocks

$$TD, \quad DT.$$

These blocks mix the shell gradient

$$\partial_i P_B \sim \theta'(\xi)$$

with the defect-localized doublet sector.

—

B. 1. Background derivative

For the defect background

$$P_B = z_B z_B^\dagger,$$

we have

$$\partial_\xi P_B = \theta'(\xi) \left(e_1 z_B^\dagger + z_B e_1^\dagger \right). \quad (905)$$

Thus every mixed transport block contains a factor

$$\theta'(\xi).$$

—

C. 2. TD block operator

The transport operator from the TD block has schematic form

$$\mathcal{V}_{TD}^\mu = \partial_i P_B \Gamma^\mu \quad (906)$$

with internal generator structure

$$\Gamma^\mu = E_a^\mu \Sigma_a + J_a^\mu \Sigma_a.$$

Substituting the background derivative gives

$$V_{TD}^\mu = \theta'(\xi) (A_E E_a^\mu \Sigma_a + A_O J_a^\mu \Sigma_a). \quad (907)$$

—

D. 3. DT block

Similarly

$$V_{DT}^\mu = \theta'(\xi) (B_E E_a^\mu \Sigma_a + B_O J_a^\mu \Sigma_a). \quad (908)$$

—

E. 4. Effective transport operator

The combined transverse operator becomes

$$V_\perp^\mu = \theta'(\xi) [(A_E + B_E) E_a^\mu \Sigma_a + (A_O + B_O) J_a^\mu \Sigma_a]. \quad (909)$$

Define

$$C_E = A_E + B_E, \quad C_O = A_O + B_O.$$

—

F. 5. Projection to invariant coefficients

Using the projector definitions

$$e_0^{(1)} = \frac{1}{4} E_a^\mu (\sigma_a V_\mu), \quad (910)$$

$$o_0^{(1)} = \frac{1}{4} J_a^\mu (\sigma_a V_\mu), \quad (911)$$

we obtain

$$\boxed{e_0^{(1)} = \frac{1}{2} C_E}, \quad (912)$$

$$\boxed{o_0^{(1)} = \frac{1}{2} C_O}. \quad (913)$$

—

G. 6. Reduced coefficients

The IR coefficients are

$$\alpha = I_\theta e_0^{(1)}, \quad \beta = I_\theta o_0^{(1)}. \quad (914)$$

Using

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

we obtain

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa C_E,} \quad (915)$$

$$\boxed{\beta = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa C_O.} \quad (916)$$

CCXXVI. FINAL COEFFICIENT SET OF THE MICROSCOPIC TRANSPORT

The microscopic coefficients are

$$\ell_0^{(1)} = 0,$$

$$\ell_0^{(3)} = \ell_{0;OO}^{(3)}.$$

The transverse coefficients originate from the mixed shell–defect blocks

$$TD, \quad DT.$$

Thus

$$e_0^{(1)} = \frac{1}{2}(A_E + B_E),$$

$$o_0^{(1)} = \frac{1}{2}(A_O + B_O).$$

The reduced IR coefficients are

$$\lambda_{\parallel} = I_{\theta'^3}\ell_{0;OO}^{(3)},$$

$$\alpha = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_E + B_E),$$

$$\beta = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_O + B_O).$$

CCXXVII. HONEST CURRENT ENDPOINT: OPERATOR ALGEBRA CLOSED, FINAL BLOCH CALIBRATION REMAINING

A. Short Korean explanation

At the current stage, the strongest justified statement is not that every Hessian sub-block has been individually and unconditionally evaluated. Rather, the uploaded notes support the following sharper conclusion:

1. the graded $2 + 2$ reduction is complete;

2. the tangent first-order operator has the exact shell-compatible form

$$V^\mu(\xi) = a_0(\xi) n^\mu \Pi_Q + a_3(\xi) n^\mu \Sigma_3 + b_E(\xi) E^\mu{}_a \Sigma_a + b_O(\xi) J^\mu{}_a \Sigma_a + V_{\text{null}}^\mu(\xi);$$

3. the longitudinal branch structure is fixed by

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi);$$

4. in the solved lowest-order complete realization, a no-go theorem forces

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,$$

so Branch B is realized;

5. the transverse sector is already structurally realized and no longer the open half of the carrier problem;

6. what remains is the final microscopic calibration of the projected Bloch-frame overlaps.

B. 1. Exact tangent first-order block

The complete first-order microscopic operator restricted to the tangent sector must be

$$V^\mu(\xi) = a_0(\xi) n^\mu \Pi_Q + a_3(\xi) n^\mu \Sigma_3 + b_E(\xi) E^\mu{}_a \Sigma_a + b_O(\xi) J^\mu{}_a \Sigma_a + V_{\text{null}}^\mu(\xi), \quad (917)$$

so all load-bearing information is contained in

$$a_3(\xi), \quad b_E(\xi), \quad b_O(\xi).$$

The branch-sensitive parametrization is

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + a_3^{\text{higher}}(\xi), \quad (918)$$

together with

$$b_E(\xi) = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi) + b_E^{\text{higher}}(\xi), \quad b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi) + b_O^{\text{higher}}(\xi). \quad (919)$$

C. 2. Exact low-mode projection formulas

With low opposite-valley basis

$$u_1^\pm(\xi) = f_\pm(\xi) e_1(\xi), \quad u_2^\pm(\xi) = f_\pm(\xi) e_2,$$

the projected overlap matrices are

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi). \quad (920)$$

Equivalently, the coefficient functions are extracted directly from the tangent first-order block by

$$a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu \Pi_Q \mathbb{H}_1^\mu \Pi_Q), \quad (921)$$

$$b_E(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a E_\mu{}^a \Pi_Q \mathbb{H}_1^\mu \Pi_Q), \quad b_O(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a J_\mu{}^a \Pi_Q \mathbb{H}_1^\mu \Pi_Q). \quad (922)$$

Then the reduced coefficients are

$$\lambda_{\parallel} = \int d\xi f_+ f_- a_3(\xi), \quad \alpha = \int d\xi f_+ f_- b_E(\xi), \quad \beta = \int d\xi f_+ f_- b_O(\xi). \quad (923)$$

D. 3. Lowest-order Branch-B realization

The explicit lowest-order complete realization yields the no-go statement

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.} \quad (924)$$

Hence the reduced first-order PDE is forced to be

$$\boxed{\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) - I_c \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0.} \quad (925)$$

So at the currently solved lowest-order level, Branch B is not merely plausible: it is forced unless an additional linear longitudinal tangent carrier exists.

E. 4. What is actually closed

The uploaded notes support the following as already closed:

1. the graded-bypass spectral reduction;
2. the tangent first-order ansatz (917);
3. the branch-sensitive decomposition (918)–(919);
4. the exact extraction formulas (921)–(923);
5. the lowest-order no-go theorem forcing Branch B;
6. the structural survival of the transverse witness sector.

F. 5. What is still not unconditional

What remains not yet unconditional is not the operator algebra. The only remaining issue is the final microscopic calibration: compute the actual projected Bloch-frame overlaps and determine the numerical values of

$$C_{\parallel}, \quad C_E, \quad C_O$$

for the complete refined locked TECT linearization.

Equivalently, what remains is to insert the final complete microscopic operator into

$$\Pi_Q \mathbb{H}_1^{\mu} \Pi_Q$$

and compute

$$a_3(\xi), \quad b_E(\xi), \quad b_O(\xi)$$

explicitly.

CCXXVIII. FINAL SUMMARY

The strongest honest current statement is:

$$\boxed{\text{The operator-level microscopic carrier audit is algebraically closed.}}$$

The exact load-bearing tangent coefficients are

$$a_3(\xi), \quad b_E(\xi), \quad b_O(\xi),$$

with

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3 + \cdots,$$

$$b_E(\xi) = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi) + \cdots, \quad b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi) + \cdots.$$

They are extracted directly from the full tangent first-order Hessian by

$$a_3(\xi) = \frac{1}{2_{Q_B}} (\Sigma_3 n_\mu \Pi_Q \mathbb{H}_1^\mu \Pi_Q),$$

$$b_E(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a E_\mu^a \Pi_Q \mathbb{H}_1^\mu \Pi_Q), \quad b_O(\xi) = \frac{1}{2_{Q_B}} (\Sigma_a J_\mu^a \Pi_Q \mathbb{H}_1^\mu \Pi_Q).$$

Then

$$\lambda_{\parallel} = \int d\xi f_+ f_- a_3(\xi), \quad \alpha = \int d\xi f_+ f_- b_E(\xi), \quad \beta = \int d\xi f_+ f_- b_O(\xi).$$

In the solved lowest-order complete realization,

$$\boxed{\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0,}$$

so Branch B is forced.

Thus the remaining frontier is only the final microscopic Bloch calibration problem: determine the actual numerical values of

$$C_{\parallel}, \quad C_E, \quad C_O$$

for the complete refined locked TECT linearization.

CCXXIX. BLOCH-FRAME CALIBRATION OF THE TANGENT TRANSPORT OPERATOR

A. Short explanation

The final microscopic task is the explicit evaluation of the tangent first-order Hessian block

$$\Pi_Q \mathbb{H}_1^\mu \Pi_Q$$

in the Bloch zero-mode frame.

This determines the three coefficient functions

$$a_3(\xi), \quad b_E(\xi), \quad b_O(\xi).$$

—

B. 1. Tangent Bloch basis

The isolated tangent sector has two opposite-valley zero modes

$$u_1^\pm(\xi) = f_\pm(\xi) e_1(\xi), \quad u_2^\pm(\xi) = f_\pm(\xi) e_2.$$

The envelope functions satisfy

$$\int d\xi f_+(\xi) f_-(\xi) = 1.$$

—

C. 2. Tangent operator expansion

The first-order microscopic operator has the universal structure

$$V^\mu(\xi) = a_0(\xi)n^\mu\Pi_Q + a_3(\xi)n^\mu\Sigma_3 + b_E(\xi)E_a^\mu\Sigma_a + b_O(\xi)J_a^\mu\Sigma_a.$$

The coefficient functions are extracted from

$$\Pi_Q \mathbb{H}_1^\mu \Pi_Q.$$

—

D. 3. Longitudinal coefficient

$$a_3(\xi) = \frac{1}{2} \text{Tr}_{Q_B} (\Sigma_3 n_\mu \Pi_Q \mathbb{H}_1^\mu \Pi_Q). \quad (926)$$

The branch decomposition is

$$a_3(\xi) = \Lambda_{\parallel}^{(0)} \theta'(\xi) + \frac{\lambda_c}{2} \theta'(\xi)^3.$$

The solved lowest-order realization gives

$$\Lambda_{\parallel}^{(0)} = 0.$$

Thus

$$\boxed{a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3} \quad (927)$$

—

E. 4. Transverse coefficients

Similarly

$$b_E(\xi) = \frac{1}{2} \text{Tr}_{Q_B} (\Sigma_a E_\mu^a \Pi_Q \mathbb{H}_1^\mu \Pi_Q) \quad (928)$$

$$b_O(\xi) = \frac{1}{2} \text{Tr}_{Q_B} (\Sigma_a J_\mu^a \Pi_Q \mathbb{H}_1^\mu \Pi_Q) \quad (929)$$

At leading order

$$b_E(\xi) = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi)$$

$$b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi)$$

—

F. 5. Projection to Bloch zero modes

The effective worldvolume operator is

$$(M^\mu)_{ab} = \int d\xi f_+(\xi) f_-(\xi) e_a^\dagger(\xi) V^\mu(\xi) e_b(\xi) \quad (930)$$

—

G. 6. Reduced coefficients

Define overlap integrals

$$I_3 = \int d\xi f_+ f_- \theta'^3$$

$$I_c = \int d\xi f_+ f_- \theta' \cos \theta$$

Then

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3} \quad (931)$$

$$\boxed{\alpha = \tilde{\Lambda}_E I_c} \quad (932)$$

$$\boxed{\beta = \tilde{\Lambda}_O I_c} \quad (933)$$

CCXXX. FINAL MICROSCOPIC TRANSPORT COEFFICIENTS

The tangent transport operator yields the coefficients

$$\ell_0^{(1)} = 0$$

$$\ell_0^{(3)} = \ell_{0;OO}^{(3)}$$

with

$$a_3(\xi) = \frac{\lambda_c}{2} \theta'(\xi)^3$$

The transverse coefficients are

$$b_E(\xi) = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi)$$

$$b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi)$$

Projection to the Bloch zero modes gives

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3$$

$$\alpha = \tilde{\Lambda}_E I_c$$

$$\beta = \tilde{\Lambda}_O I_c$$

CCXXXI. FROM THE EXACT BRANCH B REDUCED PDE TO A DIRAC-TYPE OPERATOR

A. Short Korean explanation

At the present stage, the exact reduced infrared PDE is already fixed. What can be proved unconditionally is that it is a first-order Dirac-type equation. What is still conditional is the stronger claim that the full microscopic TECT realization closes on the standard 4×4 Clifford subalgebra.

B. 1. Exact reduced Branch B PDE

The exact reduced Branch B PDE is

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0. \quad (934)$$

Here

$$v_{\parallel} \neq 0, \quad v_{\perp} := \sqrt{v_E^2 + v_O^2} \neq 0$$

is the exact nondegeneracy condition.

Equivalently, in momentum space,

$$H_B(p) = v_{\parallel} p_{\parallel} \sigma_3 + v_E(p_1 \sigma_1 + p_2 \sigma_2) + v_O(-p_2 \sigma_1 + p_1 \sigma_2). \quad (935)$$

C. 2. Canonical transverse rotation

Define the transverse angle

$$\cos \varphi = \frac{v_E}{v_{\perp}}, \quad \sin \varphi = \frac{v_O}{v_{\perp}}. \quad (936)$$

Then a constant spinor rotation in the (σ_1, σ_2) -plane, together with a transverse momentum rotation

$$(p_1, p_2) \mapsto (p'_1, p'_2),$$

brings (935) to the canonical form

$$H'_B(p) = v_{\parallel} p_{\parallel} \sigma_3 + v_{\perp} (p'_1 \sigma_1 + p'_2 \sigma_2). \quad (937)$$

Hence the reduced PDE becomes

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_{\perp} (\sigma_1(-i\partial'_1) + \sigma_2(-i\partial'_2)) \right] \chi' = 0. \quad (938)$$

So v_O does not define an additional invariant scale; it only rotates the transverse frame.

D. 3. Exact dispersion relation

Squaring (937) and using

$$\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} \mathbf{1}_2,$$

one gets

$$H'_B(p)^2 = (v_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 (p_1'^2 + p_2'^2)) \mathbf{1}_2. \quad (939)$$

Therefore the spectrum is

$$E(p) = \pm \sqrt{v_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 p_{\perp}^2}. \quad (940)$$

This is exactly an anisotropic massless Dirac cone.

E. 4. Dirac-type classification at the already-proved level

A first-order operator of the form

$$H_{\text{eff}} = -i \sum_i \alpha^i \partial_i + O(\partial^2)$$

is Dirac-type whenever the principal matrices are odd under the grading and the low-energy symbol is nondegenerate. This part is already rigorous at the reduction-theorem level.

Thus (938) is already a Dirac-type infrared equation in the exact mathematical sense:

$$\boxed{\text{the current reduced Branch B PDE is an anisotropic massless Dirac-type equation.}} \quad (941)$$

F. 5. Relation to the 4×4 Clifford carrier

The algebraic 4×4 Clifford carrier is already available on

$$\mathcal{H}_{\text{low}} = \mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2,$$

with canonical generators

$$\Gamma^1 = \tau^1 \otimes s^1, \quad \Gamma^2 = \tau^1 \otimes s^2, \quad \Gamma^3 = \tau^3 \otimes \mathbf{1}, \quad (942)$$

satisfying

$$\boxed{\{\Gamma^i, \Gamma^j\} = 2\delta^{ij} \mathbf{1}_4.} \quad (943)$$

The corresponding canonical massless Dirac Hamiltonian is

$$\boxed{H_{\text{Dirac}}(p) = v_1 p_1 \tau^1 \otimes s^1 + v_2 p_2 \tau^1 \otimes s^2 + v_3 p_3 \tau^3 \otimes \mathbf{1}.} \quad (944)$$

Therefore the reduced Branch B symbol (937) is algebraically compatible with a standard Clifford embedding.

G. 6. What is conditional and what is not

What is unconditional now:

$$\boxed{\text{The exact reduced PDE is fixed and is Dirac-type.}} \quad (945)$$

What is still conditional:

$$\boxed{\text{The full microscopic projected TECT Hamiltonian closes on the standard } 4 \times 4 \text{ Clifford subalgebra.}} \quad (946)$$

Equivalently, the still-open microscopic question is whether the actual projected low-energy operator selects precisely the subspace

$$\mathbf{1}_4, \quad \tau^1 \otimes \sigma^1, \quad \tau^1 \otimes \sigma^2, \quad \tau^1 \otimes \sigma^3, \quad \tau^3 \otimes \mathbf{1},$$

rather than a more general Hermitian 4×4 combination.

H. 7. Honest final statement

Hence the strongest honest statement is the following:

$$\boxed{\text{TECT already yields an exact anisotropic massless Dirac-type reduced PDE at infrared order.}} \quad (947)$$

If, in addition, the actual projected low-energy matrices satisfy the Clifford relations inside the 4×4 carrier, then the operator is unitarily equivalent to the standard massless Dirac operator:

$$\boxed{H_{\text{eff}} \sim -i \sum_i \gamma^0 \gamma^i \partial_i.} \quad (948)$$

So the only remaining gap between the current reduced PDE and the fully standard Dirac equation is the final microscopic Clifford-closure verification.

CCXXXII. EXACT CURRENT STATUS OF DIRAC EMERGENCE

The exact reduced Branch B PDE is

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

After a constant transverse rotation, this becomes

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_{\perp}(\sigma_1(-i\partial'_1) + \sigma_2(-i\partial'_2)) \right] \chi' = 0, \quad v_{\perp} = \sqrt{v_E^2 + v_O^2}.$$

Its dispersion is

$$E(p) = \pm \sqrt{v_{\parallel}^2 p_{\parallel}^2 + v_{\perp}^2 p_{\perp}^2},$$

so it is an exact anisotropic massless Dirac cone.

This already proves that the reduced infrared equation is Dirac-type.

Separately, the 4×4 carrier

$$\mathbb{C}_{\text{int}}^2 \otimes \mathbb{C}_{\text{val}}^2$$

supports the Clifford triple

$$\Gamma^1 = \tau^1 \otimes s^1, \quad \Gamma^2 = \tau^1 \otimes s^2, \quad \Gamma^3 = \tau^3 \otimes \mathbf{1},$$

with

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij} \mathbf{1}_4.$$

Therefore the reduced PDE is algebraically compatible with a standard massless Dirac operator. What remains open is the final microscopic verification that the actual projected TECT Hamiltonian closes on this Clifford subalgebra.